

Although this theorem will provide all the information necessary for the use of integrals in this book, it is more satisfying to have a somewhat larger supply of integrable functions. Several problems treat this question in detail. It will help to know the following three theorems, which show that f is integrable on $[a, b]$, if it is integrable on $[a, c]$ and $[c, b]$; that $f + g$ is integrable if f and g are; and that $c \cdot f$ is integrable if f is integrable and c is any number.

As a simple application of these theorems, recall that if f is 0 except at one point, where its value is 1, then f is integrable. Multiplying this function by c , it follows that the same is true if the value of f at the exceptional point is c . Adding such a function to an integrable function, we see that the value of an integrable function may be changed arbitrarily at one point without destroying integrability. By breaking up the interval into many subintervals, we see that the value can be changed at finitely many points.

The proofs of these theorems usually use the alternative criterion for integrability in Theorem 2; as some of our previous demonstrations illustrate, the details of the argument often conspire to obscure the point of the proof. It is a good idea to attempt proofs of your own, consulting those given here as a last resort, or as a check. This will probably clarify the proofs, and will certainly give good practice in the techniques used in some of the problems.

THEOREM 4 Let $a < c < b$. If f is integrable on $[a, b]$, then f is integrable on $[a, c]$ and on $[c, b]$. Conversely, if f is integrable on $[a, c]$ and on $[c, b]$, then f is integrable on $[a, b]$. Finally, if f is integrable on $[a, b]$, then

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

PROOF Suppose f is integrable on $[a, b]$. If $\varepsilon > 0$, there is a partition $P = \{t_0, \dots, t_n\}$ of $[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon.$$

We might as well assume that $c = t_j$ for some j . (Otherwise, let Q be the partition which contains $t_0, \dots, t_n$ and c ; then Q contains P , so $U(f, Q) - L(f, Q) \leq U(f, P) - L(f, P) < \varepsilon$.)

Now $P' = \{t_0, \dots, t_j\}$ is a partition of $[a, c]$ and $P'' = \{t_j, \dots, t_n\}$ is a partition of $[c, b]$ (Figure 12). Since

$$\begin{aligned} L(f, P) &= L(f, P') + L(f, P''), \\ U(f, P) &= U(f, P') + U(f, P''), \end{aligned}$$

we have

$$[U(f, P') - L(f, P')] + [U(f, P'') - L(f, P'')] = U(f, P) - L(f, P) < \varepsilon.$$

Since each of the terms in brackets is nonnegative, each is less than ε . This shows that f is integrable on $[a, c]$ and $[c, b]$. Note also that

$$\begin{aligned} L(f, P') &\leq \int_a^c f \leq U(f, P'), \\ L(f, P'') &\leq \int_c^b f \leq U(f, P''), \end{aligned}$$

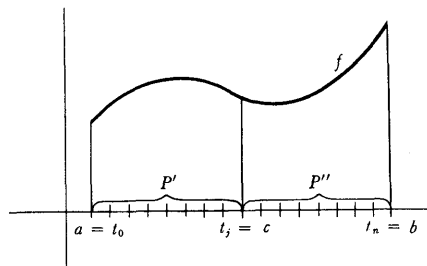


FIGURE 12

so that

$$L(f, P) \leq \int_a^b f + \int_b^c f \leq U(f, P).$$

Since this is true for any P , this proves that

$$\int_a^b f + \int_b^c f = \int_a^c f.$$

Now suppose that f is integrable on $[a, c]$ and on $[c, b]$. If $\varepsilon > 0$, there is a partition P' of $[a, c]$ and a partition P'' of $[c, b]$ such that

$$\begin{aligned} U(f, P') - L(f, P') &< \varepsilon/2, \\ U(f, P'') - L(f, P'') &< \varepsilon/2. \end{aligned}$$

If P is the partition of $[a, b]$ containing all the points of P' and P'' , then

$$\begin{aligned} L(f, P) &= L(f, P') + L(f, P''), \\ U(f, P) &= U(f, P') + U(f, P''); \end{aligned}$$

consequently,

$$U(f, P) - L(f, P) = [U(f, P') - L(f, P')] + [U(f, P'') - L(f, P'')] < \varepsilon. \blacksquare$$

Theorem 4 is the basis for some minor notational conventions. The integral $\int_a^b f$ was defined only for $a < b$. We now add the definitions

$$\int_a^a f = 0 \quad \text{and} \quad \int_a^b f = - \int_b^a f \quad \text{if } a > b.$$

With these definitions, the equation $\int_a^c f + \int_c^b f = \int_a^b f$ holds for all a, c, b even if $a < c < b$ is not true (the proof of this assertion is a rather tedious case-by-case check).

THEOREM 5 If f and g are integrable on $[a, b]$, then $f + g$ is integrable on $[a, b]$ and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

PROOF Let $P = \{t_0, \dots, t_n\}$ be any partition of $[a, b]$. Let

$$\begin{aligned} m_i &= \inf \{ (f + g)(x) : t_{i-1} \leq x \leq t_i \}, \\ m_i' &= \inf \{ f(x) : t_{i-1} \leq x \leq t_i \}, \\ m_i'' &= \inf \{ g(x) : t_{i-1} \leq x \leq t_i \}, \end{aligned}$$

and define M_i, M_i', M_i'' similarly. It is not necessarily true that

$$m_i = m_i' + m_i'',$$

but it is true (Problem 7) that

$$m_i \geq m_i' + m_i''.$$

Similarly,

$$M_i \leq M_i' + M_i''.$$

Therefore,

$$L(f, P) + L(g, P) \leq L(f + g, P)$$

and

$$U(f + g, P) \leq U(f, P) + U(g, P).$$

Thus,

$$L(f, P) + L(g, P) \leq L(f + g, P) \leq U(f + g, P) \leq U(f, P) + U(g, P).$$

Since f and g are integrable, there are partitions P' , P'' with

$$\begin{aligned} U(f, P') - L(f, P') &< \epsilon/2, \\ U(g, P'') - L(g, P'') &< \epsilon/2. \end{aligned}$$

If P contains both P' and P'' , then

$$U(f, P) + U(g, P) - [L(f, P) + L(g, P)] < \epsilon,$$

and consequently

$$U(f + g, P) - L(f + g, P) < \epsilon.$$

This proves that $f + g$ is integrable on $[a, b]$. Moreover,

$$\begin{aligned} (1) \quad L(f, P) + L(g, P) &\leq L(f + g, P) \\ &\leq \int_a^b (f + g) \\ &\leq U(f + g, P) \leq U(f, P) + U(g, P); \end{aligned}$$

and also

$$(2) \quad L(f, P) + L(g, P) \leq \int_a^b f + \int_a^b g \leq U(f, P) + U(g, P).$$

Since $U(f, P) - L(f, P)$ and $U(g, P) - L(g, P)$ can both be made as small as desired, it follows that

$$U(f, P) + U(g, P) - [L(f, P) + L(g, P)]$$

can also be made as small as desired; it therefore follows from (1) and (2) that

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g. \blacksquare$$

THEOREM 6 If f is integrable on $[a, b]$, then for any number c , the function cf is integrable on $[a, b]$ and

$$\int_a^b cf = c \cdot \int_a^b f.$$

PROOF The proof (which is much easier than that of Theorem 5) is left to you. It is a good idea to treat separately the cases $c \geq 0$ and $c \leq 0$. Why? $\blacksquare$

(Theorem 6 is just a special case of the more general theorem that $f \cdot g$ is integrable on $[a, b]$, if f and g are, but this result is quite hard to prove (see Problem 32).)

In this chapter we have acquired only one complicated definition, a few simple theorems with intricate proofs, and one theorem with no proof at all.

This is not because integrals constitute a more difficult topic than derivatives, but because powerful tools developed in previous chapters have been allowed to remain dormant. The most significant discovery of calculus is the fact that the integral and the derivative are intimately related—once we learn the connection, the integral will become as useful as the derivative, and as easy to use. The connection between derivatives and integrals deserves a separate chapter, but the preparations which we will make in this chapter may serve as a hint. We first state a simple inequality concerning integrals, which plays a role in many important theorems.

THEOREM 7 Suppose f is integrable on $[a, b]$ and that

$$m \leq f(x) \leq M \quad \text{for all } x \text{ in } [a, b].$$

Then

$$m(b-a) \leq \int_a^b f \leq M(b-a).$$

PROOF It is clear that

$$m(b-a) \leq L(f, P) \quad \text{and} \quad U(f, P) \leq M(b-a)$$

for every partition P . Since $\int_a^b f = \sup \{L(f, P)\} = \inf \{U(f, P)\}$, the desired inequality follows immediately. ■

Suppose now that f is integrable on $[a, b]$. We can define a new function F on $[a, b]$ by

$$F(x) = \int_a^x f = \int_a^x f(t) \, dt.$$

(This depends on Theorem 4.) We have seen that f may be integrable even if it is not continuous, and the Problems give examples of integrable functions which are quite pathological. The behavior of F is therefore a very pleasant surprise.

THEOREM 8 If f is integrable on $[a, b]$ and F is defined on $[a, b]$ by

$$F(x) = \int_a^x f,$$

then F is continuous on $[a, b]$.

PROOF Suppose c is in $[a, b]$. Since f is integrable on $[a, b]$ it is, by definition, bounded on $[a, b]$; let M be a number such that

$$|f(x)| \leq M \quad \text{for all } x \text{ in } [a, b].$$

If $h > 0$, then (Figure 13)

$$F(c+h) - F(c) = \int_a^{c+h} f - \int_a^c f = \int_c^{c+h} f.$$

Since

$$-M \leq f(x) \leq M \quad \text{for all } x,$$

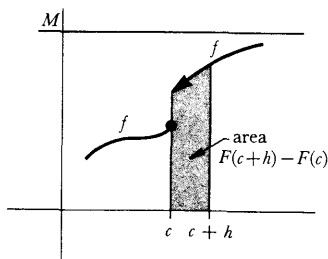


FIGURE 13

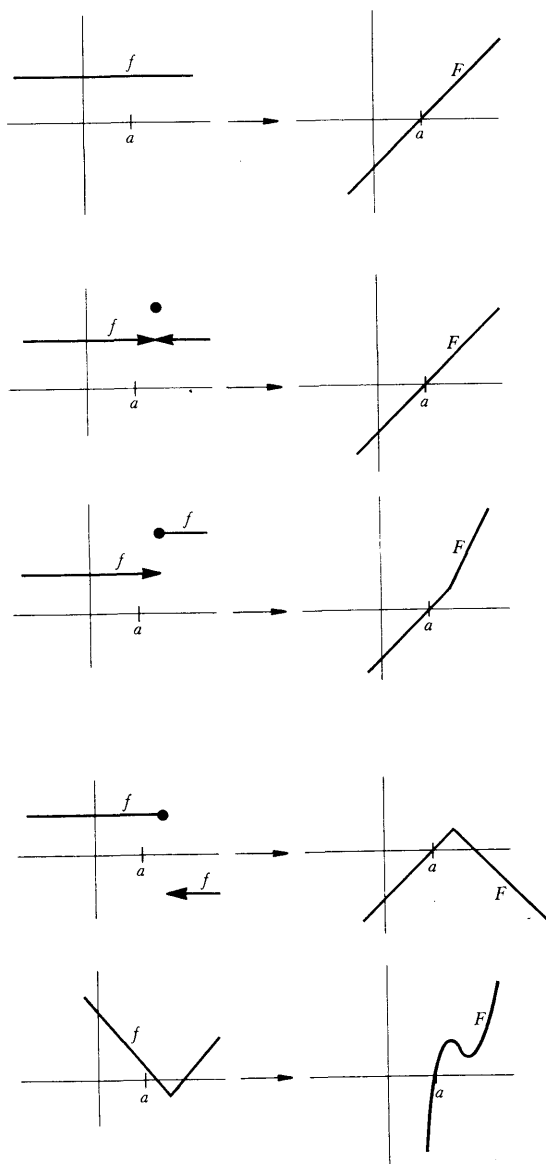


FIGURE 14

it follows from Theorem 7 that

$$-M \cdot h \leq \int_c^{c+h} f \leq Mh;$$

in other words,

$$(1) \quad -M \cdot h \leq F(c+h) - F(c) \leq M \cdot h.$$

If $h < 0$, a similar inequality can be derived. Note that

$$F(c+h) - F(c) = \int_c^{c+h} f = - \int_{c+h}^c f.$$

Applying Theorem 7 to the interval $[c+h, c]$, of length $-h$, we obtain

$$Mh \leq \int_{c+h}^c f \leq -Mh;$$

multiplying by -1 , which reverses all the inequalities, we have

$$(2) \quad Mh \leq F(c+h) - F(c) \leq -Mh.$$

Inequalities (1) and (2) can be combined:

$$|F(c+h) - F(c)| \leq M \cdot |h|.$$

Therefore, if $\varepsilon > 0$, we have

$$|F(c+h) - F(c)| < \varepsilon,$$

provided that $|h| < \varepsilon/M$. This proves that

$$\lim_{h \rightarrow 0} F(c+h) = F(c);$$

in other words F is continuous at c . ■

Figure 14 compares f and $F(x) = \int_a^x f$ for various functions f ; it appears that F is always better behaved than f . In the next chapter we will see how true this is.

PROBLEMS

1. Prove that $\int_0^b x^3 dx = b^4/4$, by considering partitions into n equal sub-intervals, using the formula for $\sum_{i=1}^n i^3$ which was found in Problem 2-5. This problem requires only a straightforward imitation of calculations in the text, but you should write it up as a formal proof to make certain that all the fine points of the argument are clear.
2. Prove, similarly, that $\int_0^b x^4 dx = b^5/5$.
- *3. (a) Using Problem 2-6, show that $\sum_{k=1}^n k^p/n^{p+1}$ can be made as close to $1/(p+1)$ as desired, by choosing n large enough.
(b) Prove that $\int_0^b x^p dx = b^{p+1}/(p+1)$.
4. Decide which of the following functions are integrable on $[0, 2]$, and calculate the integral when you can.
 - (i) $f(x) = \begin{cases} x, & 0 \leq x < 1 \\ x-2, & 1 \leq x \leq 2. \end{cases}$

- (ii) $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ x - 2, & 1 < x \leq 2. \end{cases}$
- (iii) $f(x) = x + [x]$.
- (iv) $f(x) = \begin{cases} x + [x], & x \text{ rational} \\ 0, & x \text{ irrational.} \end{cases}$
- (v) $f(x) = \begin{cases} 1, & x \text{ of the form } a + b\sqrt{2} \text{ for rational } a \text{ and } b \\ 0, & x \text{ not of this form.} \end{cases}$
- (vi) $f(x) = \begin{cases} \frac{1}{[x]}, & 0 < x \leq 1 \\ 0, & x = 0 \text{ or } x > 1. \end{cases}$
- (vii) f is the function shown in Figure 15.

5. Find the areas of the regions bounded by

- (i) the graphs of $f(x) = x^2$ and $g(x) = \frac{x^2}{2} + 2$.
- (ii) the graphs of $f(x) = x^2$ and $g(x) = -x^2$ and the vertical lines through $(-1, 0)$ and $(1, 0)$.
- (iii) the graphs of $f(x) = x^2$ and $g(x) = 1 - x^2$.
- (iv) the graphs of $f(x) = x^2$ and $g(x) = 1 - x^2$ and $h(x) = 2$.
- (v) the graphs of $f(x) = x^2$ and $g(x) = x^2 - 2x + 4$ and the vertical axis.
- (vi) the graph of $f(x) = \sqrt{x}$, the horizontal axis, and the vertical line through $(2, 0)$. (Don't try to find $\int_0^2 \sqrt{x} dx$; you should see a way of guessing the answer, using only integrals that you already know how to evaluate. The questions that this example should suggest are considered in Problem 16.)

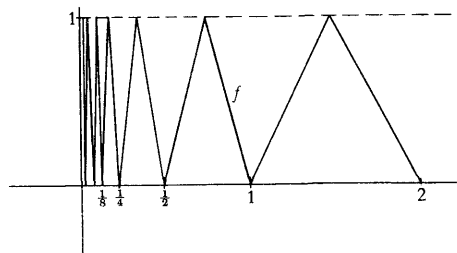


FIGURE 15

6. Find

$$\int_a^b \left(\int_c^d f(x)g(y) dy \right) dx$$

in terms of $\int_a^b f$ and $\int_c^d g$. (This problem is an exercise in notation, with a vengeance; it is crucial that you recognize a constant when it appears.)

7. Prove, using the notation of Theorem 5, that

$$m_i' + m_i'' = \inf \{f(x_1) + g(x_2) : t_{i-1} \leq x_1, x_2 \leq t_i\} \leq m_i.$$

8. (a) Which functions have the property that every lower sum equals every upper sum?
- (b) Which functions have the property that some upper sum equals some (other) lower sum?
- (c) Which continuous functions have the property that all lower sums are equal?

- (d) Which integrable functions have the property that all lower sums are equal? (Bear in mind that one such function is $f(x) = 0$ for x irrational, $f(x) = 1/q$ for $x = p/q$ in lowest terms.) Hint: You will need the notion of a dense set, introduced in Problem 8-6, as well as the results of Problem 21.
9. If $a < b < c < d$ and f is integrable on $[a, d]$, prove that f is integrable on $[b, c]$. (Don't work hard.)
10. (a) Prove that if f is integrable on $[a, b]$ and $f(x) \geq 0$ for all x in $[a, b]$, then $\int_a^b f \geq 0$.
 (b) Prove that if f and g are integrable on $[a, b]$ and $f(x) \geq g(x)$ for all x in $[a, b]$, then $\int_a^b f \geq \int_a^b g$. (By now it should be unnecessary to warn that if you work hard on part (b) you are wasting time.)

11. Prove that

$$\int_a^b f(x) dx = \int_{a+c}^{b+c} f(x-c) dx.$$

(The geometric interpretation should make this very plausible.) Hint: Every partition $P = \{t_0, \dots, t_n\}$ of $[a, b]$ gives rise to a partition $P' = \{t_0 + c, \dots, t_n + c\}$ of $[a+c, b+c]$, and conversely.

- *12. Prove that

$$\int_1^a \frac{1}{t} dt + \int_1^b \frac{1}{t} dt = \int_1^{ab} \frac{1}{t} dt.$$

Hint: This can be written $\int_1^a 1/t dt = \int_b^{ab} 1/t dt$. Every partition $P = \{t_0, \dots, t_n\}$ of $[1, a]$ gives rise to a partition $P' = \{bt_0, \dots, bt_n\}$ of $[b, ab]$, and conversely.

- *13. Prove that

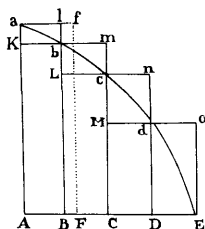
$$\int_{ca}^{cb} f(t) dt = c \int_a^b f(ct) dt.$$

(Notice that Problem 12 is a special case.)

14. Suppose that f is bounded on $[a, b]$ and that f is continuous at each point in $[a, b]$ with the exception of x_0 in (a, b) . Prove that f is integrable on $[a, b]$. Hint: Imitate one of the examples in the text.
15. Suppose that f is nondecreasing on $[a, b]$. Notice that f is automatically bounded on $[a, b]$, because $f(a) \leq f(x) \leq f(b)$ for x in $[a, b]$.
 (a) If $P = \{t_0, \dots, t_n\}$ is a partition of $[a, b]$, what is $L(f, P)$ and $U(f, P)$?
 (b) Suppose that $t_i - t_{i-1} = \delta$ for each i . Prove that $U(f, P) - L(f, P) = \delta [f(b) - f(a)]$.
 (c) Prove that f is integrable.
 (d) Give an example of a nondecreasing function on $[0, 1]$ which is discontinuous at infinitely many points.

It might be of interest to compare this problem with the following extract from Newton's *Principia*.*

LEMMA II



If in any figure AacE, terminated by the right lines Aa, AE, and the curve acE, there be inscribed any number of parallelograms Ab, Bc, Cd, &c., comprehended under equal bases AB, BC, CD, &c., and the sides, Bb, Cc, Dd, &c., parallel to one side Aa of the figure; and the parallelograms aKbl, bLcm, cMdn, &c., are completed: then if the breadth of those parallelograms be supposed to be diminished, and their number to be augmented in infinitum, I say, that the ultimate ratios which the inscribed figure AKbLcMdD, the circumscribed figure AalbmcndoE, and curvilinear figure AabcdE, will have to one another, are ratios of equality.

For the difference of the inscribed and circumscribed figures is the sum of the parallelograms Kl, Lm, Mn, Do, that is (from the equality of all their bases), the rectangle under one of their bases Kb and the sum of their altitudes Aa, that is, the rectangle ABla. But this rectangle, because its breadth AB is supposed diminished in infinitum, becomes less than any given space. And therefore (by Lem. 1) the figures inscribed and circumscribed become ultimately equal one to the other; and much more will the intermediate curvilinear figure be ultimately equal to either. Q.E.D.

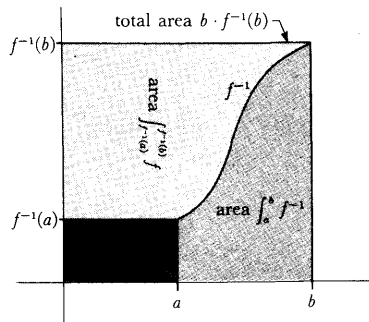


FIGURE 16

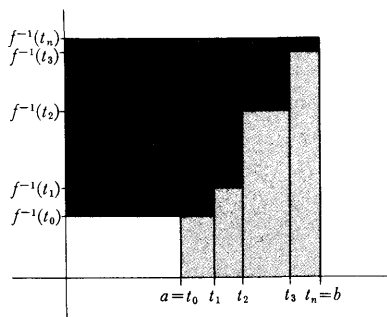


FIGURE 17

- *16. Suppose that f is increasing. Figure 16 suggests that

$$\int_a^b f^{-1} = b f^{-1}(b) - a f^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f.$$

- (a) If $P = \{t_0, \dots, t_n\}$ is a partition of $[a, b]$, let $P' = \{f^{-1}(t_0), \dots, f^{-1}(t_n)\}$. Prove that, as suggested in Figure 17,

$$L(f^{-1}, P) + U(f, P') = b f^{-1}(b) - a f^{-1}(a).$$

- (b) Now prove the formula stated above.

- (c) Find $\int_a^b \sqrt[n]{x} dx$ for $0 \leq a < b$.

17. A function s defined on $[a, b]$ is called a **step function** if there is a partition $P = \{t_0, \dots, t_n\}$ of $[a, b]$ such that s is constant on each (t_{i-1}, t_i) (the values of s at t_i may be arbitrary).

- (a) Prove that if f is integrable on $[a, b]$, then for any $\epsilon > 0$ there is a step function $s_1 \leq f$ with $\int_a^b f - \int_a^b s_1 < \epsilon$, and also a step function $s_2 \geq f$ with $\int_a^b s_2 - \int_a^b f < \epsilon$.
- (b) Suppose that for all $\epsilon > 0$ there are step functions $s_1 \leq f$ and $s_2 \geq f$ such that $\int_a^b s_2 - \int_a^b s_1 < \epsilon$. Prove that f is integrable.

* Newton's *Principia*, A Revision of Mott's Translation, by Florian Cajori. University of California Press, Berkeley, California, 1946.

- (c) Find a function f which is not a step function, but which satisfies $\int_a^b f = L(f, P)$ for some partition P of $[a, b]$.
- *18.** Prove that if f is integrable on $[a, b]$, then for any $\epsilon > 0$ there is a continuous function $g \leq f$ with $\int_a^b f - \int_a^b g < \epsilon$. Hint: First get a step function with this property, and then a continuous one. A picture will help immensely.
- 19.** (a) Show that if s_1 and s_2 are step functions on $[a, b]$, then $s_1 + s_2$ is also.
 (b) Prove, without using Theorem 5, that $\int_a^b (s_1 + s_2) = \int_a^b s_1 + \int_a^b s_2$.
 (c) Use part (b) (and Problem 17) to give an alternative proof of Theorem 5.
- 20.** Suppose that f is integrable on $[a, b]$. Prove that there is a number x in $[a, b]$ such that $\int_a^x f = \int_x^b f$. Show by example that it is *not* always possible to choose x to be in (a, b) .
- *21.** The purpose of this problem is to show that if f is integrable on $[a, b]$, then f must be continuous at many points in $[a, b]$.
- (a) Let $P = \{t_0, \dots, t_n\}$ be a partition of $[0, 1]$ with $U(f, P) - L(f, P) < b - a$. Prove that for some i we have $M_i - m_i < 1$.
 (b) Prove that there are numbers a_1 and b_1 with $a < a_1 < b_1 < b$ and $\sup \{f(x) : a_1 \leq x \leq b_1\} - \inf \{f(x) : a_1 \leq x \leq b_1\} < 1$. (You can choose $[a_1, b_1] = [t_{i-1}, t_i]$ from part (a) unless $i = 1$ or n ; and in these two cases a very simple device solves the problem.)
 (c) Prove that there are numbers a_2 and b_2 with $a_1 < a_2 < b_2 < b_1$ and $\sup \{f(x) : a_2 \leq x \leq b_2\} - \inf \{f(x) : a_2 \leq x \leq b_2\} < \frac{1}{2}$.
 (d) Continue in this way to find a sequence of intervals $I_n = [a_n, b_n]$ such that $\sup \{f(x) : x \text{ in } I_n\} - \inf \{f(x) : x \text{ in } I_n\} < 1/n$. Apply the Nested Intervals Theorem (Problem 8-14) to find a point x at which f is continuous.
 (e) Prove that f is continuous at infinitely many points in $[a, b]$.
- *22.** Recall, from Problem 10, that $\int_a^b f \geq 0$ if $f(x) \geq 0$ for all x in $[a, b]$.
- (a) Give an example where $f(x) \geq 0$ for all x , and $f(x) > 0$ for some x in $[a, b]$, and $\int_a^b f = 0$.
 (b) Suppose $f(x) \geq 0$ for all x in $[a, b]$ and f is continuous at x_0 in $[a, b]$ and $f(x_0) > 0$. Prove that $\int_a^b f > 0$. Hint: It suffices to find one lower sum $L(f, P)$ which is positive.
 (c) Suppose f is integrable on $[a, b]$ and $f(x) > 0$ for all x in $[a, b]$. Prove that $\int_a^b f > 0$. Hint: You will need Problem 21; indeed that was one reason for including Problem 21.

- *23. (a) Suppose that f is continuous on $[a, b]$ and $\int_a^b fg = 0$ for all continuous functions g on $[a, b]$. Prove that $f = 0$. (This is easy; there is an obvious g to choose.)
- (b) Suppose f is continuous on $[a, b]$ and that $\int_a^b fg = 0$ for those continuous functions g on $[a, b]$ which satisfy the extra conditions $g(a) = g(b) = 0$. Prove that $f = 0$. (This innocent looking fact is an important lemma in the calculus of variations; see the Suggested Reading for references.) Hint: Derive a contradiction from the assumption $f(x_0) > 0$ or $f(x_0) < 0$; the g you pick will depend on the behavior of f near x_0 .
24. Let $f(x) = x$ for x rational and $f(x) = 0$ for x irrational.
- (a) Compute $L(f, P)$ for all partitions P of $[0, 1]$.
- (b) Find $\inf \{U(f, P): P \text{ a partition of } [0, 1]\}$.
- *25. Let $f(x) = 0$ for irrational x , and $1/q$ if $x = p/q$ in lowest terms. Show that f is integrable on $[0, 1]$ and that $\int_0^1 f = 0$. (Every lower sum is clearly 0; you must figure out how to make upper sums small.)
- *26. Find two functions f and g which are integrable, but whose composition $g \circ f$ is not. Hint: Problem 25 is relevant.
27. (a) Prove that if $m \leq f(x) \leq M$ for all x in $[a, b]$, then
- $$\int_a^b f(x) dx = (b - a)\mu$$
- for some number μ with $m \leq \mu \leq M$.
- (b) Prove that if f is continuous on $[a, b]$, then
- $$\int_a^b f(x) dx = (b - a)f(\xi)$$
- for some ξ in $[a, b]$; and show by an example that continuity is essential. This fact is called the (First) Mean Value Theorem for Integrals.
28. (a) Suppose that f is continuous on $[a, b]$ and that g is integrable and nonnegative on $[a, b]$. Prove that
- $$\int_a^b f(x)g(x) dx = f(\xi) \int_a^b g(x) dx$$
- for some ξ in $[a, b]$. This result is called the Second (or Generalized) Mean Value Theorem for Integrals. Hint: The inequality $m \leq f(x) \leq M$ implies that $mg(x) \leq f(x)g(x) \leq Mg(x)$, since g is nonnegative.
- (b) Show that the hypothesis for g is essential. (Try $g(x) = x$ on $[-1, 1]$.)

The First and Second Mean Value Theorems for integrals will be used (in a not very essential way) in Chapter 19. The result of the next problem, which I have cribbed from several older calculus texts, is called the Third Mean Value Theorem for Integrals. To be quite frank, I haven't the slightest idea what it is good for.

- *29. (a) Suppose that g is increasing on $[a, b]$ and that f is integrable on $[a, b]$. Prove that there is a number ξ in $[a, b]$ such that

$$\int_a^b f(t)g(t) dt = g(a) \int_a^\xi f(t) dt + g(b) \int_\xi^b f(t) dt.$$

Hint: What is the value of the right side for $\xi = a$ and for $\xi = b$?

- (b) Show that the hypothesis on g is essential. (Try $g(a) = g(b) = 0$.)
- *30. Let f be a bounded function on $[a, b]$ and let P be a partition of $[a, b]$. Let M_i and m_i have their usual meanings, and let M_i' and m_i' have the corresponding meanings for the function $|f|$.

- (a) Prove that

$$\begin{aligned} M_i' &= \max(|M_i|, |m_i|), \\ m_i' &= \min(|M_i|, |m_i|). \end{aligned}$$

- (b) Prove that $M_i' - m_i' \leq M_i - m_i$. Hint: You will have to consider two separate cases, because that's how max and min are defined.
- (c) Prove that if f is integrable on $[a, b]$, then so is $|f|$.
- (d) Prove that if f and g are integrable on $[a, b]$, then so are $\max(f, g)$ and $\min(f, g)$.
- (e) Prove that f is integrable on $[a, b]$ if and only if its "positive part" $\max(f, 0)$ and its "negative part" $\min(f, 0)$ are integrable on $[a, b]$.

31. Prove that if f is integrable on $[a, b]$, then

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt.$$

Hint: This follows easily from a certain string of inequalities; Problem 1-14 is relevant.

- *32. Suppose f and g are integrable on $[a, b]$ and $f(x) \geq 0$ for all x in $[a, b]$. Let P be a partition of $[a, b]$. Let M_i' and m_i' denote the appropriate sup's and inf's for f , define M_i'' and m_i'' similarly for g , and define M_i and m_i similarly for fg .

- (a) Prove that $M_i \leq M_i' M_i''$ and $m_i \geq m_i' m_i''$.
- (b) Show that

$$U(fg, P) - L(fg, P) \leq \sum_{i=1}^n [M_i' M_i'' - m_i' m_i''] (t_i - t_{i-1}).$$

- (c) Using the fact that f and g are bounded, so that $|f(x)|, |g(x)| \leq M$ for x in $[a, b]$, show that

$$\begin{aligned} U(fg, P) - L(fg, P) &\leq M \left\{ \sum_{i=1}^n [M_i' - m_i'] (t_i - t_{i-1}) + \sum_{i=1}^n [M_i'' - m_i''] (t_i - t_{i-1}) \right\}. \end{aligned}$$

- (d) Prove that fg is integrable.
- (e) Now eliminate the restriction that $f(x) \geq 0$ for x in $[a, b]$.

33. Suppose that f and g are integrable on $[a, b]$. The *Cauchy-Schwartz inequality* states that

$$\left(\int_a^b fg\right)^2 \leq \left(\int_a^b f^2\right)\left(\int_a^b g^2\right).$$

- (a) Prove the Cauchy-Schwartz inequality by imitating one of the proofs of the Schwartz inequality in Problem 1-18.
 - (b) If equality holds, is it necessarily true that $f = \lambda g$ for some λ ? What if f and g are continuous?
 - (c) Show that the Schwartz inequality is a special case of the Cauchy-Schwartz inequality.
 - (d) Prove that $\left(\int_0^1 f\right)^2 \leq \left(\int_0^1 f^2\right)$. Is this result true if 0 and 1 are replaced by a and b ?
- *34. Suppose that f is continuous and $\lim_{x \rightarrow \infty} f(x) = a$. Prove that

$$\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(t) dt = a.$$

Hint: The condition $\lim_{x \rightarrow \infty} f(x) = a$ implies that $f(t)$ is close to a for $t \geq$ some N . This means that $\int_N^{N+M} f(t) dt$ is close to Ma . If M is large in comparison to N , then $Ma/(N + M)$ is close to a .

From the hints given in the previous chapter you may have already guessed the first theorem of this chapter. We know that if f is integrable, then $F(x) = \int_a^x f$ is continuous; it is only fitting that we ask what happens when the original function f is continuous. It turns out that F is differentiable (and its derivative is especially simple).

**THEOREM 1 (THE FIRST
FUNDAMENTAL THEOREM
OF CALCULUS)**

Let f be integrable on $[a, b]$, and define F on $[a, b]$ by

$$F(x) = \int_a^x f.$$

If f is continuous at c in $[a, b]$, then F is differentiable at c , and

$$F'(c) = f(c).$$

(If $c = a$ or b , then $F'(c)$ is understood to mean the right- or left-hand derivative of F .)

PROOF We will assume that c is in (a, b) ; the easy modifications for $c = a$ or b may be supplied by the reader. By definition,

$$F'(c) = \lim_{h \rightarrow 0} \frac{F(c+h) - F(c)}{h}.$$

Suppose first that $h > 0$. Then

$$F(c+h) - F(c) = \int_c^{c+h} f.$$

Define m_h and M_h as follows (Figure 1):

$$\begin{aligned} m_h &= \inf \{f(x): c \leq x \leq c+h\}, \\ M_h &= \sup \{f(x): c \leq x \leq c+h\}. \end{aligned}$$

It follows from Theorem 13-7 that

$$m_h \cdot h \leq \int_c^{c+h} f \leq M_h \cdot h.$$

Therefore

$$m_h \leq \frac{F(c+h) - F(c)}{h} \leq M_h.$$

If $h \leq 0$, only a few details of the argument have to be changed. Let

$$\begin{aligned} m_h &= \inf \{f(x): c+h \leq x \leq c\}, \\ M_h &= \sup \{f(x): c+h \leq x \leq c\}. \end{aligned}$$

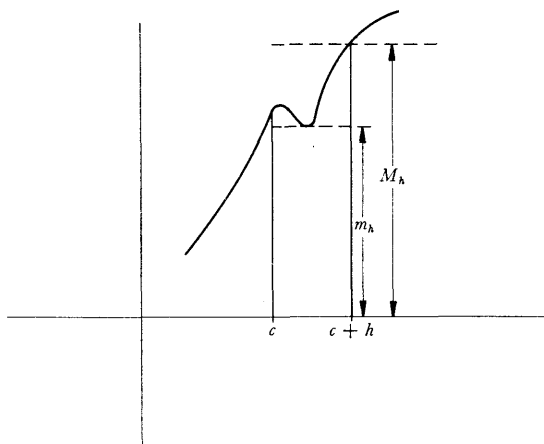


FIGURE 1

Then

$$m_h \cdot (-h) \leq \int_{c+h}^c f \leq M_h \cdot (-h).$$

Since

$$F(c+h) - F(c) = \int_c^{c+h} f = -\int_{c+h}^c f,$$

this yields

$$m_h \cdot h \geq F(c+h) - F(c) \geq M_h \cdot h.$$

Since $h < 0$, dividing by h reverses the inequality again, yielding the same result as before:

$$m_h \leq \frac{F(c+h) - F(c)}{h} \leq M_h.$$

This inequality is true for any integrable function, continuous or not. Since f is continuous at c , however,

$$\lim_{h \rightarrow 0} m_h = \lim_{h \rightarrow 0} M_h = f(c),$$

and this proves that

$$F'(c) = \lim_{h \rightarrow 0} \frac{F(c+h) - F(c)}{h} = f(c). \blacksquare$$

Although Theorem 1 deals only with the function obtained by varying the upper limit of integration, a simple trick shows what happens when the lower limit is varied. If G is defined by

$$G(x) = \int_x^b f,$$

then

$$G(x) = \int_a^b f - \int_a^x f.$$

Consequently, if f is continuous at c , then

$$G'(c) = -f(c).$$

The minus sign appearing here is very fortunate, and allows us to extend Theorem 1 to the situation where the function

$$F(x) = \int_a^x f$$

is defined even for $x < a$. In this case we can write

$$F(x) = - \int_x^a f,$$

so if $c < a$ we have

$$F'(c) = -(-f(c)) = f(c),$$

exactly as before.

Notice that in either case, differentiability of F at c is ensured by continuity of f at c alone. Nevertheless, Theorem 1 is most interesting when f is continuous at all points in $[a, b]$. In this case F is differentiable at all points in $[a, b]$ and

$$F' = f.$$

In general, it is extremely difficult to decide whether a given function f is the derivative of some other function; for this reason Theorem 11-7 and Darboux's Theorem (Problem 11-39) are particularly interesting, since they reveal certain properties which f must have. If f is continuous, however, there is no problem at all—according to Theorem 1, f is the derivative of some function, namely, the function

$$F(x) = \int_a^x f.$$

Theorem 1 has a simple corollary which frequently reduces computations of integrals to a triviality.

COROLLARY If f is continuous on $[a, b]$ and $f = g'$ for some function g , then

$$\int_a^b f = g(b) - g(a).$$

PROOF Let

$$F(x) = \int_a^x f.$$

Then $F' = f = g'$ on $[a, b]$. Consequently, there is a number c such that

$$F = g + c.$$

The number c can be evaluated easily: note that

$$0 = F(a) = g(a) + c,$$

so $c = -g(a)$; thus

$$F(x) = g(x) - g(a).$$

This is true, in particular, for $x = b$. Thus

$$\int_a^b f = F(b) = g(b) - g(a). \blacksquare$$

The proof of this corollary tends, at first sight, to make the corollary seem useless: after all, what good is it to know that

$$\int_a^b f = g(b) - g(a)$$

if g is, for example, $g(x) = \int_a^x f$? The point, of course, is that one might happen to know a quite different function g with this property. For example, if

$$g(x) = \frac{x^3}{3} \quad \text{and} \quad f(x) = x^2,$$

then $g'(x) = f(x)$ so we obtain, without ever computing lower and upper sums:

$$\int_a^b x^2 dx = \frac{b^3}{3} - \frac{a^3}{3}.$$

One can treat other powers similarly; if n is a natural number and $g(x) = x^{n+1}/(n+1)$, then $g'(x) = x^n$, so

$$\int_a^b x^n dx = \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1}.$$

For any natural number n , the function $f(x) = x^{-n}$ is not bounded on any interval containing 0, but if a and b are both positive or both negative, then

$$\int_a^b x^{-n} dx = \frac{b^{-n+1}}{-n+1} - \frac{a^{-n+1}}{-n+1}.$$

Naturally this formula is only true for $n \neq -1$. We do not know a simple expression for

$$\int_a^b \frac{1}{x} dx.$$

The problem of computing this integral is discussed later, but it provides a good opportunity to warn against a serious error. The conclusion of Corollary 1 is often confused with the definition of integrals—many students think that $\int_a^b f$ is defined as: “ $g(b) - g(a)$, where g is a function whose derivative is f .” This “definition” is not only wrong—it is useless. One reason is that a function f may be integrable without being the derivative of another function. For example, if $f(x) = 0$ for $x \neq 1$ and $f(1) = 1$, then f is integrable, but f cannot be a derivative (why not?). There is also another reason that is much more important: If f is continuous, then we know that $f = g'$ for some function g ; but we know this *only because of Theorem 1*. The function $f(x) = 1/x$ provides an excel-

lent illustration: if $x > 0$, then $f(x) = g'(x)$, where

$$g(x) = \int_1^x \frac{1}{t} dt,$$

and we know of no simpler function g with this property.

The corollary to Theorem 1 is so useful that it is frequently called the Second Fundamental Theorem of Calculus. In this book, that name is reserved for a somewhat stronger result (which in practice, however, is not much more useful). As we have just mentioned, a function f might be of the form g' even if f is not continuous. If f is integrable, then it is still true that

$$\int_a^b f = g(b) - g(a).$$

The proof, however, must be entirely different—we cannot use Theorem 1, so we must return to the definition of integrals.

THEOREM 2 (THE SECOND FUNDAMENTAL THEOREM OF CALCULUS)

If f is integrable on $[a, b]$ and $f = g'$ for some function g , then

$$\int_a^b f = g(b) - g(a).$$

PROOF

Let $P = \{t_0, \dots, t_n\}$ be any partition of $[a, b]$. By the Mean Value Theorem there is a point x_i in $[t_{i-1}, t_i]$ such that

$$\begin{aligned} g(t_i) - g(t_{i-1}) &= g'(x_i)(t_i - t_{i-1}) \\ &= f(x_i)(t_i - t_{i-1}). \end{aligned}$$

If

$$\begin{aligned} m_i &= \inf \{f(x): t_{i-1} \leq x \leq t_i\}, \\ M_i &= \sup \{f(x): t_{i-1} \leq x \leq t_i\}, \end{aligned}$$

then clearly

$$m_i(t_i - t_{i-1}) \leq f(x_i)(t_i - t_{i-1}) \leq M_i(t_i - t_{i-1}),$$

that is,

$$m_i(t_i - t_{i-1}) \leq g(t_i) - g(t_{i-1}) \leq M_i(t_i - t_{i-1}).$$

Adding these equations for $i = 1, \dots, n$ we obtain

$$\sum_{i=1}^n m_i(t_i - t_{i-1}) \leq g(b) - g(a) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

so that

$$L(f, P) \leq g(b) - g(a) \leq U(f, P)$$

for every partition P . But this means that

$$g(b) - g(a) = \int_a^b f. \blacksquare$$

We have already used the corollary to Theorem 1 (or, equivalently, Theorem 2) to find the integrals of a few elementary functions:

$$\int_a^b x^n dx = \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1}, \quad n \neq -1. \quad (a \text{ and } b \text{ both positive or both negative if } n < 0.)$$

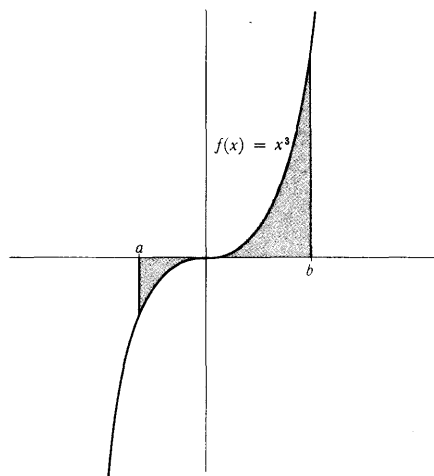


FIGURE 2

As we pointed out in Chapter 13, this integral does not always represent the area bounded by the graph of the function, the horizontal axis, and the vertical lines through $(a, 0)$ and $(b, 0)$. For example, if $a < 0 < b$, then

$$\int_a^b x^3 dx$$

does not represent the area of the region shown in Figure 2, which is given instead by

$$\begin{aligned} -\left(\int_a^0 x^3 dx\right) + \int_0^b x^3 dx &= -\left(\frac{0^4}{4} - \frac{a^4}{4}\right) + \left(\frac{b^4}{4} - \frac{0^4}{4}\right) \\ &= \frac{a^4}{4} + \frac{b^4}{4}. \end{aligned}$$

Similar care must be exercised in finding the areas of regions which are bounded by the graphs of more than one function—a problem which may frequently involve considerable ingenuity in any case. Suppose, to take a simple example first, that we wish to find the area of the region, shown in Figure 3, between the graphs of the functions

$$f(x) = x^2 \quad \text{and} \quad g(x) = x^3$$

on the interval $[0, 1]$. If $0 \leq x \leq 1$, then $0 \leq x^3 \leq x^2$, so that the graph of g lies below that of f . The area of the region of interest to us is therefore

$$\text{area } R(f, 0, 1) - \text{area } R(g, 0, 1),$$

which is

$$\int_0^1 x^2 dx - \int_0^1 x^3 dx = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

This area could have been expressed as

$$\int_a^b (f - g).$$

If $g(x) \leq f(x)$ for all x in $[a, b]$, then this integral always gives the area bounded by f and g , even if f and g are sometimes negative. The easiest way to see this is shown in Figure 4. If c is a number such that $f + c$ and $g + c$ are nonnegative on $[a, b]$, then the region R_1 , bounded by f and g , has the same area as the region R_2 , bounded by $f + c$ and $g + c$. Consequently,

$$\begin{aligned} \text{area } R_1 &= \text{area } R_2 = \int_a^b (f + c) - \int_a^b (g + c) \\ &= \int_a^b [(f + c) - (g + c)] \\ &= \int_a^b (f - g). \end{aligned}$$

This observation is useful in the following problem: Find the area of the region bounded by the graphs of

$$f(x) = x^3 - x \quad \text{and} \quad g(x) = x^2.$$

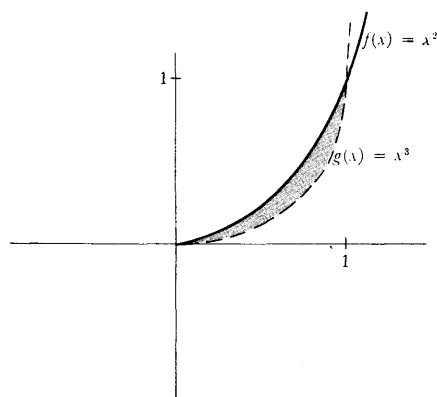


FIGURE 3

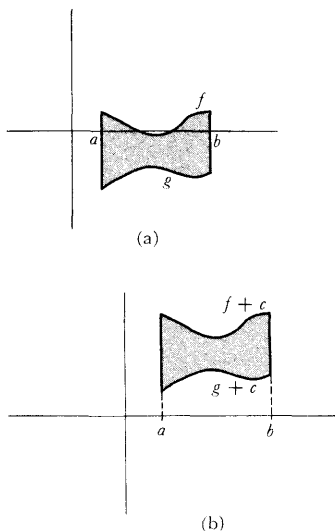


FIGURE 4

The first necessity is to determine this region more precisely. The graphs of f and g intersect when

$$\begin{aligned} x^3 - x &= x^2, \\ \text{or } x^3 - x^2 - x &= 0, \\ \text{or } x(x^2 - x - 1) &= 0, \\ \text{or } x &= 0, \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2}. \end{aligned}$$

On the interval $([1 - \sqrt{5}]/2, 0)$ we have $x^3 - x \geq x^2$ and on the interval $(0, [1 + \sqrt{5}]/2)$ we have $x^2 \geq x^3 - x$. These assertions are apparent from the graphs (Figure 5), but they can also be checked easily, as follows. Since $f(x) = g(x)$ only if $x = 0$, $[1 + \sqrt{5}]/2$, or $[1 - \sqrt{5}]/2$, the function $f - g$ does not change sign on the intervals $([1 - \sqrt{5}]/2, 0)$ and $[0, [1 + \sqrt{5}]/2)$; it is therefore only necessary to observe, for example, that

$$\begin{aligned} \left(-\frac{1}{2}\right)^3 - \left(-\frac{1}{2}\right) - \left(-\frac{1}{2}\right)^2 &= \frac{1}{8} > 0, \\ 1^3 - 1 - 1^2 &= -1 < 0, \end{aligned}$$

to conclude that

$$\begin{aligned} f - g &\geq 0 \quad \text{on } ([1 - \sqrt{5}]/2, 0), \\ f - g &\leq 0 \quad \text{on } (0, [1 + \sqrt{5}]/2). \end{aligned}$$

The area of the region in question is thus

$$\int_{\frac{1-\sqrt{5}}{2}}^0 (x^3 - x - x^2) dx + \int_0^{\frac{1+\sqrt{5}}{2}} [x^2 - (x^3 - x)] dx.$$

As this example reveals, one of the major problems involved in finding the areas of a region may be the exact determination of the region. There are, however, more substantial problems of a logical nature—we have thus far defined the areas of some very special regions only, which do not even include some of the regions whose areas have just been computed! We have simply assumed that area made sense for these regions, and that certain reasonable properties of “area” do hold. These remarks are not meant to suggest that you should regard exercising ingenuity to compute areas as beneath you, but are meant to indicate that a better approach to the definition of area is available, although its proper place is somewhere in advanced calculus. The desire to define area was the motivation, both in this book and historically, for the definition of the integral, but the integral does not really provide the best method of *defining* areas, although it is frequently the proper tool for *computing* them.

It may be discouraging to learn that integrals are not suitable for the very purpose for which they were invented, but we will soon see how essential they are for other purposes. The most important use of integrals has already been

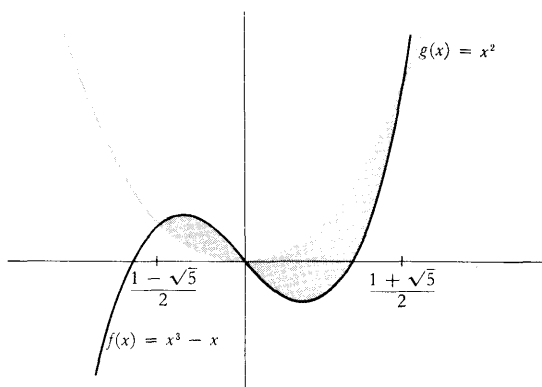


FIGURE 5

emphasized: if f is continuous, the integral provides a function y such that

$$y'(x) = f(x).$$

This equation is the simplest example of a “differential equation” (an equation for a function y which involves derivatives of y). The Fundamental Theorem of Calculus says that this differential equation has a solution, if f is continuous. In succeeding chapters, and in various problems, we will solve more complicated equations, but the solution almost always depends somehow on the integral; in order to solve a differential equation it is necessary to construct a new function, and the integral is one of the best ways of doing this.

Since the differentiable functions provided by the Fundamental Theorem of Calculus will play such a prominent role in later work, it is very important to realize that these functions may be combined, like less esoteric functions, to yield still more functions, whose derivatives can be found by the Chain Rule.

Suppose, for example, that

$$f(x) = \int_a^x \frac{1}{1 + \sin^2 t} dt.$$

Although the notation tends to disguise the fact somewhat, f is the composition of the functions

$$C(x) = x^3 \quad \text{and} \quad F(x) = \int_a^x \frac{1}{1 + \sin^2 t} dt.$$

In fact, $f(x) = F(C(x))$; in other words, $f = F \circ C$. Therefore, by the Chain Rule,

$$\begin{aligned} f'(x) &= F'(C(x)) \cdot C'(x) \\ &= F'(x^3) \cdot 3x^2 \\ &= \frac{1}{1 + \sin^2 x^3} \cdot 3x^2. \end{aligned}$$

If f is defined, instead, as

$$f(x) = \int_x^a \frac{1}{1 + \sin^2 t} dt,$$

then

$$f'(x) = -\frac{1}{1 + \sin^2 x} \cdot 3x^2.$$

If f is defined as the *reverse* composition,

$$f(x) = \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)^3,$$

then

$$\begin{aligned} f'(x) &= C'(F(x)) \cdot F'(x) \\ &= 3 \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)^2 \cdot \frac{1}{1 + \sin^2 x}. \end{aligned}$$

Similarly, if

$$\begin{aligned} f(x) &= \int_a^{\sin x} \frac{1}{1 + \sin^2 t} dt, \\ g(x) &= \int_{\sin x}^a \frac{1}{1 + \sin^2 t} dt, \\ h(x) &= \sin \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right), \end{aligned}$$

then

$$\begin{aligned} f'(x) &= \frac{1}{1 + \sin^2(\sin x)} \cdot \cos x, \\ g'(x) &= \frac{-1}{1 + \sin^2(\sin x)} \cdot \cos x, \\ h'(x) &= \cos \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right) \cdot \frac{1}{1 + \sin^2 x}. \end{aligned}$$

The formidable appearing function

$$f(x) = \int_a^{\left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)} \frac{1}{1 + \sin^2 t} dt$$

is also a composition; in fact, $f = F \circ F$. Therefore

$$\begin{aligned} f'(x) &= F'(F(x)) \cdot F'(x) \\ &= \frac{1}{1 + \sin^2 \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)} \cdot \frac{1}{1 + \sin^2 x}. \end{aligned}$$

As these examples reveal, the expression occurring above (or below) the

integral sign indicates the function which will appear on the *right* when f is written as a composition. As a final example, consider the triple compositions

$$f(x) = \int_a^{\left(\int_a^{\left(\int_a^x \frac{1}{1+\sin^2 t} dt\right)} \frac{1}{1+\sin^2 t} dt\right)} \frac{1}{1+\sin^2 t} dt, \quad g(x) = \int_a^{\left[\left(\int_a^x \frac{1}{1+\sin^2 t} dt\right)\right]} \frac{1}{1+\sin^2 t} dt,$$

which can be written

$$f = F \circ F \circ C \quad \text{and} \quad g = F \circ F \circ F.$$

Omitting the intermediate steps (which you may supply, if you still feel insecure), we obtain

$$\begin{aligned} f'(x) &= \frac{1}{1 + \sin^2 \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)} \cdot \frac{1}{1 + \sin^2 x^3} \cdot 3x^2, \\ g'(x) &= \frac{1}{1 + \sin^2 \left[\int_a^{\left(\int_a^{\left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)} \frac{1}{1 + \sin^2 t} dt \right)} \frac{1}{1 + \sin^2 t} dt \right]} \cdot \frac{1}{1 + \sin^2 \left(\int_a^x \frac{1}{1 + \sin^2 t} dt \right)} \cdot \frac{1}{1 + \sin^2 x}. \end{aligned}$$

Like the simpler differentiations of Chapter 10, these manipulations should become much easier after the practice provided by some of the problems, and, like the problems of Chapter 10, these differentiations are simply a test of your understanding of the Chain Rule, in the somewhat unfamiliar context provided by the Fundamental Theorem of Calculus.

The powerful uses to which the integral will be put in the following chapters all depend on the Fundamental Theorem of Calculus, yet the proof of that theorem was quite easy—it seems that all the real work went into the definition of the integral. Actually, this is not quite true. In order to apply Theorem 1 to a continuous function we need the one theorem whose proof has not yet been given: If f is continuous on $[a, b]$, then f is integrable on $[a, b]$. The proof which will be given depends upon a trick which is not very illuminating, but which at least serves to fill this gap in our knowledge.

If f is any bounded function on $[a, b]$, then

$$\sup \{L(f, P)\} \quad \text{and} \quad \inf \{U(f, P)\}$$

will both exist, even if f is not integrable. These numbers are called the **lower integral** of f on $[a, b]$ and the **upper integral** of f on $[a, b]$, respectively, and will be denoted by

$$\mathbf{L} \int_a^b f \quad \text{and} \quad \mathbf{U} \int_a^b f.$$

The lower and upper integrals both have several properties which the integral

possesses. In particular, if $a < c < b$, then

$$\mathbf{L} \int_a^b f = \mathbf{L} \int_a^c f + \mathbf{L} \int_c^b f \quad \text{and} \quad \mathbf{U} \int_a^b f = \mathbf{U} \int_a^c f + \mathbf{U} \int_c^b f,$$

and if $m \leq f(x) \leq M$ for all x in $[a, b]$, then

$$m(b-a) \leq \mathbf{L} \int_a^b f \leq \mathbf{U} \int_a^b f \leq M(b-a).$$

The proofs of these facts are left as an exercise, since they are quite similar to the corresponding proofs for integrals. The results for integrals are actually a corollary of the results for upper and lower integrals, because f is integrable precisely when

$$\mathbf{L} \int_a^b f = \mathbf{U} \int_a^b f.$$

We will prove that a continuous function f is integrable by showing that this equality always holds for continuous functions. It is actually easier to show that

$$\mathbf{L} \int_a^x f = \mathbf{U} \int_a^x f$$

for all x in $[a, b]$; the trick is to note that most of the proof of Theorem 1 didn't even depend on the fact that f was integrable!

THEOREM 13-3 If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

PROOF Define functions L and U on $[a, b]$ by

$$L(x) = \mathbf{L} \int_a^x f \quad \text{and} \quad U(x) = \mathbf{U} \int_a^x f.$$

Let x be in (a, b) . If $h > 0$ and

$$\begin{aligned} m_h &= \inf \{f(t) : x \leq t \leq x+h\}, \\ M_h &= \sup \{f(t) : x \leq t \leq x+h\}, \end{aligned}$$

then

$$m_h \cdot h \leq \mathbf{L} \int_x^{x+h} f \leq \mathbf{U} \int_x^{x+h} f \leq M_h \cdot h,$$

so

$$m_h \cdot h \leq L(x+h) - L(x) \leq U(x+h) - U(x) \leq M_h \cdot h$$

or

$$m_h \leq \frac{L(x+h) - L(x)}{h} \leq \frac{U(x+h) - U(x)}{h} \leq M_h.$$

If $h < 0$ and

$$\begin{aligned} m_h &= \inf \{f(t) : x+h \leq t \leq x\}, \\ M_h &= \sup \{f(t) : x+h \leq t \leq x\}, \end{aligned}$$

one obtains the same inequality, precisely as in the proof of Theorem 1.

Since f is continuous at x , we have

$$\lim_{h \rightarrow 0} m_h = \lim_{h \rightarrow 0} M_h = f(x),$$

and this proves that

$$L'(x) = U'(x) = f(x) \quad \text{for } x \text{ in } (a, b).$$

This means that there is a number c such that

$$U(x) = L(x) + c \quad \text{for all } x \text{ in } [a, b].$$

Since

$$U(a) = L(a) = 0,$$

the number c must equal 0, so

$$U(x) = L(x) \quad \text{for all } x \text{ in } [a, b].$$

In particular,

$$\mathbf{U} \int_a^b f = U(b) = L(b) = \mathbf{L} \int_a^b f,$$

and this means that f is integrable on $[a, b]$. ■

PROBLEMS

1. Find the derivatives of each of the following functions.

$$(i) \quad F(x) = \int_a^{x^3} \sin^3 t \, dt.$$

$$(ii) \quad F(x) = \int_3^x \left(\int_1^{\sin^3 t} dt \right) \frac{1}{1 + \sin^6 t + t^2} dt.$$

$$(iii) \quad F(x) = \int_{15}^x \left(\int_8^y \frac{1}{1 + t^2 + \sin^2 t} dt \right) dy.$$

$$(iv) \quad F(x) = \int_x^b \frac{1}{1 + t^2 + \sin^2 t} dt.$$

$$(v) \quad F(x) = \int_a^b \frac{x}{1 + t^2 + \sin^2 t} dt.$$

$$(vi) \quad F(x) = \sin \left(\int_0^x \sin \left(\int_0^y \sin^3 t \, dt \right) dy \right).$$

$$(vii) \quad F^{-1}, \text{ where } F(x) = \int_1^x \frac{1}{t} dt. \quad \left\{ \begin{array}{l} \text{(Find } (F^{-1})'(x) \text{ in terms of} \\ F^{-1}(x).) \end{array} \right.$$

$$(viii) \quad F^{-1}, \text{ where } F(x) = \int_0^x \frac{1}{\sqrt{1-t^2}} dt. \quad \left\{ \begin{array}{l} \text{(Find } (F^{-1})'(x) \text{ in terms of} \\ F^{-1}(x).) \end{array} \right.$$

2. For each of the following f , if $F(x) = \int_0^x f$, at which points x is $F'(x) = f(x)$? (Caution: it might happen that $F'(x) = f(x)$, even if f is not continuous at x .)

$$(i) \quad f(x) = 0 \text{ if } x \leq 1, f(x) = 1 \text{ if } x > 1.$$

$$(ii) \quad f(x) = 0 \text{ if } x < 1, f(x) = 1 \text{ if } x \geq 1.$$

$$(iii) \quad f(x) = 0 \text{ if } x \neq 1, f(x) = 1 \text{ if } x = 1.$$

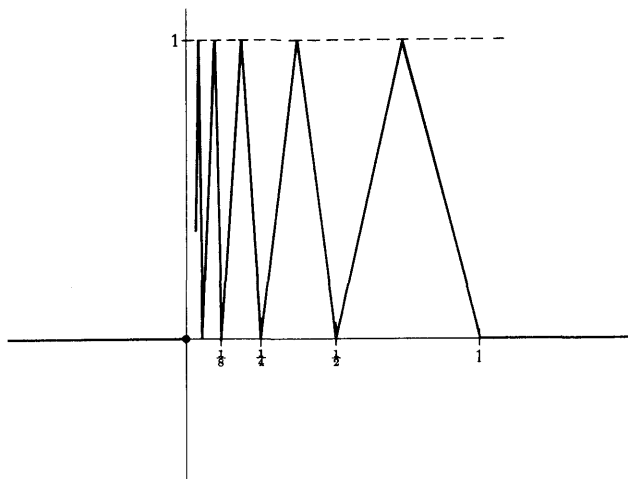


FIGURE 6

- (iv) $f(x) = 0$ if x is irrational, $f(x) = 1/q$ if $x = p/q$ in lowest terms.
- (v) $f(x) = 0$ if $x \leq 0$, $f(x) = x$ if $x \geq 0$.
- (vi) $f(x) = 0$ if $x \leq 0$, $f(x) = 1/[1/x]$ if $x \geq 0$.
- (vii) f is the function shown in Figure 6.
- (viii) $f(x) = 1$ if $x = 1/n$ for some n in $\mathbf{N}$, $f(x) = 0$ otherwise.

3. Find $(f^{-1})'(0)$ if

(i) $f(x) = \int_0^x 1 + \sin(\sin t) dt.$

(ii) $f(x) = \int_1^x \sin(\sin t) dt.$

(Don't try to evaluate f explicitly.)

4. Find $F'(x)$ if $F(x) = \int_0^x xf(t) dt$. (The answer is *not* $xf(x)$; you should perform an obvious manipulation on the integral before trying to find F' .)
5. Prove that if f is continuous, then

$$\int_0^x f(u)(x-u) du = \int_0^x \left(\int_0^u f(t) dt \right) du.$$

Hint: Differentiate both sides, making use of Problem 4.

****6.** Use Problem 5 to prove that

$$\int_0^x f(u)(x-u)^2 du = 2 \int_0^x \left(\int_0^{u_1} \left(\int_0^{u_1} f(t) dt \right) du_1 \right) du_2$$

7. Find a function f such that $f'''(x) = 1/\sqrt{1 + \sin^2 x}$. (This problem is supposed to be easy; don't misinterpret the word "find.")

***8.** A function f is **periodic**, with **period** a , if $f(x+a) = f(x)$ for all x .

(a) If f is periodic with period a and integrable on $[0, a]$, show that

$$\int_0^a f = \int_b^{b+a} f \quad \text{for all } b.$$

- (b) Find a function f such that f is not periodic, but f' is. Hint: Choose a periodic g for which it can be guaranteed that $f(x) = \int_0^x g$ is not periodic.
- (c) Suppose that f' is periodic with period a . Prove that f is periodic if and only if $f(a) = f(0)$.
9. Find $\int_0^b \sqrt[n]{x} dx$, by simply guessing a function f with $f'(x) = \sqrt[n]{x}$, and using the Second Fundamental Theorem of Calculus. Then check with Problem 13-16.
- *10. Use the Fundamental Theorem of Calculus and Problem 13-16 to derive the result stated in Problem 12-15.
11. (a) Find the derivatives of $F(x) = \int_1^x 1/t dt$ and $G(x) = \int_b^{bx} 1/t dt$.
(b) Now give a new proof for Problem 13-12.
- *12. Use the Fundamental Theorem of Calculus and Darboux's Theorem (Problem 11-39) to give another proof of the Intermediate Value Theorem.
13. Prove that if h is continuous, f and g are differentiable, and

$$F(x) = \int_{f(x)}^{g(x)} h(t) dt,$$

then $F'(x) = h(g(x)) \cdot g'(x) - h(f(x)) \cdot f'(x)$. Hint: Try to reduce this to the two cases you can already handle, with a constant either as the lower or the upper limit of integration.

- *14. (a) Suppose $G' = g$ and $F' = f$. Prove that if the function y satisfies the differential equation
- $$(*) \quad g(y(x)) \cdot y'(x) = f(x) \quad \text{for all } x \text{ in some interval,}$$
- then there is a number c such that
- $$(**) \quad G(y(x)) = F(x) + c \quad \text{for all } x \text{ in this interval.}$$
- (b) Show, conversely, that if y satisfies (**), then y is a solution of (*).
- (c) Find what condition y must satisfy if

$$y'(x) = \frac{1 + x^2}{1 + y(x)}.$$

(In this case $g(t) = 1 + t$ and $f(t) = 1 + t^2$.) Then "solve" the resulting equations to find all possible solutions y (no solution will have $\mathbf{R}$ as its domain).

- (d) Find what condition y must satisfy if

$$y'(x) = \frac{-1}{1 + 5[y(x)]^4}.$$

(An appeal to Problem 12-10 will show that there are functions satisfying the resulting equation.)

- (e) Find all functions
- y
- satisfying

$$y(x)y'(x) = -x.$$

Find the solution y satisfying $y(0) = -1$.

- *15.** The limit $\lim_{N \rightarrow \infty} \int_a^N f$, if it exists, is denoted by $\int_a^\infty f$ (or $\int_a^\infty f(x) dx$), and called an “improper integral.”
- (a) Determine $\int_1^\infty x^r dx$, if $r < -1$.
- (b) Use Problem 13-12 to show that $\int_1^\infty 1/x dx$ does not exist. Hint: What can you say about $\int_1^{2^n} 1/x dx$?
- (c) Suppose that $f(x) \geq 0$ for $x \geq 0$ and that $\int_0^\infty f$ exists. Prove that if $0 \leq g(x) \leq f(x)$ for all $x \geq 0$, then $\int_0^\infty g$ also exists.
- (d) Explain why $\int_0^\infty 1/(1+x^2) dx$ exists. Hint: Split this integral up at 1.
- *16.** The improper integral $\int_{-\infty}^a f$ is defined in the obvious way, as $\lim_{N \rightarrow -\infty} \int_N^a f$. But another kind of improper integral $\int_{-\infty}^\infty f$ is defined in a nonobvious way: it is $\int_0^\infty f + \int_{-\infty}^0 f$, provided these improper integrals both exist.
- (a) Explain why $\int_{-\infty}^\infty 1/(1+x^2) dx$ exists.
- (b) Explain why $\int_{-\infty}^\infty x dx$ does not exist. (But notice that $\lim_{N \rightarrow \infty} \int_{-N}^N x dx$ does exist.)
- (c) Prove that if $\int_{-\infty}^\infty f$ exists, then $\lim_{N \rightarrow \infty} \int_{-N}^N f$ exists and equals $\int_{-\infty}^\infty f$. Show moreover, that $\lim_{N \rightarrow \infty} \int_{-N}^{N+1} f$ and $\lim_{N \rightarrow \infty} \int_{-N^2}^N f$ both exist and equal $\int_{-\infty}^\infty f$. Can you state a reasonable generalization of these facts? (If you can't, you will have a miserable time trying to do these special cases!)
- *17.** There is another kind of “improper integral” in which the interval is bounded, but the function is unbounded:
- (a) If $a > 0$, find $\lim_{\epsilon \rightarrow 0^+} \int_\epsilon^a 1/\sqrt{x} dx$. This limit is denoted by $\int_0^a 1/\sqrt{x} dx$, even though the function $f(x) = 1/\sqrt{x}$ is not bounded on $[0, a]$, no matter how we define $f(0)$.
- (b) Find $\int_0^a x^r dx$ if $-1 < r < 0$.
- (c) Use Problem 13-12 to show that $\int_0^a x^{-1} dx$ does not make sense, even as a limit.
- (d) Invent a reasonable definition of $\int_a^0 |x|^r dx$ for $a < 0$ and compute it for $-1 < r < 0$.

- (e) Invent a reasonable definition of $\int_{-1}^1 (1 - x^2)^{-1/2} dx$, as a sum of two limits, and show that the limits exists. Hint: Why does $\int_{-1}^0 (1 + x)^{-1/2} dx$ exist? How does $(1 + x)^{-1/2}$ compare with $(1 - x^2)^{-1/2}$ for $-1 < x < 0$?
- *18.** It is possible, finally, to combine the two possible extensions of the notion of the integral.
- (a) If $f(x) = 1/\sqrt{x}$ for $0 \leq x \leq 1$ and $f(x) = 1/x^2$ for $x \geq 1$, find $\int_0^\infty f(x) dx$ (after deciding what this should mean).
- (b) Show that $\int_0^\infty x^r dx$ never makes sense. (Distinguish the cases $-1 < r < 0$ and $r < -1$. In one case things go wrong at 0, in the other case at ∞ ; for $r = -1$ things go wrong at both places.)

The definitions of the functions $\sin$ and $\cos$ are considerably more subtle than one might suspect. For this reason, this chapter begins with some informal and intuitive definitions, which should not be scrutinized too carefully, as they shall soon be replaced by the formal definitions which we really intend to use.

In elementary geometry an angle is simply the union of two half-lines with a common initial point (Figure 1).

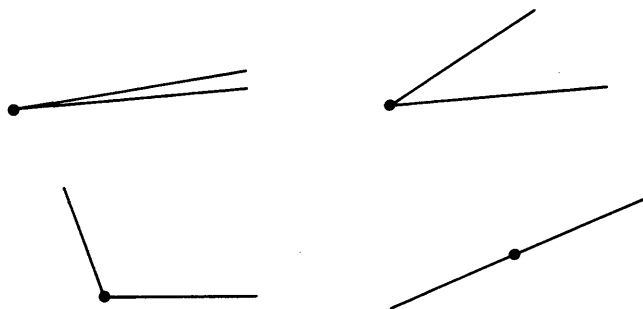


FIGURE 1

More useful for trigonometry are “directed angles,” which may be regarded as pairs (l_1, l_2) of half-lines with the same initial point, visualized as in Figure 2.

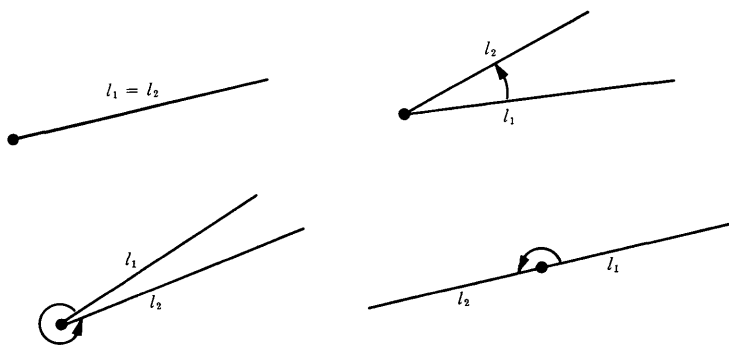


FIGURE 2

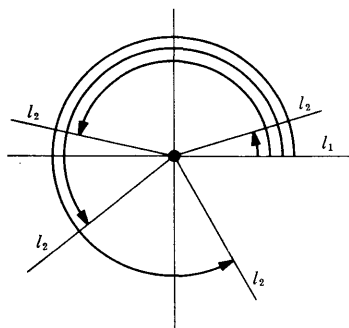


FIGURE 3

If for l_1 we always choose the positive half of the horizontal axis, a directed angle is described completely by the second half-line (Figure 3).

Since each half-line intersects the unit circle precisely once, a directed angle is described, even more simply, by a point on the unit circle (Figure 4), that is, by a point (x, y) with $x^2 + y^2 = 1$.

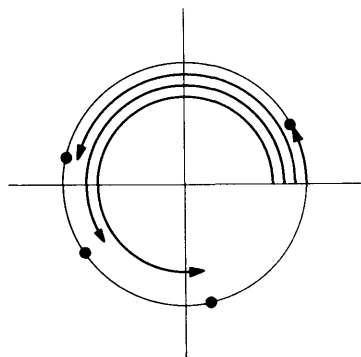


FIGURE 4

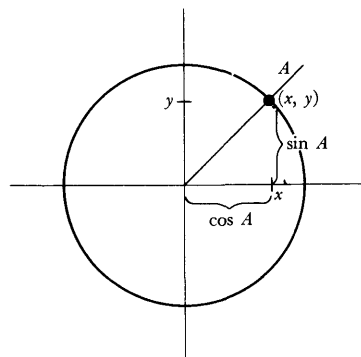


FIGURE 5

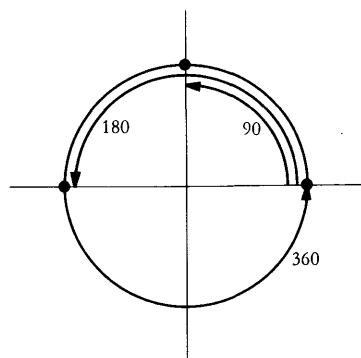


FIGURE 6

The sine and cosine of a directed angle can now be defined as follows (Figure 5): a directed angle is determined by a point (x, y) with $x^2 + y^2 = 1$; the sine of the angle is defined as y , and the cosine as x .

Despite the aura of precision surrounding the previous paragraph, we are not yet finished with the definitions of $\sin$ and $\cos$. Indeed, we have barely begun. What we have defined is the sine and cosine of a directed angle; what we *want* to define is $\sin x$ and $\cos x$ for each *number* x . The usual procedure for doing this depends on associating an angle to every number. The oldest method is to “measure angles in degrees.” An angle “all the way around” is associated to 360, an angle “half-way around” is associated to 180, an angle “a quarter way around” to 90, etc. (Figure 6). The angle associated, in this manner, to the number x , is called “the angle of x degrees.” The angle of 0 degrees is the same as the angle of 360 degrees, and this ambiguity is purposely extended further, so that an angle of 90 degrees is also an angle of $360 + 90$ degrees, etc. One can now define a function, which we will denote by $\sin^\circ$, as follows:

$$\sin^\circ(x) = \text{sine of the angle of } x \text{ degrees.}$$

There are two difficulties with this approach. Although it may be clear what we mean by an angle of 90 or 45 degrees, it is not quite clear what an angle of $\sqrt{2}$ degrees is, for example. Even if this difficulty could be circumvented, it is unlikely that this system, depending as it does on the arbitrary choice of 360, will lead to elegant results—it would be sheer luck if the function $\sin^\circ$ had mathematically pleasing properties.

“Radian measure” appears to offer a remedy for both these defects. Given any number x , choose a point P on the unit circle such that x is the length of the arc of the circle beginning at $(1, 0)$ and running counterclockwise to P (Figure 7). The directed angle determined by P is called “the angle of x radians.” Since the length of the whole circle is 2π , the angle of x radians and the angle of $2\pi + x$ radians are identical. A function $\sin^r$ can now be defined as follows:

$$\sin^r(x) = \text{sine of the angle of } x \text{ radians.}$$

This same method can easily be adopted to define $\sin^\circ$; since we want to have $\sin^\circ 360 = \sin^r 2\pi$, we can define

$$\sin^\circ x = \sin^r \frac{2\pi x}{360} = \sin^r \frac{\pi x}{180}.$$

We shall soon drop the superscript r in $\sin^r$, since $\sin^r$ (and not $\sin^\circ$) is the only function which will interest us; before we do, a few words of warning are advisable.

The expressions $\sin^\circ x$ and $\sin^r x$ are sometimes written

$$\begin{aligned} &\sin x^\circ \\ &\sin x \text{ radians,} \end{aligned}$$

but this notation is quite misleading; a number x is simply a number—it does

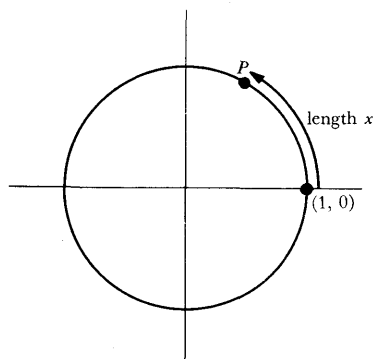


FIGURE 7

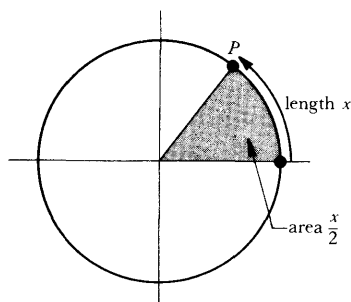


FIGURE 8

DEFINITION

$$\pi = 2 \cdot \int_{-1}^1 \sqrt{1-x^2} dx.$$

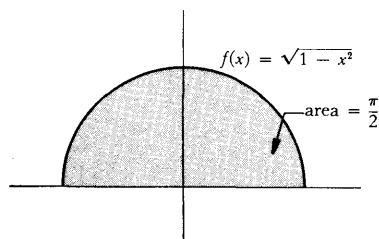


FIGURE 9

not carry a banner indicating that it is “in degrees” or “in radians.” If the meaning of the notation “ $\sin x$ ” is in doubt one usually asks:

“Is x in degrees or radians?”

but what one means is:

“Do you mean ‘ $\sin^\circ$ ’ or ‘ $\sin^r$ ’?”

Even for mathematicians, addicted to precision, these remarks might be dispensable, were it not for the fact that failure to take them into account will lead to incorrect answers to certain problems (an example is given in Problem 18).

Although the function $\sin^r$ is the function which we wish to denote simply by $\sin$ (and use exclusively henceforth), there is a difficulty involved even in the definition of $\sin^r$. Our proposed definition depends on the concept of length of a curve. Although we could digress, and define length, at the present time it is much easier to reformulate the definition in terms of areas, which we can treat by means of the integral. (A treatment in terms of length is outlined in Problems 32 to 34.)

Suppose that x is the length of the arc of the unit circle from $(1, 0)$ to P ; this arc thus contains $x/2\pi$ of the total length 2π of the circumference of the unit circle. Let S denote the “sector” shown in Figure 8; S is bounded by the unit circle, the horizontal axis, and the half-line through $(0, 0)$ and P . The area of S should be $x/2\pi$ times the area inside the unit circle, which we expect to be π ; thus S should have area

$$\frac{x}{2\pi} \cdot \pi = \frac{x}{2}.$$

We can therefore define $\cos x$ and $\sin x$ as the coordinates of the point P which determines a sector of area $x/2$.

With these remarks as background, the rigorous definition of the functions $\sin$ and $\cos$ now begins. The first definition identifies π as the area of the unit circle—more precisely, as twice the area of a semicircle (Figure 9).

(This definition is not offered simply as an embellishment; to define the trigonometric functions it will be necessary to first define $\sin x$ and $\cos x$ only for $0 \leq x \leq \pi$.)

The second definition is meant to describe, for $-1 \leq x \leq 1$, the area $A(x)$ of the sector bounded by the unit circle, the horizontal axis, and the half-line through $(x, \sqrt{1-x^2})$. If $0 \leq x \leq 1$, this area can be expressed (Figure 10) as the sum of the area of a triangle and the area of a region under the unit circle:

$$\frac{x\sqrt{1-x^2}}{2} + \int_x^1 \sqrt{1-t^2} dt.$$

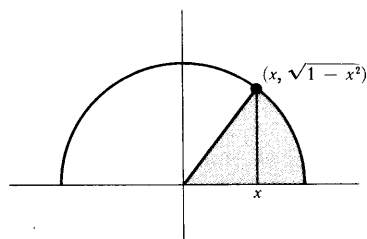


FIGURE 10

DEFINITION

 If $-1 \leq x \leq 1$, then

$$A(x) = \frac{x\sqrt{1-x^2}}{2} + \int_x^1 \sqrt{1-t^2} dt.$$

Notice that if $-1 < x < 1$, then A is differentiable at x and (using the Fundamental Theorem of Calculus),

$$\begin{aligned} A'(x) &= \frac{1}{2} \left[x \cdot \frac{-2x}{2\sqrt{1-x^2}} + \sqrt{1-x^2} \right] - \sqrt{1-x^2} \\ &= \frac{1}{2} \left[\frac{-x^2 + (1-x^2)}{\sqrt{1-x^2}} \right] - \sqrt{1-x^2} \\ &= \frac{1-2x^2}{2\sqrt{1-x^2}} - \sqrt{1-x^2} \\ &= \frac{1-2x^2-2(1-x^2)}{2\sqrt{1-x^2}} \\ &= \frac{-1}{2\sqrt{1-x^2}}. \end{aligned}$$

Notice also (Figure 12) that on the interval $[-1, 1]$ the function A decreases from

$$A(-1) = 0 + \int_{-1}^1 \sqrt{1-t^2} dt = \frac{\pi}{2}$$

to $A(1) = 0$. This follows directly from the definition of A , and also from the fact that its derivative is negative on $(-1, 1)$.

For $0 \leq x \leq \pi$ we wish to define $\cos x$ and $\sin x$ as the coordinates of a point $P = (\cos x, \sin x)$ on the unit circle which determines a sector whose area is $x/2$ (Figure 13). In other words:

DEFINITION

 If $0 \leq x \leq \pi$, then $\cos x$ is the unique number in $[-1, 1]$ such that

$$A(\cos x) = \frac{x}{2};$$

and

$$\sin x = \sqrt{1 - (\cos x)^2}.$$

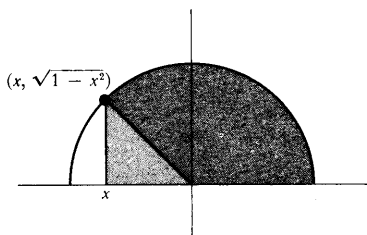


FIGURE 11

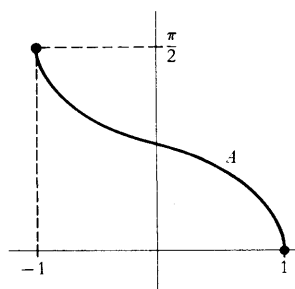


FIGURE 12

This definition actually requires a few words of justification. In order to know that there *is* a number y satisfying $A(y) = x/2$, we use the fact that A is continuous, and that A takes on the values 0 and $\pi/2$. This tacit appeal to the Intermediate Value Theorem is crucial, if we want to make our preliminary definition precise. Having made, and justified, our definition, we can now proceed quite rapidly.

THEOREM 1 If $0 < x < \pi$, then

$$\begin{aligned}\cos'(x) &= -\sin x, \\ \sin'(x) &= \cos x.\end{aligned}$$

PROOF If $B = 2A$, then the definition $A(\cos x) = x/2$ can be written

$$B(\cos x) = x;$$

in other words, $\cos$ is just the inverse of B . We have already computed that

$$A'(x) = -\frac{1}{2\sqrt{1-x^2}},$$

from which we conclude that

$$B'(x) = -\frac{1}{\sqrt{1-x^2}}.$$

Consequently,

$$\begin{aligned}\cos'(x) &= (B^{-1})'(x) \\ &= \frac{1}{B'(B^{-1}(x))} \\ &= \frac{1}{-\frac{1}{\sqrt{1-[B^{-1}(x)]^2}}} \\ &= -\sqrt{1-(\cos x)^2} \\ &= -\sin x.\end{aligned}$$

Since

$$\sin x = \sqrt{1-(\cos x)^2},$$

we also obtain

$$\begin{aligned}\sin'(x) &= \frac{1}{2} \cdot \frac{-2 \cos x \cdot \cos'(x)}{\sqrt{1-(\cos x)^2}} \\ &= \frac{\cos x \sin x}{\sin x} \\ &= \cos x. \blacksquare\end{aligned}$$

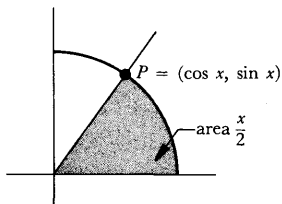


FIGURE 13

The information contained in Theorem 1 can be used to sketch the graphs

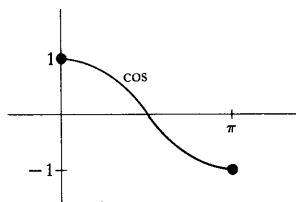


FIGURE 14

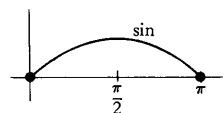
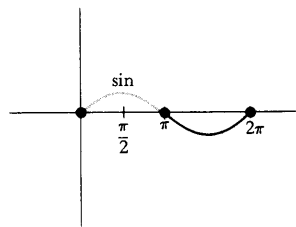
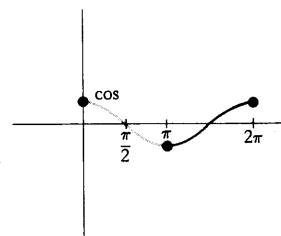


FIGURE 15



(a)



(b)

FIGURE 16

of $\sin$ and $\cos$ on the interval $[0, \pi]$. Since

$$\cos'(x) = -\sin x < 0, \quad 0 < x < \pi,$$

the function $\cos$ decreases from $\cos 0 = 1$ to $\cos \pi = -1$ (Figure 14). Consequently, $\cos y = 0$ for a unique y in $[0, \pi]$. To find y , we note that the definition of $\cos$,

$$A(\cos x) = \frac{x}{2},$$

means that

$$A(0) = \frac{y}{2},$$

so

$$y = 2 \int_0^1 \sqrt{1-t^2} dt.$$

It is easy to see that

$$\int_{-1}^0 \sqrt{1-t^2} dt = \int_0^1 \sqrt{1-t^2} dt,$$

so we can also write

$$y = \int_{-1}^1 \sqrt{1-t^2} dt = \frac{\pi}{2}.$$

Now we have

$$\sin'(x) = \cos x \begin{cases} > 0, & 0 < x < \pi/2 \\ < 0, & \pi/2 < x < \pi, \end{cases}$$

so $\sin$ increases on $[0, \pi/2]$ from $\sin 0 = 0$ to $\sin \pi/2 = 1$, and then decreases on $[\pi/2, \pi]$ to $\sin \pi = 0$ (Figure 15).

The values of $\sin x$ and $\cos x$ for x not in $[0, \pi]$ are most easily defined by a two-step piecing together process:

(1) If $\pi \leq x \leq 2\pi$, then

$$\begin{aligned} \sin x &= -\sin(2\pi - x), \\ \cos x &= \cos(2\pi - x). \end{aligned}$$

Figure 16 shows the graphs of $\sin$ and $\cos$ on $[0, 2\pi]$.

(2) If $x = 2\pi k + x'$ for some integer k , and some x' in $[0, 2\pi]$, then

$$\begin{aligned} \sin x &= \sin x', \\ \cos x &= \cos x', \end{aligned}$$

Figure 17 shows the graphs of $\sin$ and $\cos$, now defined on all of $\mathbf{R}$.

Having extended the functions $\sin$ and $\cos$ to $\mathbf{R}$, we must now check that the basic properties of these functions continue to hold. In most cases this is easy. For example, it is clear that the equation

$$\sin^2 x + \cos^2 x = 1$$

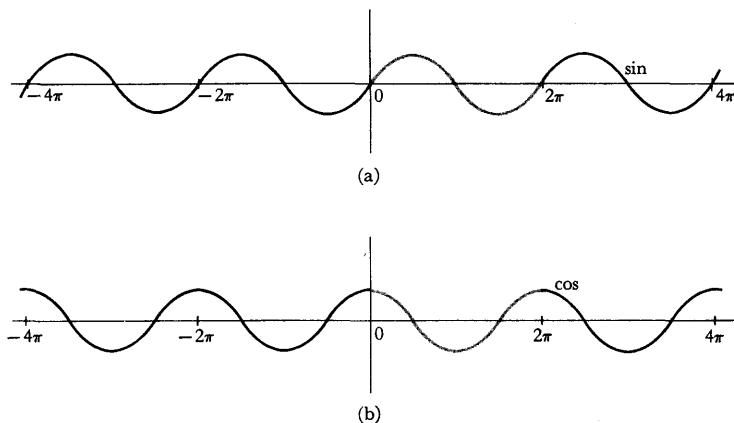


FIGURE 17

holds for all x . It is also not hard to prove that

$$\begin{aligned}\sin'(x) &= \cos x, \\ \cos'(x) &= -\sin x,\end{aligned}$$

if x is not a multiple of π . For example, if $\pi < x < 2\pi$, then

$$\sin x = -\sin(2\pi - x),$$

so

$$\begin{aligned}\sin'(x) &= -\sin'(2\pi - x) \cdot (-1) \\ &= \cos(2\pi - x) \\ &= \cos x.\end{aligned}$$

If x is a multiple of π we resort to a trick; it is only necessary to apply Theorem 11-7 to conclude that the same formulas are true in this case also.

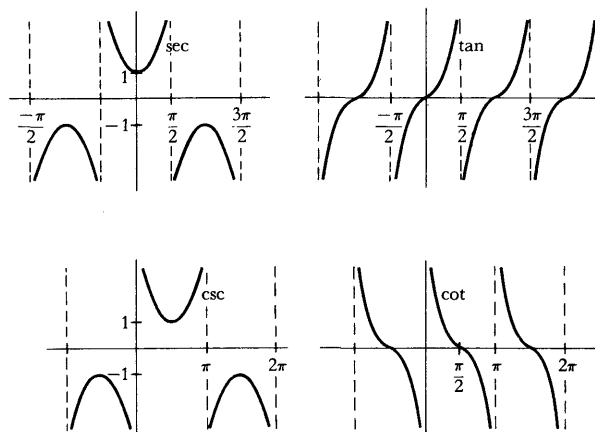


FIGURE 18

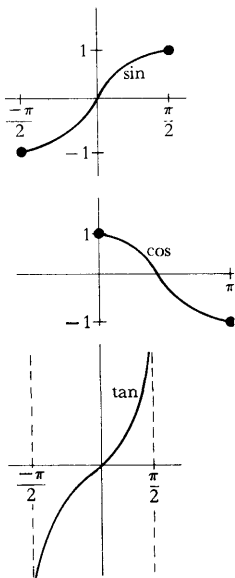


FIGURE 19

The other standard trigonometric functions present no difficulty at all. We define

$$\left. \begin{aligned} \sec x &= \frac{1}{\cos x} \\ \tan x &= \frac{\sin x}{\cos x} \end{aligned} \right\} x \neq k\pi + \pi/2,$$

$$\left. \begin{aligned} \csc x &= \frac{1}{\sin x} \\ \cot x &= \frac{\cos x}{\sin x} \end{aligned} \right\} x \neq k\pi.$$

The graphs are sketched in Figure 18. It is a good idea to convince yourself that the general features of these graphs can be predicted from the derivatives of these functions, which are listed in the next theorem (there is no need to memorize the statement of the theorem, since the results can be rederived whenever needed.)

THEOREM 2

If $x \neq k\pi + \pi/2$, then

$$\begin{aligned} \sec'(x) &= \sec x \tan x, \\ \tan'(x) &= \sec^2 x. \end{aligned}$$

If $x \neq k\pi$, then

$$\begin{aligned} \csc'(x) &= -\csc x \cot x, \\ \cot'(x) &= -\csc^2 x. \end{aligned}$$

PROOF

Left to you (a straightforward computation). ■

The inverses of the trigonometric functions are also easily differentiated. The trigonometric functions are not one-one, so it is first necessary to restrict them to suitable intervals; the largest possible length obtainable is π , and the intervals usually chosen are (Figure 19)

$$\begin{aligned} &[-\pi/2, \pi/2] \text{ for sin,} \\ &[0, \pi] \text{ for cos,} \\ &(-\pi/2, \pi/2) \text{ for tan.} \end{aligned}$$

(The inverses of the other trigonometric functions are so rarely used that they will not even be discussed here.)

The inverse of the function

$$f(x) = \sin x, \quad -\pi/2 \leq x \leq \pi/2$$

is denoted by **arcsin** (Figure 20); the domain of arcsin is $[-1, 1]$. The notation $\sin^{-1}$ has been avoided because arcsin is not the inverse of sin (which is not one-one), but of the restricted function f ; the notation $\sin^{-1}$ has the additional disadvantage that $\sin^{-1}(x)$ might be construed as $1/\sin x$.

The inverse of the function

$$g(x) = \cos x, \quad 0 \leq x \leq \pi$$

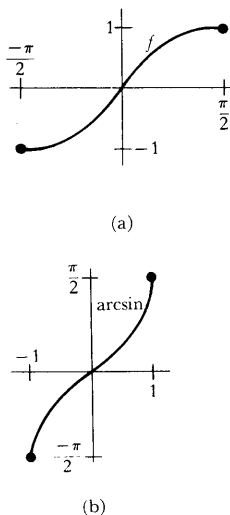


FIGURE 20

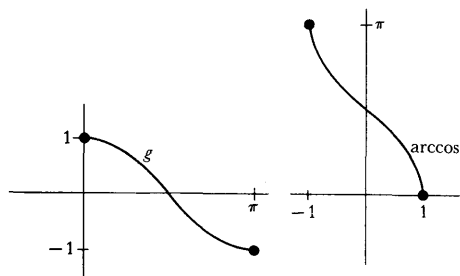


FIGURE 21

is denoted by **arccos** (Figure 21); the domain of arccos is $[-1, 1]$.

The inverse of the function

$$h(x) = \tan x, \quad -\pi/2 < x < \pi/2$$

is denoted by **arctan** (Figure 22); arctan is one of the simplest examples of a differentiable function which is bounded even though it is one-one on all of $\mathbf{R}$.

The derivatives of the inverse trigonometric functions are surprisingly simple, and do not involve trigonometric functions at all. Finding the derivatives is a simple matter, but to express them in a suitable form we will have to simplify expressions like

$$\cos(\arcsin x), \\ \sec(\arctan x).$$

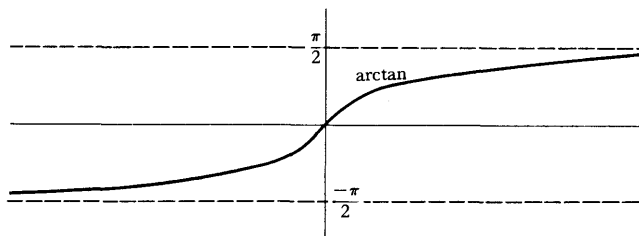


FIGURE 22

A little picture is the best way to remember the correct simplifications. For example, Figure 23 shows a directed angle whose sine is x —the angle shown is thus an angle of $(\arcsin x)$ radians; consequently $\cos(\arcsin x)$ is the length of the other side, namely, $\sqrt{1 - x^2}$. However, in the proof of the next theorem we will not resort to such pictures.

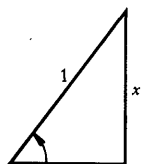


FIGURE 23

THEOREM 3 If $-1 < x < 1$, then

$$\arcsin'(x) = \frac{1}{\sqrt{1 - x^2}},$$

$$\arccos'(x) = \frac{-1}{\sqrt{1 - x^2}}.$$

Moreover, for all x we have

$$\arctan'(x) = \frac{1}{1 + x^2}.$$

PROOF

$$\begin{aligned} \arcsin'(x) &= (f^{-1})'(x) \\ &= \frac{1}{f'(f^{-1}(x))} \\ &= \frac{1}{\sin'(\arcsin x)} \\ &= \frac{1}{\cos(\arcsin x)}. \end{aligned}$$

Now

$$[\sin(\arcsin x)]^2 + [\cos(\arcsin x)]^2 = 1,$$

that is,

$$x^2 + [\cos(\arcsin x)]^2 = 1;$$

therefore,

$$\cos(\arcsin x) = \sqrt{1 - x^2}.$$

(The positive square root is to be taken because $\arcsin x$ is in $(-\pi/2, \pi/2)$, so $\cos(\arcsin x) > 0$.) This proves the first formula.

The second formula has already been established (in the proof of Theorem 1). It is also possible to imitate the proof for the first formula, a valuable exercise if that proof presented any difficulties. The third formula is proved as follows.

$$\begin{aligned} \arctan'(x) &= (h^{-1})'(x) \\ &= \frac{1}{h'(h^{-1}(x))} \\ &= \frac{1}{\tan'(\arctan x)} \\ &= \frac{1}{\sec^2(\arctan x)}. \end{aligned}$$

Dividing both sides of the identity

$$\sin^2 a + \cos^2 a = 1$$

by $\cos^2 a$ yields

$$\tan^2 a + 1 = \sec^2 a.$$

It follows that

$$[\tan(\arctan x)]^2 + 1 = \sec^2(\arctan x),$$

or

$$x^2 + 1 = \sec^2(\arctan x),$$

which proves the third formula. ■

The traditional proof of the formula $\sin'(x) = \cos x$ (quite different from the one given here) is outlined in Problem 26. This proof depends upon first establishing the limit

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1,$$

and the “addition formula”

$$\sin(x + y) = \sin x \cos y + \cos x \sin y.$$

Both of these formulas can be derived easily now that the derivative of $\sin$ and $\cos$ are known. The first is just the special case $\sin'(0) = \cos 0$. The second depends on a beautiful characterization of the functions $\sin$ and $\cos$. In order to derive this result we need a lemma whose proof involves a clever trick; a more straightforward proof will be supplied in Part IV.

LEMMA Suppose f has a second derivative everywhere and that

$$\begin{aligned}f'' + f &= 0, \\f(0) &= 0, \\f'(0) &= 0.\end{aligned}$$

Then $f = 0$.

PROOF Multiplying both sides of the first equation by f' yields

$$f'f'' + ff' = 0.$$

Thus

$$[(f')^2 + f^2]' = 2(f'f'' + ff') = 0,$$

so $(f')^2 + f^2$ is a constant function. From $f(0) = 0$ and $f'(0) = 0$ it follows that the constant is 0; thus

$$[f'(x)]^2 + [f(x)]^2 = 0 \quad \text{for all } x.$$

This implies that

$$f(x) = 0 \quad \text{for all } x. \blacksquare$$

THEOREM 4 If f has a second derivative everywhere and

$$\begin{aligned}f'' + f &= 0, \\f(0) &= a, \\f'(0) &= b,\end{aligned}$$

then

$$f = b \cdot \sin + a \cdot \cos.$$

(In particular, if $f(0) = 0$ and $f'(0) = 1$, then $f = \sin$; if $f(0) = 1$ and $f'(0) = 0$, then $f = \cos$.)

PROOF Let

$$g(x) = f(x) - b \sin x - a \cos x.$$

Then

$$\begin{aligned}g'(x) &= f'(x) - b \cos x + a \sin x, \\g''(x) &= f''(x) + b \sin x + a \cos x.\end{aligned}$$

Consequently,

$$\begin{aligned}g'' + g &= 0, \\g(0) &= 0, \\g'(0) &= 0,\end{aligned}$$

which shows that

$$0 = g(x) = f(x) - b \sin x - a \cos x, \quad \text{for all } x. \blacksquare$$

THEOREM 5 If x and y are any two numbers, then

$$\begin{aligned}\sin(x + y) &= \sin x \cos y + \cos x \sin y, \\\cos(x + y) &= \cos x \cos y - \sin x \sin y.\end{aligned}$$

PROOF For any particular number y we can define a function f by

$$f(x) = \sin(x + y).$$

Then

$$\begin{aligned} f'(x) &= \cos(x + y), \\ f''(x) &= -\sin(x + y). \end{aligned}$$

Consequently,

$$\begin{aligned} f'' + f &= 0, \\ f(0) &= \sin y, \\ f'(0) &= \cos y. \end{aligned}$$

It follows from Theorem 4 that

$$f = (\cos y) \cdot \sin + (\sin y) \cdot \cos;$$

that is,

$$\sin(x + y) = \cos y \sin x + \sin y \cos x, \quad \text{for all } x.$$

Since any number y could have been chosen to begin with, this proves the first formula for all x and y .

The second formula is proved similarly. ■

As a conclusion to this chapter, and as a prelude to Chapter 17, we will mention an alternative approach to the definition of the function $\sin$. Since

$$\arcsin'(x) = \frac{1}{\sqrt{1-x^2}} \quad \text{for } -1 < x < 1,$$

it follows from the Second Fundamental Theorem of Calculus that

$$\arcsin x = \arcsin x - \arcsin 0 = \int_0^x \frac{1}{\sqrt{1-t^2}} dt.$$

This equation could have been taken as the *definition* of $\arcsin$. It would follow immediately that

$$\arcsin'(x) = \frac{1}{\sqrt{1-x^2}};$$

the function $\sin$ could then be defined as $(\arcsin)^{-1}$ and the formula for the derivative of an inverse function would show that

$$\sin'(x) = \sqrt{1 - \sin^2 x},$$

which could be defined as $\cos x$. Eventually, one could show that $A(\cos x) = x/2$, recovering at the very end of the development the definition with which we started. While much of this presentation would proceed more rapidly, the definition would be utterly unmotivated; the reasonableness of the definitions would be known to the author, but not to the student, for whom it was intended! Nevertheless, as we shall see in Chapter 17, an approach of this sort is sometimes very reasonable indeed.

PROBLEMS

1. Differentiate each of the following functions.

(i) $f(x) = \arctan(\arctan(\arctan x))$.

(ii) $f(x) = \arcsin(\arctan(\arccos x))$.

(iii) $f(x) = \arctan(\tan x \arctan x)$.

(iv) $f(x) = \arcsin\left(\frac{1}{\sqrt{1+x^2}}\right)$.

2. Find the following limits by l'Hôpital's Rule.

(i) $\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6}{x^3}$.

(ii) $\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6}{x^4}$.

(iii) $\lim_{x \rightarrow 0} \frac{\cos x - 1 + x^2/2}{x^2}$.

(iv) $\lim_{x \rightarrow 0} \frac{\cos x - 1 + x^2/2}{x^4}$.

(v) $\lim_{x \rightarrow 0} \frac{\arctan x - x + x^3/3}{x^3}$.

(vi) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$.

3. Let $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0. \end{cases}$

(a) Find $f'(0)$.

(b) Find $f''(0)$.

At this point, you will almost certainly have to use l'Hôpital's Rule, but in Chapter 23 we will be able to find $f^{(k)}(0)$ for all k , with almost no work at all.

4. Graph the following functions.

(a) $f(x) = \sin 2x$.

(b) $f(x) = \sin(x^2)$. (A pretty respectable sketch of this graph can be obtained using only a picture of the graph of $\sin$. Indeed, pure thought is your only hope in this problem, because determining the sign of the derivative $f'(x) = \cos(x^2) \cdot 2x$ is no easier than determining the behavior of f directly. The formula for $f'(x)$ does indicate one important fact, however— $f'(0) = 0$, which must be true since f is even, and which should be clear in your graph.)

(c) $f(x) = \sin x + \sin 2x$. (It will probably be instructive to first draw the graphs of $g(x) = \sin x$ and $h(x) = \sin 2x$ carefully on the same

set of axes, from 0 to 2π , and guess what the sum will look like. You can easily find out how many critical points f has on $[0, 2\pi]$ by considering the derivative of f . You can then determine the nature of these critical points by finding out the sign of f at each point; your sketch will probably suggest the answer.)

- (d) $f(x) = \tan x - x$. (First determine the behavior of f in $(-\pi/2, \pi/2)$; in the intervals $(k\pi - \pi/2, k\pi + \pi/2)$ the graph of f will look exactly the same, except moved up a certain amount. Why?)
- (e) $f(x) = \sin x - x$. (The material in the Appendix to Chapter 11 will be particularly helpful for this function.)
- (f) $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0. \end{cases}$

(Part (d) should enable you to determine approximately where the zeros of f' are located. Notice that f is even and continuous at 0; also consider the size of f for large x .)

5. Prove the addition formula for cos.
6. (a) From the addition formula for sin and cos derive formulas for $\sin 2x$, $\cos 2x$, $\sin 3x$, and $\cos 3x$.
- (b) Use these formulas to find the following values of the trigonometric functions (usually deduced by geometric arguments in elementary trigonometry):

$$\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2},$$

$$\tan \frac{\pi}{4} = 1,$$

$$\sin \frac{\pi}{6} = \frac{1}{2},$$

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}.$$

7. (a) Show that $A \sin(x + B)$ can be written as $a \sin x + b \cos x$ for suitable a and b . (One of the theorems in this chapter provides a one-line proof. You should also be able to figure out what a and b are.)
- (b) Conversely, given a and b , find numbers A and B such that $a \sin x + b \cos x = A \sin(x + B)$ for all x .
- (c) Use part (b) to graph $f(x) = \sqrt{3} \sin x + \cos x$.
8. (a) Prove that

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

provided that x , y , and $x + y$ are not of the form $k\pi + \pi/2$. (Use the addition formulas for sin and cos.)

(b) Prove that

$$\arctan x + \arctan y = \arctan \left(\frac{x+y}{1-xy} \right),$$

indicating any necessary restrictions on x and y . Hint: Replace x by $\arctan x$ and y by $\arctan y$ in part (a).

9. Prove that

$$\arcsin \alpha + \arcsin \beta = \arcsin(\alpha\sqrt{1-\beta^2} + \beta\sqrt{1-\alpha^2}),$$

indicating any restrictions on α and β .

10. Prove that if m and n are any numbers, then

$$\sin mx \sin nx = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x],$$

$$\sin mx \cos nx = \frac{1}{2}[\sin(m+n)x + \sin(m-n)x],$$

$$\cos mx \cos nx = \frac{1}{2}[\cos(m+n)x + \cos(m-n)x].$$

11. Prove that if m and n are natural numbers, then

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n, \end{cases}$$

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n, \end{cases}$$

$$\int_{-\pi}^{\pi} \sin mx \cos nx \, dx = 0.$$

These relations are particularly important in the theory of Fourier series. Although this topic will receive serious attention only in the Suggested Reading, the next problem provides a hint as to their importance.

12. (a) If f is integrable on $[-\pi, \pi]$, show that the minimum value of

$$\int_{-\pi}^{\pi} (f(x) - a \cos nx)^2 \, dx$$

occurs when

$$a = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx,$$

and the minimum value of

$$\int_{-\pi}^{\pi} (f(x) - a \sin nx)^2 \, dx$$

when

$$a = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

(In each case, bring a outside the integral sign, obtaining a quadratic expression in a .)

(b) Define

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad n = 0, 1, 2, \dots,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3, \dots$$

Show that if c_i and d_i are any numbers, then

$$\begin{aligned}
 & \int_{-\pi}^{\pi} \left(f(x) - \left[\frac{c_0}{2} + \sum_{n=1}^N c_n \cos nx + d_n \sin nx \right] \right)^2 dx \\
 &= \int_{-\pi}^{\pi} [f(x)]^2 dx - 2\pi \left(\frac{a_0 c_0}{2} + \sum_{n=1}^N a_n c_n + b_n d_n \right) + \pi \left(\frac{c_0^2}{2} + \sum_{n=1}^N c_n^2 + d_n^2 \right) \\
 &= \int_{-\pi}^{\pi} [f(x)]^2 dx - \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N a_n^2 + b_n^2 \right) \\
 &\quad + \pi \left(\left(\frac{c_0}{\sqrt{2}} - \frac{a_0}{\sqrt{2}} \right)^2 + \sum_{n=1}^N (c_n - a_n)^2 + (d_n - b_n)^2 \right),
 \end{aligned}$$

thus showing that the first integral is smallest when $a_i = c_i$ and $b_i = d_i$. In other words, among all “linear combinations” of the functions $s_n(x) = \sin nx$ and $c_n(x) = \cos nx$ for $1 \leq n \leq N$, the particular function

$$g(x) = \frac{a_0}{2} + \sum_{n=1}^N a_n \cos nx + b_n \sin nx$$

has the “closest fit” to f on $[-\pi, \pi]$.

13. Find a formula for $\sin x + \sin y$ and for $\cos x + \cos y$. Hint: First find a formula for $\sin(a+b) + \sin(a-b)$. What good does that do?
14. (a) Starting from the formula for $\cos 2x$, derive formulas for $\sin^2 x$ and $\cos^2 x$ in terms of $\cos 2x$.
(b) Prove that

$$\cos \frac{x}{2} = \sqrt{\frac{1 + \cos x}{2}} \quad \text{and} \quad \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}}$$

for $0 \leq x \leq \pi/2$.

- (c) Use part (a) to find $\int_a^b \sin^2 x \, dx$ and $\int_a^b \cos^2 x \, dx$.
15. Find $\sin(\arctan x)$ and $\cos(\arctan x)$ as expressions not involving trigonometric functions. Hint: $y = \arctan x$ means that $x = \tan y = \sin y / \cos y = \sin y / \sqrt{1 - \sin^2 y}$.
16. If $x = \tan u/2$, express $\sin u$ and $\cos u$ in terms of x . (Use Problem 15; the answers should be very simple expressions.)
17. (a) Prove that $\sin(x + \pi/2) = \cos x$. (All along we have been drawing the graphs of $\sin$ and $\cos$ as if this were the case.)
(b) What is $\arcsin(\cos x)$ and $\arccos(\sin x)$?
18. (a) Find $\int_0^1 \frac{1}{1+t^2} dt$. Hint: The answer is not 45.

(b) Find $\int_0^\infty \frac{1}{1+t^2} dt$.

19. Find $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$.

20. (a) Define functions $\sin^\circ$ and $\cos^\circ$ by $\sin^\circ(x) = \sin(\pi x/180)$ and $\cos^\circ(x) = \cos(\pi x/180)$. Find $(\sin^\circ)'$ and $(\cos^\circ)'$ in terms of these same functions.

(b) Find $\lim_{x \rightarrow 0} \frac{\sin^\circ x}{x}$ and $\lim_{x \rightarrow \infty} x \sin^\circ \frac{1}{x}$.

21. Prove that every point on the unit circle is of the form $(\cos \theta, \sin \theta)$ for at least one (and hence for infinitely many) numbers θ .

22. (a) Prove that π is the maximum possible length of an interval on which $\sin$ is one-one, and that such an interval must be of the form $(2k\pi - \pi/2, 2k\pi + \pi/2)$.

- (b) Suppose we let $g(x) = \sin x$ for x in $(2k\pi - \pi/2, 2k\pi + \pi/2)$. What is $(g^{-1})'$?

23. Let $f(x) = \sec x$ for $0 \leq x \leq \pi$. Find the domain of f^{-1} and sketch its graph.

24. Prove that $|\sin x - \sin y| < |x - y|$ for all numbers x and y . Hint: The same statement, with $<$ replaced by $\leq$, is a very straightforward consequence of a well-known theorem; simple supplementary considerations then allow $\leq$ to be improved to $<$.

- *25. It is an excellent test of intuition to predict the value of

$$\lim_{\lambda \rightarrow \infty} \int_a^b f(x) \sin \lambda x \, dx.$$

Continuous functions should be most accessible to intuition, but once you get the right idea for a proof the limit can easily be established for any integrable f .

- (a) Show that $\lim_{\lambda \rightarrow \infty} \int_c^d \sin \lambda x \, dx = 0$, by computing the integral explicitly.

- (b) Show that if s is a step function on $[a, b]$ (terminology from Problem 13-17), then $\lim_{\lambda \rightarrow \infty} \int_a^b s(x) \sin \lambda x \, dx = 0$.

- (c) Finally, use Problem 13-17 to show that $\lim_{\lambda \rightarrow \infty} \int_a^b f(x) \sin \lambda x \, dx = 0$ for any function f which is integrable on $[a, b]$.

This result, like Problem 11, plays an important role in the theory of Fourier series; it is known as the Riemann-Lebesgue Lemma.

26. This problem outlines the classical approach to the trigonometric functions. The shaded sector in Figure 24 has area $x/2$.

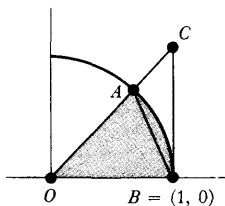


FIGURE 24

- (a) By considering the triangles OAB and OCB prove that if $0 < x < \pi/4$, then

$$\frac{\sin x}{2} < \frac{x}{2} < \frac{\sin x}{2 \cos x}.$$

- (b) Conclude that

$$\cos x < \frac{\sin x}{x} < 1,$$

and prove that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

- (c) Use this limit to find

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}.$$

- (d) Using parts (b) and (c), and the addition formula for $\sin$, find $\sin'(x)$, starting from the definition of the derivative.

***27.** Yet another development of the trigonometric functions was briefly mentioned in the text—starting with inverse functions defined by integrals. It is convenient to begin with $\arctan$, since this function is defined for all x . To do this problem, pretend that you have never heard of the trigonometric functions.

- (a) Let $\alpha(x) = \int_0^x (1+t^2)^{-1} dt$. Prove that α is odd and increasing, and that $\lim_{x \rightarrow \infty} \alpha(x)$ and $\lim_{x \rightarrow -\infty} \alpha(x)$ both exist, and are negatives of each other. If we define $\pi = 2 \lim_{x \rightarrow \infty} \alpha(x)$, then α^{-1} is defined on $(-\pi/2, \pi/2)$.

- (b) Show that $(\alpha^{-1})'(x) = 1 + [\alpha^{-1}(x)]^2$.

- (c) For $x = k\pi + x'$ with $x' \neq \pi/2$ or $-\pi/2$, define $\tan x = \alpha^{-1}(x')$. Then define $\cos x = 1/\sqrt{1 + \tan^2 x}$, for x not of the form $k\pi + \pi/2$ or $k\pi - \pi/2$, and $\cos(k\pi \pm \pi/2) = 0$. Prove first that $\cos'(x) = -\tan x \cos x$, and then that $\cos''(x) = -\cos x$ for all x .

***28.** If we are willing to assume that certain differential equations have solutions, another approach to the trigonometric functions is possible. Suppose, in particular, that there is some function y_0 which is not always 0 and which satisfies $y_0'' + y_0 = 0$.

- (a) Prove that $y_0^2 + (y_0')^2$ is constant, and conclude that either $y_0(0) \neq 0$ or $y_0'(0) \neq 0$.
- (b) Prove that there is a function s satisfying $s'' + s = 0$ and $s(0) = 0$ and $s'(0) = 1$. Hint: Try s of the form $ay_0 + by_0'$.

If we define $\sin = s$ and $\cos = s'$, then almost all facts about trigonometric functions become trivial. There is one point which requires work, however—producing the number π . This is most easily done using an exercise from the Appendix to Chapter 11:

- (c) Use Problem 6 of the Appendix to Chapter 11 to prove that $\cos x$ cannot be positive for all $x > 0$. It follows that there is a smallest $x_0 > 0$ with $\cos x_0 = 0$, and we can define $\pi = 2x_0$.
 - (d) Prove that $\sin \pi/2 = 1$. (Since $\sin^2 + \cos^2 = 1$, we have $\sin \pi/2 = \pm 1$; the problem is to decide why $\sin \pi/2$ is positive.)
 - (e) Find $\cos \pi$, $\sin \pi$, $\cos 2\pi$, and $\sin 2\pi$. (Naturally you may use any addition formulas, since these can be derived once we know that $\sin' = \cos$ and $\cos' = -\sin$.)
 - (f) Prove that $\cos$ and $\sin$ are periodic with period 2π .
29. (a) After all the work involved in the definition of $\sin$, it would be disconcerting to find that $\sin$ is actually a rational function. Prove that it isn't. (There is a simple property of $\sin$ which a rational function cannot possibly have.)
- (b) Prove that $\sin$ isn't even defined implicitly by an algebraic equation; that is, there do not exist rational functions $f_0, \dots, f_{n-1}$ such that

$$(\sin x)^n + f_{n-1}(x)(\sin x)^{n-1} + \dots + f_0(x) = 0 \quad \text{for all } x.$$

Hint: Prove that $f_0 = 0$, so that $\sin x$ can be factored out. The remaining factor is 0 except perhaps at multiples of 2π . But this implies that it is 0 for all x . (Why?) You are now set up for a proof by induction.

- *30. Suppose that ϕ_1 and ϕ_2 satisfy

$$\begin{aligned}\phi_1'' + g_1\phi_1 &= 0, \\ \phi_2'' + g_2\phi_2 &= 0,\end{aligned}$$

and that $g_2 > g_1$.

- (a) Show that

$$\phi_1''\phi_2 - \phi_2''\phi_1 - (g_2 - g_1)\phi_1\phi_2 = 0.$$

- (b) Show that if $\phi_1(x) > 0$ and $\phi_2(x) > 0$ for all x in (a, b) , then

$$\int_a^b [\phi_1''\phi_2 - \phi_2''\phi_1] > 0,$$

and conclude that

$$[\phi_1'(b)\phi_2(b) - \phi_1'(a)\phi_2(a)][\phi_1(b)\phi_2'(b) - \phi_1(a)\phi_2'(a)] > 0.$$

- (c) Show that in this case we cannot have $\phi_1(a) = \phi_1(b) = 0$. Hint: Consider the sign of $\phi_1'(a)$ and $\phi_1'(b)$.
- (d) Show that the equations $\phi_1(a) = \phi_1(b) = 0$ are also impossible if $\phi_1 > 0$, $\phi_2 < 0$ or $\phi_1 < 0$, $\phi_2 > 0$, or $\phi_1 < 0$, $\phi_2 < 0$ on (a, b) . (You should be able to do this with almost no extra work.)

The net result of this problem may be stated as follows: if a and b are consecutive zeros of ϕ_1 , then ϕ_2 must have a zero somewhere between a and b . This result, in a slightly more general form, is known as the Sturm Comparison Theorem. As a particular example, any solution of the differential equation

$$y'' + (x + 1)y = 0$$

must have zeros on the positive horizontal axis which are within π of each other.

31. Prove that if $\sin x/2 \neq 0$, then

$$\frac{1}{2} + \cos x + \cos 2x + \cdots + \cos nx = \frac{\sin(n + \frac{1}{2})x}{2 \sin \frac{x}{2}}.$$

At this stage of the game, this problem is a mathematical idiot's delight. A proof by induction depends on establishing the formula

$$\frac{\sin(n + \frac{1}{2})x}{2 \sin \frac{x}{2}} + \cos(n + 1)x = \frac{\sin(n + \frac{3}{2})x}{2 \sin \frac{x}{2}}.$$

This will require repeated use of addition formulas, in just the right places, but very little thinking. A more reasonable derivation of the formula will occur in Problem 26-13. Like two other results in this problem set, this equation is very important in the study of Fourier series, and we also make use of it in Problems 18-28 and 22-17.

The remaining problems in this chapter form a brief introduction to the topic of arc length, and show how trigonometric functions can be defined in terms of arc length.

- *32. Let f be a continuous function on $[a, b]$. If $P = \{t_0, \dots, t_n\}$ is a partition of $[a, b]$, define

$$\ell(f, P) = \sum_{i=1}^n \sqrt{(t_i - t_{i-1})^2 + [f(t_i) - f(t_{i-1})]^2}.$$

The number $\ell(f, P)$ represents the length of a polygonal curve inscribed in the graph of f (see Figure 25). We define the **length** of f on $[a, b]$ to be the least upper bound of all $\ell(f, P)$ for all partitions P (provided that the set of all such $\ell(f, P)$ is bounded above).

- If f is a linear function on $[a, b]$, prove that the length of f is the distance from $(a, f(a))$ to $(b, f(b))$.
- If f is not linear, prove that there is a partition $P = \{a, t, b\}$ of $[a, b]$ such that $\ell(f, P)$ is greater than the distance from $(a, f(a))$ to $(b, f(b))$. (You will need Problem 4-8.)

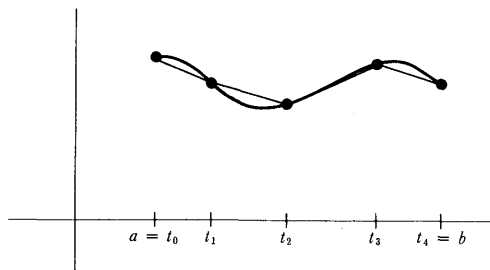


FIGURE 25

- (c) Conclude that of all functions f on $[a, b]$ with $f(a) = c$ and $f(b) = d$, the length of the linear function is less than the length of any other. (Or, in conventional but hopelessly muddled terminology: "A straight line is the shortest distance between two points.")
- *33.** (a) Suppose that f' is bounded on $[a, b]$. If P is any partition of $[a, b]$ show that

$$L(\sqrt{1 + (f')^2}, P) \leq \ell(f, P) \leq U(\sqrt{1 + (f')^2}, P).$$

Hint: Use the Mean Value Theorem.

- (b) Why is $\sup \{L(\sqrt{1 + (f')^2}, P)\} \leq \sup \{\ell(f, P)\}$? (This is easy.)
- (c) Now show that $\sup \{\ell(f, P)\} \leq \inf \{U(\sqrt{1 + (f')^2}, P)\}$, thereby proving that the length of f on $[a, b]$ is $\int_a^b \sqrt{1 + (f')^2}$, if $\sqrt{1 + (f')^2}$ is integrable on $[a, b]$. Hint: It suffices to show that if P' and P'' are any two partitions, then $\ell(f, P') \leq U(\sqrt{1 + (f')^2}, P'')$. If P contains the points of both P' and P'' , how does $\ell(f, P')$ compare to $\ell(f, P)$?
- *34.** Let $f(x) = \sqrt{1 - x^2}$ for $-1 \leq x \leq 1$. Define $\mathfrak{L}(x)$ to be the length of f on $[x, 1]$.

- (a) Show that

$$\mathfrak{L}(x) = \int_x^1 \frac{1}{\sqrt{1 - t^2}} dt.$$

(This is actually an improper integral, as defined in Problem 14-17.)

- (b) Show that

$$\mathfrak{L}'(x) = -\frac{1}{\sqrt{1 - x^2}} \quad \text{for } -1 < x < 1.$$

- (c) Define π as $\mathfrak{L}(-1)$. For $0 \leq x \leq \pi$, define $\cos x$ by $\mathfrak{L}(\cos x) = x$, and define $\sin x = \sqrt{1 - \cos^2 x}$. Prove that $\cos'(x) = -\sin x$ and $\sin'(x) = \cos x$ for $0 < x < \pi$.

This short chapter, diverging from the main stream of the book, is included to demonstrate that we are already in a position to do some sophisticated mathematics. This entire chapter is devoted to an elementary proof that π is irrational. Like many “elementary” proofs of deep theorems, the motivation for many steps in our proof cannot be supplied; nevertheless, it is still quite possible to follow the proof step-by-step.

Two observations must be made before the proof. The first concerns the function

$$f_n(x) = \frac{x^n(1-x)^n}{n!},$$

which clearly satisfies

$$0 < f_n(x) < \frac{1}{n!} \quad \text{for } 0 < x < 1.$$

An important property of the function f_n is revealed by considering the expression obtained by actually multiplying out $x^n(1-x)^n$. The lowest power of x appearing will be n and the highest power will be $2n$. Thus f_n can be written in the form

$$f_n(x) = \frac{1}{n!} \sum_{i=n}^{2n} c_i x^i,$$

where the numbers c_i are integers. It is clear from this expression that

$$f_n^{(k)}(0) = 0 \quad \text{if } k < n \text{ or } k > 2n.$$

Moreover,

$$\begin{aligned} f_n^{(n)}(x) &= \frac{1}{n!} [n! c_n + \text{terms involving } x] \\ f_n^{(n+1)}(x) &= \frac{1}{n!} [(n+1)! c_{n+1} + \text{terms involving } x] \\ &\vdots \\ f_n^{(2n)}(x) &= \frac{1}{n!} [(2n)! c_{2n}]. \end{aligned}$$

This means that

$$\begin{aligned} f_n^{(n)}(0) &= c_n, \\ f_n^{(n+1)}(0) &= (n+1)c_{n+1}, \\ &\vdots \\ f_n^{(2n)}(0) &= (2n)(2n-1) \cdot \dots \cdot (n+1)c_{2n}, \end{aligned}$$

where the numbers on the right are all integers. Thus

$$f_n^{(k)}(0) \text{ is an integer for all } k.$$

The relation

$$f_n(x) = f_n(1-x)$$

implies that

$$f_n^{(k)}(x) = (-1)^k f_n^{(k)}(1-x);$$

therefore,

$$f_n^{(k)}(1) \text{ is also an integer for all } k.$$

The proof that π is irrational requires one further observation: if a is any number, and $\varepsilon > 0$, then for sufficiently large n we will have

$$\frac{a^n}{n!} < \varepsilon.$$

To prove this, notice that if $n \geq 2a$, then

$$\frac{a^{n+1}}{(n+1)!} = \frac{a}{n+1} \cdot \frac{a^n}{n!} < \frac{1}{2} \cdot \frac{a^n}{n!}.$$

Now let n_0 be any natural number with $n_0 \geq 2a$. Then, whatever value

$$\frac{a^{n_0}}{(n_0)!}$$

may have, the succeeding values satisfy

$$\begin{aligned} \frac{a^{(n_0+1)}}{(n_0+1)!} &< \frac{1}{2} \cdot \frac{a^{n_0}}{(n_0)!} \\ \frac{a^{(n_0+2)}}{(n_0+2)!} &< \frac{1}{2} \cdot \frac{a^{(n_0+1)}}{(n_0+1)!} < \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{a^{n_0}}{(n_0)!} \\ &\vdots \\ \frac{a^{(n_0+k)}}{(n_0+k)!} &< \frac{1}{2^k} \cdot \frac{a^{n_0}}{(n_0)!}. \end{aligned}$$

If k is so large that $\frac{a^{n_0}}{(n_0)!} < 2^k$, then

$$\frac{a^{(n_0+k)}}{(n_0+k)!} < \varepsilon,$$

which is the desired result. Having made these observations, we are ready for the one theorem in this chapter.

THEOREM 1 The number π is irrational; in fact, π^2 is irrational. (Notice that the irrationality of π^2 implies the irrationality of π , for if π were rational, then π^2 certainly would be.)

PROOF Suppose π^2 were rational, so that

$$\pi^2 = \frac{a}{b}$$

for some positive integers a and b . Let

$$(1) \quad G(x) = b^n [\pi^{2n} f_n(x) - \pi^{2n-2} f_n''(x) + \pi^{2n-4} f_n^{(4)}(x) - \cdots + (-1)^n f_n^{(2n)}(x)].$$

Notice that each of the factors

$$b^n \pi^{2n-2k} = b^n (\pi^2)^{n-k} = b^n \left(\frac{a}{b}\right)^{n-k} = a^{n-k} b^k$$

is an integer. Since $f_n^{(k)}(0)$ and $f_n^{(k)}(1)$ are integers, this shows that

$$G(0) \text{ and } G(1) \text{ are integers.}$$

Differentiating G twice yields

$$(2) \quad G''(x) = b^n [\pi^{2n} f_n''(x) - \pi^{2n-2} f_n^{(4)}(x) + \cdots + (-1)^n f_n^{(2n+2)}(x)].$$

The last term, $(-1)^n f_n^{(2n+2)}(x)$, is zero. Thus, adding (1) and (2) gives

$$(3) \quad G''(x) + \pi^2 G(x) = b^n \pi^{2n+2} f_n(x) = \pi^2 a^n f_n(x).$$

Now let

$$H(x) = G'(x) \sin \pi x - \pi G(x) \cos \pi x.$$

Then

$$\begin{aligned} H'(x) &= \pi G'(x) \cos \pi x + G''(x) \sin \pi x - \pi G'(x) \cos \pi x + \pi^2 G(x) \sin \pi x \\ &= [G''(x) + \pi^2 G(x)] \sin \pi x \\ &= \pi^2 a^n f_n(x) \sin \pi x, \text{ by (3).} \end{aligned}$$

By the Second Fundamental Theorem of Calculus,

$$\begin{aligned} \pi^2 \int_0^1 a^n f_n(x) \sin \pi x \, dx &= H(1) - H(0) \\ &= G'(1) \sin \pi - \pi G(1) \cos \pi - G'(0) \sin 0 + \pi G(0) \cos 0 \\ &= \pi [G(1) + G(0)]. \end{aligned}$$

Thus

$$\pi \int_0^1 a^n f_n(x) \sin \pi x \, dx \text{ is an integer.}$$

On the other hand, $0 < f_n(x) < 1/n!$ for $0 < x < 1$, so

$$0 < \pi a^n f_n(x) \sin \pi x < \frac{\pi a^n}{n!} \quad \text{for } 0 < x < 1.$$

Consequently,

$$0 < \pi \int_0^1 a^n f_n(x) \sin \pi x \, dx < \frac{\pi a^n}{n!}.$$

This reasoning was completely independent of the value of n . Now if n is large enough, then

$$0 < \pi \int_0^1 a^n f_n(x) \sin \pi x \, dx < \frac{\pi a^n}{n!} < 1.$$

But this is absurd, because the integral is an integer, and there is no integer between 0 and 1. Thus our original assumption must have been incorrect: π^2 is irrational. ■

This proof is admittedly mysterious; perhaps most mysterious of all is the way that π enters the proof—it almost looks as if we have proved π irrational without ever mentioning a definition of π . A close reexamination of the proof will show that precisely one property of π is essential—

$$\sin(\pi) = 0.$$

The proof really depends on the properties of the function $\sin$, and proves the irrationality of the smallest positive number x with $\sin x = 0$. In fact, very few properties of $\sin$ are required, namely,

$$\begin{aligned} \sin' &= \cos, \\ \cos' &= -\sin, \\ \sin(0) &= 0, \\ \cos(0) &= 1. \end{aligned}$$

Even this list could be shortened; as far as the proof is concerned, $\cos$ might just as well be defined as $\sin'$. The properties of $\sin$ required in the proof may then be written

$$\begin{aligned} \sin'' + \sin &= 0, \\ \sin(0) &= 0, \\ \sin'(0) &= 1. \end{aligned}$$

Of course, this is not really very surprising at all, since, as we have seen in the previous chapter, these properties characterize the function $\sin$ completely.

PROBLEMS

1. (a) Prove that the areas of triangles OAB and OAC in Figure 1 are related by the equation

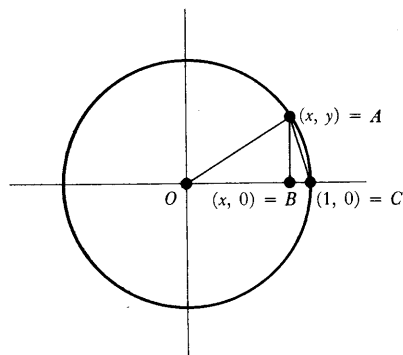


FIGURE 1

$$\text{area } OAC = \frac{1}{2} \sqrt{\frac{1 - \sqrt{1 - 16(\text{area } OAB)^2}}{2}}.$$

Hint: Solve the equations $xy = 2(\text{area } OAB)$, $x^2 + y^2 = 1$, for y .

- (b) Let P_m be the regular polygon of m sides inscribed in the unit circle. If A_m is the area of P_m show that

$$A_{2m} = \frac{m}{2} \sqrt{2 - 2 \sqrt{1 - (2A_m/m)^2}}.$$

This result allows one to obtain (more and more complicated) expressions for A_{2^n} , starting with $A_4 = 2$, and thus to compute π as accurately as desired (according to Problem 8-11). Although better methods will appear in Chapter 19, a slight variant of this approach yields a very interesting expression for π :

2. (a) Using the fact that

$$\frac{\text{area}(OAB)}{\text{area}(OAC)} = OB,$$

show that if α_m is the distance from O to one side of P_m , then

$$\frac{A_m}{A_{2m}} = \alpha_m.$$

- (b) Show that

$$\frac{2}{A_{2^k}} = \alpha_4 \cdot \alpha_8 \cdot \dots \cdot \alpha_{2^{k-1}}.$$

- (c) Using the fact that

$$\alpha_m = \cos \frac{\pi}{m},$$

and the formula $\cos x/2 = \sqrt{\frac{1 + \cos x}{2}}$ (Problem 15-14), prove that

$$\alpha_4 = \sqrt{\frac{1}{2}},$$

$$\alpha_8 = \sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{1}{2}}},$$

$$\alpha_{16} = \sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{1}{2}}}},$$

etc.

Together with part (b), this shows that $2/\pi$ can be written as an “infinite product”

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2}} \cdot \dots ;$$

to be precise, this equation means that the product of the first n factors can be made as close to $2/\pi$ as desired, by choosing n sufficiently large. This product was discovered by Francois Viète in 1579, and is only one of many fascinating expressions for π , some of which are mentioned later.

In Chapter 15 the integral provided a rigorous formulation for a preliminary definition of the functions $\sin$ and $\cos$. In this chapter the integral plays a more essential role. For certain functions even a preliminary definition presents difficulties. For example, consider the function

$$f(x) = 10^x.$$

This function is assumed to be defined for all x and to have an inverse function, defined for positive x , which is the “logarithm to the base 10,”

$$f^{-1}(x) = \log_{10} x.$$

In algebra, 10^x is usually defined only for *rational* x , while the definition for irrational x is quietly ignored. A brief review of the definition for rational x will not only explain this omission, but also recall an important principle behind the definition of 10^x .

The symbol 10^n is first defined for natural numbers n . This notation turns out to be extremely convenient, especially for multiplying very large numbers, because

$$10^n \cdot 10^m = 10^{n+m}.$$

The extension of the definition of 10^x to rational x is motivated by the desire to preserve this equation; this requirement actually forces upon us the customary definition. Since we want the equation

$$10^0 \cdot 10^n = 10^{0+n} = 10^n$$

to be true, we must define $10^0 = 1$; since we want the equation

$$10^{-n} \cdot 10^n = 10^0 = 1$$

to be true, we must define $10^{-n} = 1/10^n$; since we want the equation

$$\underbrace{10^{1/n} \cdot \dots \cdot 10^{1/n}}_{n \text{ times}} = 10^{1/n + \dots + 1/n} = 10^1 = 10$$

to be true, we must define $10^{1/n} = \sqrt[n]{10}$; and since we want the equation

$$\underbrace{10^{1/n} \cdot \dots \cdot 10^{1/n}}_{m \text{ times}} = 10^{1/n + \dots + 1/n} = 10^{m/n}$$

to be true, we must define $10^{m/n} = (\sqrt[n]{10})^m$.

Unfortunately, at this point the program comes to a dead halt. We have been guided by the principle that 10^x should be defined so as to ensure that $10^{x+y} = 10^x 10^y$; but this principle does not suggest any simple algebraic way of defining 10^x for irrational x . For this reason we will try some more sophisti-

cated ways of finding a function f such that

$$(*) \quad f(x + y) = f(x) \cdot f(y) \quad \text{for all } x \text{ and } y.$$

Of course, we are interested in a function which is not always zero, so we might add the condition $f(1) \neq 0$. If we add the more specific condition $f(1) = 10$, then $(*)$ will imply that $f(x) = 10^x$ for rational x , and 10^x could be *defined* as $f(x)$ for other x ; in general $f(x)$ will equal $[f(1)]^x$ for rational x .

One way to find such a function is suggested if we try to solve an apparently more difficult problem: find a *differentiable* function f such that

$$\begin{aligned} f(x + y) &= f(x) \cdot f(y) \quad \text{for all } x \text{ and } y, \\ f(1) &= 10. \end{aligned}$$

Assuming that such a function exists, we can try to find f' —knowing the derivative of f might provide a clue to the definition of f itself. Now

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x) \cdot f(h) - f(x)}{h} \\ &= f(x) \cdot \lim_{h \rightarrow 0} \frac{f(h) - 1}{h}. \end{aligned}$$

The answer thus depends on

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - 1}{h};$$

for the moment assume this limit exists, and denote it by α . Then

$$f'(x) = \alpha \cdot f(x) \quad \text{for all } x.$$

Even if α could be computed, this approach seems self-defeating. The derivative of f has been expressed in terms of f again.

If we examine the inverse function $f^{-1} = \log_{10}$, the whole situation appears in a new light:

$$\begin{aligned} \log_{10}'(x) &= \frac{1}{f'(f^{-1}(x))} \\ &= \frac{1}{\alpha \cdot f(f^{-1}(x))} = \frac{1}{\alpha x}. \end{aligned}$$

The derivative of f^{-1} is about as simple as one could ask! And, what is even more interesting, of all the integrals $\int_a^b x^n dx$ examined previously, the integral $\int_a^b x^{-1} dx$ is the only one which we cannot evaluate. Since $\log_{10} 1 = 0$ we should have

$$\frac{1}{\alpha} \int_1^x \frac{1}{t} dt = \log_{10} x - \log_{10} 1 = \log_{10} x.$$

This suggests that we define $\log_{10} x$ as $(1/\alpha) \int_1^x t^{-1} dt$. The difficulty is that α is unknown. One way of evading this difficulty is to define

$$\log x = \int_1^x \frac{1}{t} dt,$$

and hope that this integral will be the logarithm to *some* base, which might be determined later. In any case, the function defined in this way is surely more reasonable, from a mathematical point of view, than $\log_{10}$. The usefulness of $\log_{10}$ depends on the important role of the number 10 in arabic notation (and thus ultimately on the fact that we have ten fingers), while the function $\log$ provides a notation for an extremely simple integral which cannot be evaluated in terms of any functions already known to us.

DEFINITION

If $x > 0$, then

$$\log x = \int_1^x \frac{1}{t} dt.$$

The graph of $\log$ is shown in Figure 1. Notice that if $x > 1$, then $\log x > 0$, and if $0 < x < 1$, then $\log x < 0$, since, by our conventions,

$$\int_1^x \frac{1}{t} dt = - \int_x^1 \frac{1}{t} dt < 0.$$

For $x \leq 0$, a number $\log x$ cannot be defined in this way, because $f(t) = 1/t$ is not bounded on $[x, 1]$.

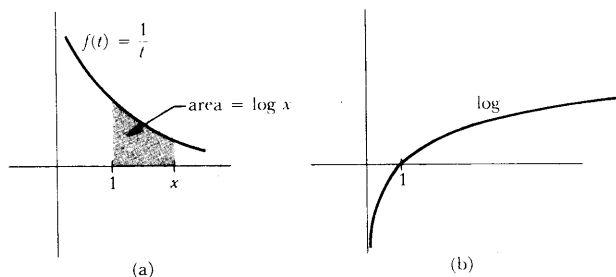


FIGURE 1

The justification for the notation “ $\log$ ” comes from the following theorem.

THEOREM 1 If $x, y > 0$, then

$$\log(xy) = \log x + \log y.$$

PROOF Notice first that $\log'(x) = 1/x$, by the Fundamental Theorem of Calculus. Now choose a number $y > 0$ and let

$$f(x) = \log(xy).$$

Then

$$f'(x) = \log'(xy) \cdot y = \frac{1}{xy} \cdot y = \frac{1}{x}.$$

Thus $f' = \log'$. This means that there is a number c such that

$$f(x) = \log x + c \quad \text{for all } x > 0,$$

that is,

$$\log(xy) = \log x + c \quad \text{for all } x > 0.$$

The number c can be evaluated by noting that when $x = 1$ we obtain

$$\begin{aligned} \log(1 \cdot y) &= \log 1 + c \\ &= c. \end{aligned}$$

Thus

$$\log(xy) = \log x + \log y \quad \text{for all } x.$$

Since this is true for all $y > 0$, the theorem is proved. ■

COROLLARY 1 If n is a natural number and $x > 0$, then

$$\log(x^n) = n \log x.$$

PROOF Left to you (use induction). ■

COROLLARY 2 If $x, y > 0$, then

$$\log\left(\frac{x}{y}\right) = \log x - \log y.$$

PROOF This follows from the equations

$$\log x = \log\left(\frac{x}{y} \cdot y\right) = \log\left(\frac{x}{y}\right) + \log y. \quad \blacksquare$$

Theorem 1 provides some important information about the graph of $\log$. The function $\log$ is clearly increasing, but since $\log'(x) = 1/x$, the derivative becomes very small as x becomes large, and $\log$ consequently grows more and more slowly. It is not immediately clear whether $\log$ is bounded or unbounded on $\mathbf{R}$. Observe, however, that for a natural number n ,

$$\log(2^n) = n \log 2 \quad (\text{and } \log 2 > 0);$$

it follows that $\log$ is, in fact, not bounded above. Similarly,

$$\log\left(\frac{1}{2^n}\right) = \log 1 - \log 2^n = -n \log 2;$$

therefore $\log$ is not bounded below on $(0, 1)$. Since $\log$ is continuous, it actually takes on all values. Therefore $\mathbf{R}$ is the domain of the function $\log^{-1}$. This important function has a special name, whose appropriateness will soon become clear.

DEFINITION

The “exponential function,” **exp**, is defined as $\log^{-1}$.

The graph of $\exp$ is shown in Figure 2. Since $\log x$ is defined only for $x > 0$, we always have $\exp(x) > 0$. The derivative of the function $\exp$ is easy to determine.

THEOREM 2 For all numbers x ,

$$\exp'(x) = \exp(x).$$

PROOF

$$\begin{aligned}\exp'(x) &= (\log^{-1})'(x) = \frac{1}{\log'(\log^{-1}(x))} \\ &= \frac{1}{\frac{1}{\log^{-1}(x)}} \\ &= \log^{-1}(x) = \exp(x). \blacksquare\end{aligned}$$

A second important property of $\exp$ is an easy consequence of Theorem 1.

THEOREM 3 If x and y are any two numbers, then

$$\exp(x + y) = \exp(x) \cdot \exp(y).$$

PROOF Let $x' = \exp(x)$ and $y' = \exp(y)$, so that

$$\begin{aligned}x &= \log x', \\ y &= \log y'.$$

Then

$$x + y = \log x' + \log y' = \log(x'y').$$

This means that

$$\exp(x + y) = x'y' = \exp(x) \cdot \exp(y). \blacksquare$$

This theorem, and the discussion at the beginning of this chapter, suggest that $\exp(1)$ is particularly important. There is, in fact, a special symbol for this number.

DEFINITION

$$e = \exp(1).$$

This definition is equivalent to the equation

$$1 = \log e = \int_1^e \frac{1}{t} dt.$$

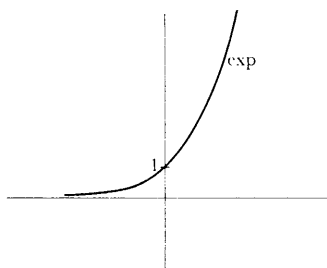


FIGURE 2

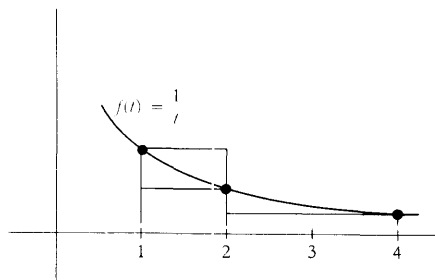


FIGURE 3

As illustrated in Figure 3,

$$\int_1^2 \frac{1}{t} dt < 1, \quad \text{since } 1 \cdot (2 - 1) \text{ is an upper sum for}$$

$$f(t) = 1/t \text{ on } [1, 2],$$

and

$$\int_1^4 \frac{1}{t} dt > 1, \quad \text{since } \frac{1}{2} \cdot (2 - 1) + \frac{1}{4} \cdot (4 - 2) = 1 \text{ is a lower}$$

sum for $f(t) = 1/t$ on $[1, 4]$.

Thus

$$\int_1^2 \frac{1}{t} dt < \int_1^e \frac{1}{t} dt < \int_1^4 \frac{1}{t} dt,$$

which shows that

$$2 < e < 4.$$

In Chapter 19 we will find much better approximations for e , and also prove that e is irrational (the proof is much easier than the proof that π is irrational!).

As we remarked at the beginning of the chapter, the equation

$$\exp(x + y) = \exp(x) \cdot \exp(y)$$

implies that

$$\begin{aligned} \exp(x) &= [\exp(1)]^x \\ &= e^x, \text{ for all rational } x. \end{aligned}$$

Since $\exp$ is defined for all x and $\exp(x) = e^x$ for rational x , it is consistent with our earlier use of the exponential notation to *define* e^x as $\exp(x)$ for all x .

DEFINITION

For any number x ,

$$e^x = \exp(x).$$

The terminology “exponential function” should now be clear. We have succeeded in defining e^x for an arbitrary (even irrational) exponent x . We have not yet defined a^x , if $a \neq e$, but there is a reasonable principle to guide us in the attempt. If x is *rational*, then

$$a^x = (e^{\log a})^x = e^{x \log a}.$$

But the last expression is defined for *all* x , so we can use it to define a^x .

DEFINITION

If $a > 0$, then, for any real number x ,

$$a^x = e^{x \log a}.$$

(If $a = e$ this definition clearly agrees with the previous one.)

The requirement $a > 0$ is necessary, in order that $\log a$ be defined. This is

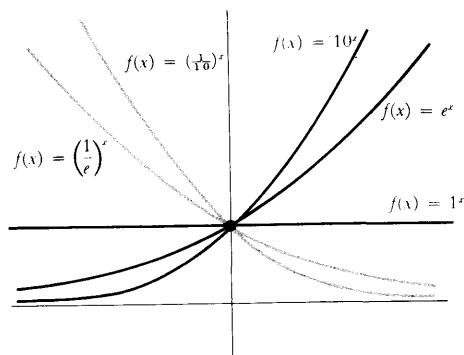


FIGURE 4

THEOREM 4

 If $a > 0$, then

$$(1) \quad (a^b)^c = a^{bc} \quad \text{for all } b, c.$$

 (Notice that a^b will automatically be positive, so $(a^b)^c$ will be defined);

$$(2) \quad a^1 = a \quad \text{and} \quad a^{x+y} = a^x \cdot a^y \quad \text{for all } x, y.$$

 (Notice that (2) implies that this definition of a^x agrees with the old one for rational x .)

PROOF

$$(1) \quad (a^b)^c = e^{c \log a^b} = e^{c \log (e^{b \log a})} = e^{c(b \log a)} = e^{cb \log a} = a^{bc}.$$

 (Each of the steps in this string of equalities depends upon our last definition, or the fact that $\exp = \log^{-1}$.)

$$(2) \quad a^1 = e^{1 \log a} = e^{\log a} = a, \\ a^{x+y} = e^{(x+y) \log a} = e^{x \log a + y \log a} = e^{x \log a} \cdot e^{y \log a} = a^x \cdot a^y. \quad \blacksquare$$

 Figure 4 shows the graphs of $f(x) = a^x$ for several different a . The behavior of the function depends on whether $a < 1$, $a = 1$, or $a > 1$. If $a = 1$, then $f(x) = 1^x = 1$. Suppose $a > 1$. In this case $\log a > 0$. Thus,

$$\begin{aligned} &\text{if} && x < y, \\ &\text{then} && x \log a < y \log a, \\ &\text{so} && e^{x \log a} < e^{y \log a}, \\ &\text{i.e.,} && a^x < a^y. \end{aligned}$$

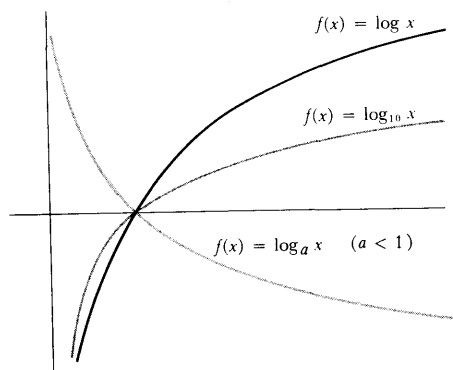
 Thus the function $f(x) = a^x$ is increasing. On the other hand, if $0 < a < 1$, so that $\log a < 0$, the same sort of reasoning shows that the function $f(x) = a^x$ is decreasing. In either case, if $a > 0$ and $a \neq 1$, then $f(x) = a^x$ is one-one. Since $\exp$ takes on every positive value it is also easy to see that a^x takes on every positive value. Thus the inverse function is defined for all positive numbers, and takes on all values. If $f(x) = a^x$, then f^{-1} is the function usually denoted by $\log_a$ (Figure 5).


FIGURE 5

Just as a^x can be expressed in terms of $\exp$, so $\log_a$ can be expressed in terms of $\log$. Indeed,

$$\begin{array}{ll} \text{if} & y = \log_a x, \\ \text{then} & x = a^y = e^{y \log a}, \\ \text{so} & \log x = y \log a, \\ \text{or} & y = \frac{\log x}{\log a}. \end{array}$$

In other words,

$$\log_a x = \frac{\log x}{\log a}.$$

The derivatives of $f(x) = a^x$ and $g(x) = \log_a x$ are both easy to find:

$$\begin{aligned} f(x) &= e^{x \log a}, \quad \text{so } f'(x) = \log a \cdot e^{x \log a} = \log a \cdot a^x, \\ g(x) &= \frac{\log x}{\log a}, \quad \text{so } g'(x) = \frac{1}{x \log a}. \end{aligned}$$

A more complicated function like

$$f(x) = g(x)^{h(x)}$$

is also easy to differentiate, if you remember that, *by definition*,

$$f(x) = e^{h(x) \log g(x)};$$

it follows from the Chain Rule that

$$\begin{aligned} f'(x) &= e^{h(x) \log g(x)} \cdot \left[h'(x) \log g(x) + h(x) \frac{g'(x)}{g(x)} \right] \\ &= g(x)^{h(x)} \cdot \left[h'(x) \log g(x) + h(x) \frac{g'(x)}{g(x)} \right]. \end{aligned}$$

There is no point in remembering this formula—simply apply the principle behind it in any specific case that arises; it does help, however, to remember that the first factor in the derivative will be $g(x)^{h(x)}$.

There is one special case of the above formula which is worth remembering. The function $f(x) = x^a$ was previously defined only for rational a . We can now define and find the derivative of the function $f(x) = x^a$ for any number a ; the result is just what we would expect:

$$f(x) = x^a = e^{a \log x},$$

so

$$f'(x) = \frac{a}{x} \cdot e^{a \log x} = \frac{a}{x} \cdot x^a = ax^{a-1}.$$

Algebraic manipulations with the exponential functions will become second nature after a little practice—just remember that all the rules which ought to work actually do. The basic properties of $\exp$ are still those stated in Theorems 2 and 3:

$$\begin{aligned}\exp'(x) &= \exp(x), \\ \exp(x+y) &= \exp(x) \cdot \exp(y).\end{aligned}$$

In fact, each of these properties comes close to characterizing the function $\exp$. Naturally, $\exp$ is not the only function f satisfying $f' = f$, for if $f = ce^x$, then $f'(x) = ce^x = f(x)$; these functions are the only ones with this property, however.

THEOREM 5 If f is differentiable and

$$f'(x) = f(x) \quad \text{for all } x,$$

then there is a number c such that

$$f(x) = ce^x \quad \text{for all } x.$$

PROOF Let

$$g(x) = \frac{f(x)}{e^x}.$$

(This is permissible, since $e^x \neq 0$ for all x .) Then

$$g'(x) = \frac{e^x f'(x) - f(x)e^x}{(e^x)^2} = 0.$$

Therefore there is a number c such that

$$g(x) = \frac{f(x)}{e^x} = c \quad \text{for all } x. \blacksquare$$

The second basic property of $\exp$ requires a more involved discussion. The function $\exp$ is clearly not the only function f which satisfies

$$f(x+y) = f(x) \cdot f(y).$$

In fact, $f(x) = 0$ or any function of the form $f(x) = a^x$ also satisfies this equation. But the true story is much more complex than this—there are infinitely many other functions which satisfy this property, but it is impossible, without appealing to more advanced mathematics, to prove that there is even one function other than those already mentioned! It is for this reason that the definition of 10^x is so difficult: there are infinitely many functions f which satisfy

$$\begin{aligned}f(x+y) &= f(x) \cdot f(y), \\ f(1) &= 10,\end{aligned}$$

but which are *not* the function $f(x) = 10^x$! One thing is true however—any *continuous* function f satisfying

$$f(x+y) = f(x) \cdot f(y)$$

must be of the form $f(x) = a^x$ or $f(x) = 0$. (Problem 27 indicates the way to

prove this, and also has a few words to say about discontinuous functions with this property.)

In addition to the two basic properties stated in Theorems 2 and 3, the function $\exp$ has one further property which is very important— $\exp$ “grows faster than any polynomial.” In other words,

THEOREM 6 For any natural number n ,

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty.$$

PROOF The proof consists of several steps.

Step 1. $e^x > x$ for all x , and consequently $\lim_{x \rightarrow \infty} e^x = \infty$ (this may be considered to be the case $n = 0$).

To prove this statement (which is clear for $x \leq 0$) it suffices to show that

$$x > \log x \quad \text{for all } x > 0.$$

If $x < 1$ this is clearly true, since $\log x < 0$. If $x > 1$, then (Figure 6) $x - 1$ is an upper sum for $f(t) = 1/t$ on $[1, x]$, so $\log x < x - 1 < x$.

Step 2. $\lim_{x \rightarrow \infty} \frac{e^x}{x} = \infty$.

To prove this, note that

$$\frac{e^x}{x} = \frac{e^{x/2} \cdot e^{x/2}}{\frac{x}{2} \cdot 2} = \frac{1}{2} \left(\frac{e^{x/2}}{\frac{x}{2}} \right) \cdot e^{x/2}.$$

By Step 1, the expression in parentheses is greater than 1, and $\lim_{x \rightarrow \infty} e^{x/2} = \infty$; this shows that $\lim_{x \rightarrow \infty} e^x/x = \infty$.

Step 3. $\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$.

Note that

$$\frac{e^x}{x^n} = \frac{(e^{x/n})^n}{\left(\frac{x}{n}\right)^n \cdot n^n} = \frac{1}{n^n} \cdot \left(\frac{e^{x/n}}{\frac{x}{n}} \right)^n.$$

The expression in parentheses becomes arbitrarily large, by Step 2, so the n th power certainly becomes arbitrarily large. ■

It is now possible to examine carefully the following very interesting function: $f(x) = e^{-1/x^2}$, $x \neq 0$. We have

$$f'(x) = e^{-1/x^2} \cdot \frac{2}{x^3}.$$

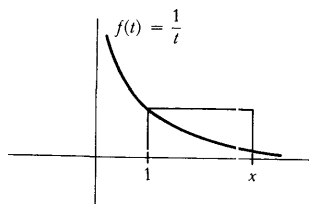


FIGURE 6

Therefore,

$$\begin{aligned} f'(x) &< 0 & \text{for } x < 0, \\ f'(x) &> 0 & \text{for } x > 0, \end{aligned}$$

so f is decreasing for negative x and increasing for positive x . Moreover, if $|x|$ is large, then x^2 is large, so $-1/x^2$ is close to 0, so e^{-1/x^2} is close to 1 (Figure 7).

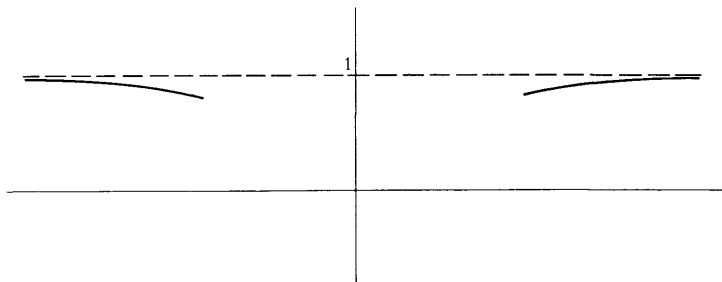


FIGURE 7

The behavior of f near 0 is more interesting. If x is small, then $1/x^2$ is large, so e^{1/x^2} is large, so $e^{-1/x^2} = 1/(e^{1/x^2})$ is small. This argument, suitably stated with ϵ 's and δ 's, shows that

$$\lim_{x \rightarrow 0} e^{-1/x^2} = 0.$$

Therefore, if we define

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

then the function f is continuous (Figure 8). In fact, f is actually differentiable

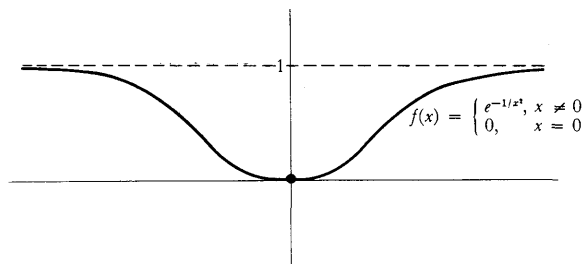


FIGURE 8

at 0: Indeed

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{e^{-1/h^2}}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h e^{(1/h)^2}} \\ &= \lim_{x \rightarrow \infty} \frac{x}{e^{(x^2)}}. \end{aligned}$$

We already know that

$$\lim_{x \rightarrow \infty} \frac{e^x}{x} = \infty;$$

it is all the more true that

$$\lim_{x \rightarrow \infty} \frac{e^{(x^2)}}{x} = \infty,$$

and this means that

$$\lim_{x \rightarrow \infty} \frac{x}{e^{(x^2)}} = 0.$$

Thus

$$f'(x) = \begin{cases} e^{-1/x^2} \cdot \frac{2}{x^3}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

We can now compute that

$$\begin{aligned} f''(0) &= \lim_{h \rightarrow 0} \frac{f'(h) - f'(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{-1/h^2} \cdot \frac{2}{h^3}}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \cdot e^{-1/h^2}}{h^4} = \lim_{h \rightarrow 0} \frac{2 \cdot \frac{1}{h^4}}{e^{1/h^2}} = \lim_{x \rightarrow \infty} \frac{2x^4}{e^{(x^2)}}; \end{aligned}$$

an argument similar to the one above shows that $f''(0) = 0$. Thus

$$f''(x) = \begin{cases} e^{-1/x^2} \cdot \frac{-6}{x^4} + e^{-1/x^2} \cdot \frac{4}{x^6}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

This argument can be continued. In fact, using induction it can be shown (Problem 29) that $f^{(k)}(0) = 0$ for every k . The function f is *extremely flat* at 0,

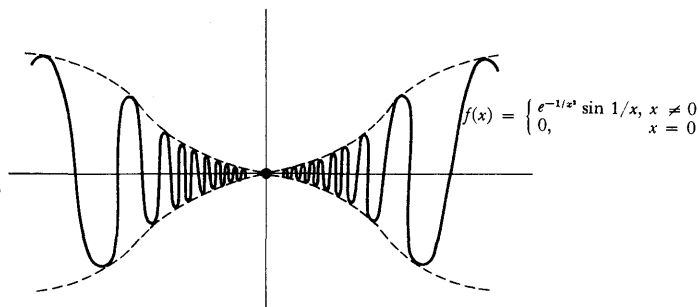


FIGURE 9

and approaches 0 so quickly that it can mask many irregularities of other functions. For example (Figure 9), suppose that

$$f(x) = \begin{cases} e^{-1/x^2} \cdot \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

It can be shown (Problem 30) that for this function it is also true that $f^{(k)}(0) = 0$ for all k . This example shows, perhaps more strikingly than any other, just how bad a function can be, and still be infinitely differentiable. In Part IV we will investigate even more restrictive conditions on a function, which will finally rule out behavior of this sort.

PROBLEMS

1. Differentiate each of the following functions (remember that a^{b^c} always denotes $a^{(b^c)}$).

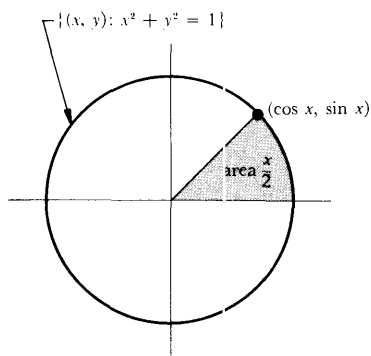
- (i) $f(x) = e^{e^{e^x}}$.
- (ii) $f(x) = \log(1 + \log(1 + \log(1 + e^{1+e^{1+x}})))$.
- (iii) $f(x) = (\sin x)^{\sin(\sin x)}$.
- (iv) $f(x) = e^{\left(\int_0^x e^{-t^2} dt\right)}$.
- (v) $f(x) = \sin x^{\sin x \sin x}$.
- (vi) $f(x) = \log_{(e^x)} \sin x$.
- (vii) $f(x) = \left[\arcsin \left(\frac{x}{\sin x} \right) \right]^{\log(\sin e^x)}$.
- (viii) $f(x) = (\log(3 + e^4))e^{4x} + (\arcsin x)^{\log 3}$.
- (ix) $f(x) = (\log x)^{\log x}$.
- (x) $f(x) = x^x$.

2. Graph each of the following functions.

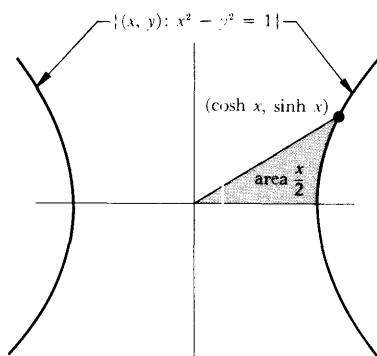
- (a) $f(x) = e^{x+1}$.
 - (b) $f(x) = e^{\sin x}$.
 - (c) $f(x) = e^x + e^{-x}$.
 - (d) $f(x) = e^x - e^{-x}$.
- } (Compare the graphs with the graphs of exp and 1/exp.)
- (e) $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}$.

3. Find the following limits by l'Hôpital's Rule.

- (i) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2/2}{x^2}$.
- (ii) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2/2 - x^3/6}{x^3}$.
- (iii) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2/2}{x^3}$.



(a)



(b)

FIGURE 10

$$(iv) \lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2}{x^2}.$$

$$(v) \lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2}{x^3}.$$

$$(vi) \lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2 - x^3/3}{x^3}.$$

4. The functions

$$\sinh x = \frac{e^x - e^{-x}}{2},$$

$$\cosh x = \frac{e^x + e^{-x}}{2},$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = 1 - \frac{2}{e^{2x} + 1},$$

are called the **hyperbolic sine**, **hyperbolic cosine**, and **hyperbolic tangent**, respectively (but usually read *sinch*, *cosh*, and *tanch*). There are many analogies between these functions and their ordinary trigonometric counterparts. One analogy is illustrated in Figure 10; a proof that the region shown in Figure 10(b) really has area $x/2$ is best deferred until the next chapter, when we will develop methods of computing integrals. Other analogies are discussed in the following three problems, but the deepest analogies must wait until Chapter 26. If you have not already done Problem 2, graph the functions $\sinh$, $\cosh$, and $\tanh$.

5. Prove that

$$(a) \cosh^2 - \sinh^2 = 1.$$

$$(b) \tanh^2 + 1/\cosh^2 = 1.$$

$$(c) \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y.$$

$$(d) \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

$$(e) \sinh' = \cosh.$$

$$(f) \cosh' = \sinh.$$

$$(g) \tanh' = \frac{1}{\cosh^2}.$$

6. The functions $\sinh$ and $\tanh$ are one-one; their inverses, denoted by $\arg \sinh$ and $\arg \tanh$ (the “argument” of the hyperbolic sine and tangent), are defined on $\mathbf{R}$ and $(-1, 1)$, respectively. If $\cosh$ is restricted to $[0, \infty)$ it has an inverse, denoted by $\arg \cosh$, which is defined on $[1, \infty)$. Prove, using the information in Problem 5, that

$$(a) \sinh(\cosh^{-1} x) = \sqrt{x^2 - 1}.$$

$$(b) \cosh(\sinh^{-1} x) = \sqrt{1 + x^2}.$$

$$(c) (\sinh^{-1})'(x) = \frac{1}{\sqrt{1 + x^2}}.$$

$$(d) (\cosh^{-1})'(x) = \frac{1}{\sqrt{x^2 - 1}} \quad \text{for } x > 1.$$

$$(e) (\tanh^{-1})'(x) = \frac{1}{1 - x^2} \quad \text{for } |x| < 1.$$

7. (a) Find an explicit formula for $\sinh^{-1}$, $\cosh^{-1}$, and $\tanh^{-1}$ (by solving the equation $y = \sinh^{-1} x$ for x in terms of y , etc.).

- (b) Find

$$\int_a^b \frac{1}{\sqrt{1 + x^2}} dx,$$

$$\int_a^b \frac{1}{\sqrt{x^2 - 1}} dx \quad \text{for } a, b > 1 \text{ or } a, b < -1,$$

$$\int_a^b \frac{1}{1 - x^2} dx \quad \text{for } |a|, |b| < 1.$$

Compare your answer for the third integral with that obtained by writing

$$\frac{1}{1 - x^2} = \frac{1}{2} \left[\frac{1}{1 - x} + \frac{1}{1 + x} \right].$$

8. Find

$$(a) \lim_{x \rightarrow \infty} a^x \text{ for } 0 < a < 1. \text{ (Remember the definition!)}$$

$$(b) \lim_{x \rightarrow \infty} \frac{x}{(\log x)^n}.$$

$$(c) \lim_{x \rightarrow \infty} \frac{(\log x)^n}{x}.$$

$$(d) \lim_{x \rightarrow 0^+} x(\log x)^n. \text{ Hint: } x(\log x)^n = \frac{(-1)^n \left(\log \frac{1}{x} \right)^n}{\frac{1}{x}}.$$

$$(e) \lim_{x \rightarrow 0^+} x^x.$$

9. Graph $f(x) = x^x$ for $x > 0$. (Use Problem 8(e).)

10. (a) Find the minimum value of $f(x) = e^x/x^n$ for $x > 0$, and conclude that $f(x) > e^n/n^n$ for $x > n$.

(b) Using the expression $f'(x) = e^x(x - n)/x^{n+1}$, prove that $f'(x) > e^{n+1}/(n + 1)^{n+1}$ for $x > n + 1$, and thus obtain another proof that $\lim_{x \rightarrow \infty} f(x) = \infty$.

11. Graph $f(x) = e^x/x^n$.

12. (a) Find $\lim_{y \rightarrow 0} \log(1 + y)/y$. (You can use l'Hôpital's Rule, but that would be silly.)

(b) Find $\lim_{x \rightarrow \infty} x \log(1 + 1/x)$.

- (c) Prove that $e = \lim_{x \rightarrow \infty} (1 + 1/x)^x$.
- (d) Prove that $e^a = \lim_{x \rightarrow \infty} (1 + a/x)^x$. (It is possible to derive this from part (c) with just a little algebraic fiddling.)
- *(e) Prove that $\log b = \lim_{x \rightarrow \infty} x(b^{1/x} - 1)$.
13. Graph $f(x) = (1 + 1/x)^x$ for $x > 0$. (Use Problem 12(c).)
14. If a bank gives a percent interest per annum, then an initial investment I yields $I(1 + a/100)$ after 1 year. If the bank compounds the interest (counts the accrued interest as part of the capital for computing interest the next year), then the initial investment grows to $I(1 + a/100)^n$ after n years. Now suppose that interest is given twice a year. The final amount after n years is, alas, not $I(1 + a/100)^{2n}$, but merely $I(1 + a/200)^{2n}$ —although interest is awarded twice as often, the interest must be halved in each calculation, since the interest is $a/2$ per half year. This amount is larger than $I(1 + a/100)^n$, but not that much larger. Suppose that the bank now compounds the interest continuously, i.e., the bank considers what the investment would yield when compounding k times a year, and then takes the least upper bound of all these numbers. How much will an initial investment of 1 dollar yield after 1 year?
15. Let $f(x) = \log |x|$ for $x \neq 0$. Prove that $f'(x) = 1/x$ for $x \neq 0$.
- *16. Prove that if $f' = cf$ for some number c , then $f(x) = ke^{cx}$ for some number k .
- *17. A radioactive substance diminishes at a rate proportional to the amount present (since all atoms have equal probability of disintegrating, the total disintegration is proportional to the number of atoms remaining). If $A(t)$ is the amount at time t , this means that $A'(t) = cA(t)$ for some c (which represents the probability that an atom will disintegrate).
- (a) Find $A(t)$ in terms of the amount $A_0 = A(0)$ present at time 0.
- (b) Show that there is a number τ (the “half-life” of the radioactive element) with the property that $A(t + \tau) = A(t)/2$.
- *18. *Newton's law of cooling* states that an object cools at a rate proportional to the difference of its temperature and the temperature of the surrounding medium. Find the temperature $T(t)$ of the object at time t , in terms of its temperature T_0 at time 0, assuming that the temperature of the surrounding medium is kept at a constant, M . Hint: To solve the differential equation expressing Newton's law, remember that $T' = (T - M)'$.
- *19. Prove that if $f(x) = \int_0^x f(t) dt$, then $f = 0$.
- *20. Find all functions f satisfying $f'(t) = f(t) + \int_0^1 f(t) dt$.
21. (a) Prove that

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} \leq e^x \quad \text{for } x \geq 0.$$

Hint: Use induction on n , and compare derivatives.

(b) Give a new proof that $\lim_{x \rightarrow \infty} e^x/x^n = \infty$.

22. Give yet another proof of this fact, using the appropriate form of l'Hôpital's Rule. (See Problem 11-38.)

*23. A point P is moving along a line segment AB of length 10^7 while another point Q moves along an infinite ray (Figure 11). The velocity of P is always equal to the distance from P to B (in other words, if $P(t)$ is the position of P at time t , then $P'(t) = 10^7 - P(t)$), while Q moves with constant velocity $Q'(t) = 10^7$. The distance traveled by Q after time t is defined to be the *Napierian logarithm* of the distance from P to B at time t . Thus

$$10^7 t = \text{Nap log}[10^7 - P(t)].$$

This was the definition of logarithms given by Napier (1550–1617) in his publication of 1614, *Mirifici logarithmonum canonis description* (A Description of the Wonderful Law of Logarithms); work which was done *before* the use of exponents was invented! The number 10^7 was chosen because Napier's tables (intended for astronomical and navigational calculations), listed the logarithms of sines of angles, for which the best possible available tables extended to seven decimal places, and Napier wanted to avoid fractions. Prove that

$$\text{Nap log } x = 10^7 \log \frac{10^7}{x}.$$

Hint: Use the same trick as in Problem 18 to solve the equation for P .

- *24. (a) Sketch the graph of $f(x) = (\log x)/x$ (paying particular attention to the behavior near 0 and ∞).
 (b) Which is larger, e^x or π^e ?
 (c) Prove that if $0 < x < 1$, or $x = e$, then the only number y satisfying $x^y = y^x$ is $y = x$; but if $x > 1$, $x \neq e$, then there is precisely one number $y \neq x$ satisfying $x^y = y^x$; moreover, if $x < e$, then $y > e$, and if $x > e$, then $y < e$. (Interpret these statements in terms of the graph in part (a)!)
 (d) Prove that if x and y are natural numbers and $x^y = y^x$, then $x = y$ or $x = 2$, $y = 4$, or $x = 4$, $y = 2$.
 (e) Show that the set of all pairs (x, y) with $x^y = y^x$ consists of a curve and a straight line which intersect; find the intersection and draw a rough sketch.
 ** (f) For $1 < x < e$ let $g(x)$ be the unique number $> e$ with $x^{g(x)} = g(x)^x$. Prove that g is differentiable. (It is a good idea to consider separate functions,

$$f_1(x) = \frac{\log x}{x}, \quad 0 < x < e$$

$$f_2(x) = \frac{\log x}{x}, \quad e < x$$

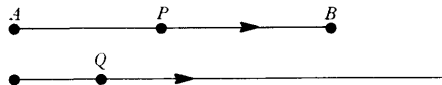


FIGURE 11

and write g in terms of f_1 and f_2 . If you do this part properly you should be able to show that

$$g'(x) = \frac{[g(x)]^2}{1 - \log g(x)} \cdot \frac{1 - \log x}{x^2}.$$

***25.** This problem uses the material from the Appendix to Chapter 11.

(a) Prove that $\exp$ is convex and $\log$ is concave.

(b) Prove that if $\sum_{i=1}^n p_i = 1$ and all $p_i > 0$, then

$$z_1^{p_1} \cdot \dots \cdot z_n^{p_n} < p_1 z_1 + \dots + p_n z_n.$$

(Use Problem 8 from the Appendix to Chapter 11.)

(c) Deduce another proof that $G_n \leq A_n$ (Problem 2-20).

26. Suppose f satisfies $f' = f$ and $f(x+y) = f(x)f(y)$ for all x and y . Prove that $f = \exp$ or $f = 0$.

***27.** Prove that if f is continuous and $f(x+y) = f(x)f(y)$ for all x and y , then either $f = 0$ or $f(x) = [f(1)]^x$ for all x . Hint: Show that $f(x) = [f(1)]^x$ for rational x , and then use Problem 8-6. This problem is closely related to Problem 8-7, and the information mentioned at the end of Problem 8-7 can be used to show that there are discontinuous functions f satisfying $f(x+y) = f(x)f(y)$.

***28.** Prove that if f is a continuous function defined on the positive reals, and $f(xy) = f(x) + f(y)$ for all positive x and y , then $f = 0$ or $f(x) = f(e) \log x$ for all $x > 0$. Hint: Consider $g(x) = f(e^x)$.

***29.** Prove that if $f(x) = e^{-1/x^2}$ for $x \neq 0$, and $f(0) = 0$, then $f^{(k)}(0) = 0$ for all k .

***30.** Prove that if $f(x) = e^{-1/x^2} \sin 1/x$ for $x \neq 0$, and $f(0) = 0$, then $f^{(k)}(0) = 0$ for all k .

31. (a) Prove that if α is a root of the equation

$$(*) \quad a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0,$$

then the function $y(x) = e^{\alpha x}$ satisfies the differential equation

$$(**) \quad a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0.$$

*** (b)** Prove that if α is a double root of $(*)$, then $y(x) = x e^{\alpha x}$ also satisfies $(**)$. Hint: Remember that if α is a double root of a polynomial equation $f(x) = 0$, then $f'(\alpha) = 0$.

*** (c)** Prove that if α is a root of $(*)$ of order r , then $y(x) = x^k e^{\alpha x}$ is a solution for $0 \leq k \leq r-1$.

If $(*)$ has n real numbers as roots (counting multiplicities), part (c) gives n solutions $y_1, \dots, y_n$ of $(**)$.

(d) Prove that in this case the function $c_1 y_1 + \dots + c_n y_n$ also satisfies $(**)$.

It is a theorem that in this case these are the only solutions of (**). Problem 16 and the next two problems prove special cases of this theorem, and the general case is considered in Problem 19-17. In Chapter 26 we will see what to do when (*) does not have n real numbers as roots.

- *32.** Suppose that f satisfies $f'' - f = 0$ and $f(0) = f'(0) = 0$. Prove that $f = 0$ as follows.
- Show that $f^2 - (f')^2 = 0$.
 - Suppose that $f(x) \neq 0$ for all x in some interval (a, b) . Show that either $f(x) = ce^x$ or else $f(x) = ce^{-x}$ for all x in (a, b) , for some constant c .
 - If $f(x_0) \neq 0$ for $x_0 > 0$, say, then there would be a number a such that $0 \leq a < x_0$ and $f(a) = 0$, while $f(x) \neq 0$ for $a < x < x_0$. Why? Use this fact and part (b) to deduce a contradiction.
- *33.** (a) Show that if f satisfies $f'' - f = 0$, then $f(x) = ae^x + be^{-x}$ for some a and b . (First figure out what a and b should be in terms of $f(0)$ and $f'(0)$, and then use Problem 32.)
- (b) Show also that $f = a \sinh + b \cosh$ for some (other) a and b .
- 34.** Find all functions f satisfying
- $f^{(n)} = f^{(n-1)}$.
 - $f^{(n)} = f^{(n-2)}$.
- *35.** This problem, a companion to Problem 15-28, outlines a treatment of the exponential function starting from the assumption that the differential equation $f' = f$ has a nonzero solution.
- Suppose there is a function $f \neq 0$ with $f' = f$. Prove that $f(x) \neq 0$ for each x by considering the function $g(x) = f(x_0 + x)f(x_0 - x)$, where $f(x_0) \neq 0$.
 - Show that there is a function f satisfying $f' = f$ and $f(0) = 1$.
 - For this f show that $f(x + y) = f(x) \cdot f(y)$ by considering the function $g(x) = f(x + y)/f(x)$.
 - Prove that f is one-one and that $(f^{-1})'(x) = 1/x$.
- **36.** A function f is said to **grow faster than** g if $\lim_{x \rightarrow \infty} f(x)/g(x) = \infty$. For example, $\exp$ grows faster than any polynomial function. Suppose that $g_1, g_2, g_3, \dots$ are continuous functions. Show that there is a continuous function f which grows faster than each g_i .
- 37.** Prove that $\log_{10} 2$ is irrational.

Every computation of a derivative yields, according to the Second Fundamental Theorem of Calculus, a formula about integrals. For example,

$$\text{if } F(x) = x(\log x) - x, \text{ then } F'(x) = \log x;$$

consequently,

$$\int_a^b \log x \, dx = F(b) - F(a) = b(\log b) - b - [a(\log a) - a], \quad 0 < a, b.$$

Formulas of this sort are simplified considerably if we adopt the notation

$$F(x) \Big|_a^b = F(b) - F(a).$$

We may then write

$$\int_a^b \log x \, dx = x(\log x) - x \Big|_a^b.$$

This evaluation of $\int_a^b \log x \, dx$ depended on the lucky guess that $\log$ is the derivative of the function $F(x) = x(\log x) - x$. In general, a function F satisfying $F' = f$ is called a **primitive** of f . Of course, a **continuous function f always has a primitive**, namely,

$$F(x) = \int_a^x f,$$

but in this chapter we will try to find a primitive which can be written in terms of familiar functions like $\sin$, $\log$, etc. A function which can be written in this way is called an elementary function. To be precise,* an **elementary function** is one which can be obtained by addition, multiplication, division, and composition from the rational functions, the trigonometric functions and their inverses, and the functions $\log$ and $\exp$.

It should be stated at the very outset that elementary primitives usually cannot be found. For example, there is no *elementary* function F such that

$$F'(x) = e^{-x^2} \quad \text{for all } x$$

(this is not merely a report on the present state of mathematical ignorance; it is a (difficult) theorem that no such function exists). And, what is even worse, you will have no way of knowing whether or not an elementary primitive *can* be found (you will just have to hope that the problems for this chapter contain

* The definition which we will give is precise, but not really accurate, or at least not quite standard. Usually the elementary functions are defined to include "algebraic" functions, that is, functions g satisfying an equation

$$(g(x))^n + f_{n-1}(x)(g(x))^{n-1} + \cdots + f_0(x) = 0,$$

where the f_i are rational functions. But for our purposes these functions can be ignored.

no misprints). Because the search for elementary primitives is so uncertain, finding one is often peculiarly satisfying. If we observe that the function

$$F(x) = x \arctan x - \frac{\log(1 + x^2)}{2}$$

satisfies

$$F'(x) = \arctan x$$

(just how we would ever be led to such an observation is quite another matter), so that

$$\int_a^b \arctan x \, dx = x \arctan x - \frac{\log(1 + x^2)}{2} \Big|_a^b,$$

then we may feel that we have “really” evaluated $\int_a^b \arctan x \, dx$.

This chapter consists of little more than methods for finding elementary primitives of given elementary functions (a process known simply as “integration”), together with some notation, abbreviations, and conventions designed to facilitate this procedure. This preoccupation with elementary functions can be justified by three considerations:

- (1) Integration is a standard topic in calculus, and everyone should know about it.
- (2) Every once in a while you might actually need to evaluate an integral, under conditions which do not allow you to consult any of the standard integral tables (for example, you might take a (physics) course in which you are expected to be able to integrate).
- (3) The most useful “methods” of integration are actually very important theorems (that apply to all functions, not just elementary ones).

Naturally, the last reason is the crucial one. Even if you intend to forget how to integrate (and you probably will forget some details the first time through), you must never forget the basic methods.

These basic methods are theorems which allow us to express primitives of one function in terms of primitives of other functions. To begin integrating we will therefore need a list of primitives for *some* functions; such a list can be obtained simply by differentiating various well-known functions. The list given below makes use of a standard symbol which requires some explanation. The symbol

$$\int f \quad \text{or} \quad \int f(x) \, dx$$

means “a primitive of f ” or, more precisely, “the collection of all primitives of f .” The symbol $\int f$ will often be used in stating theorems, while $\int f(x) \, dx$ is most useful in formulas like the following:

$$\int x^3 \, dx = \frac{x^4}{4}.$$

This “equation” means that the function $F(x) = x^4/4$ satisfies $F'(x) = x^3$. It cannot be interpreted literally because the right side is a number, not a function, but in this one context we will allow such discrepancies; our aim is to make the integration process as mechanical as possible, and we will resort to any possible device. Another feature of the equation deserves mention. Most people write

$$\int x^3 dx = \frac{x^4}{4} + C,$$

to emphasize that the primitives of $f(x) = x^3$ are precisely the functions of the form $F(x) = x^4/4 + C$ for some number C . Although it is possible (Problem 11) to obtain contradictions if this point is disregarded, in practice such difficulties do not arise, and concern for this constant is merely an annoyance.

There is one important convention accompanying this notation: the letter appearing on the right side of the equation should match with the letter appearing after “ d ” on the left side—thus

$$\begin{aligned}\int u^3 du &= \frac{u^4}{4}, \\ \int tx dx &= \frac{tx^2}{2}, \\ \int tx dt &= \frac{xt^2}{2}.\end{aligned}$$

A function in $\int f(x) dx$, i.e., a primitive of f , is often called an “indefinite integral” of f , while $\int_a^b f(x) dx$ is called, by way of contrast, a “definite integral.” This suggestive notation works out quite well in practice, but it is important not to be led astray. At the risk of boring you, the following fact is emphasized once again: the integral $\int_a^b f(x) dx$ is *not* defined as “ $F(b) - F(a)$,” where F is an indefinite integral of f (if you do not find this statement repetitious, it is time to reread Chapter 13).

We can verify the formulas in the following short table of indefinite integrals simply by differentiating the functions indicated on the right side.

$$\begin{aligned}\int a dx &= ax \\ \int x^n dx &= \frac{x^{n+1}}{n+1}, \quad n \neq -1 \\ \int \frac{1}{x} dx &= \log x \quad \left(\int \frac{1}{x} dx \text{ is often written } \int \frac{dx}{x} \text{ for convenience; similar abbreviations are used in the last two examples of this table.} \right) \\ \int e^x dx &= e^x\end{aligned}$$

$$\begin{aligned}
\int \sin x \, dx &= -\cos x \\
\int \cos x \, dx &= \sin x \\
\int \sec^2 x \, dx &= \tan x \\
\int \sec x \tan x \, dx &= \sec x \\
\int \frac{dx}{1+x^2} &= \arctan x \\
\int \frac{dx}{\sqrt{1-x^2}} &= \arcsin x
\end{aligned}$$

Two general formulas of the same nature are consequences of theorems about differentiation:

$$\begin{aligned}
\int [f(x) + g(x)] \, dx &= \int f(x) \, dx + \int g(x) \, dx, \\
\int c \cdot f(x) \, dx &= c \cdot \int f(x) \, dx.
\end{aligned}$$

These equations should be interpreted as meaning that a primitive of $f + g$ can be obtained by adding a primitive of f to a primitive of g , while a primitive of $c \cdot f$ can be obtained by multiplying a primitive of f by c .

Notice the consequences of these formulas for definite integrals: If f and g are continuous, then

$$\begin{aligned}
\int_a^b [f(x) + g(x)] \, dx &= \int_a^b f(x) \, dx + \int_a^b g(x) \, dx, \\
\int_a^b c \cdot f(x) \, dx &= c \cdot \int_a^b f(x) \, dx.
\end{aligned}$$

These follow from the previous formulas, since each definite integral may be written as the difference of the values at a and b of a corresponding primitive. Continuity is required in order to know that these primitives exist. (Of course, the formulas are also true when f and g are merely integrable, but recall how much more difficult the proofs are in this case.)

The product formula for the derivative yields a more interesting theorem, which will be written in several different ways.

THEOREM 1 (INTEGRATION BY PARTS)

If f' and g' are continuous, then

$$\begin{aligned}
\int fg' &= fg - \int f'g, \\
\int f(x)g'(x) \, dx &= f(x)g(x) - \int f'(x)g(x) \, dx, \\
\int_a^b f(x)g'(x) \, dx &= f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) \, dx.
\end{aligned}$$

(Notice that in the second equation $f(x)g(x)$ denotes the function $f \cdot g$.)

PROOF The formula

$$(fg)' = f'g + fg'$$

can be written

$$fg' = (fg)' - f'g.$$

Thus

$$\int fg' = \int (fg)' - \int f'g,$$

and fg can be chosen as one of the functions denoted by $\int (fg)'$. This proves the first formula.

The second formula is merely a restatement of the first, and the third formula follows immediately from either of the first two. ■

As the following examples illustrate, integration by parts is useful when the function to be integrated can be considered as a product of a function f , whose derivative is simpler than f , and another function which is obviously of the form g' .

$$\begin{array}{ccccccc} \int x e^x dx & = & x e^x & - & \int 1 \cdot e^x dx \\ \downarrow \downarrow & & \downarrow \downarrow & & \downarrow & \downarrow \\ fg' & & fg & & f' & g \\ & & & & & \\ & & & & & = x e^x - e^x. \end{array}$$

$$\begin{array}{cccccccc} \int x \sin x dx & = & x \cdot (-\cos x) & - & \int 1 \cdot (-\cos x) dx \\ \downarrow \downarrow & & \downarrow & \downarrow & & \downarrow & \downarrow \\ f & g' & f & g & & f' & g \\ & & & & & & \\ & & & & & & = -x \cos x + \sin x. \end{array}$$

There are two special tricks which often work with integration by parts. The first is to consider the function g' to be the factor 1, which can always be written in.

$$\begin{array}{ccccccc} \int \log x dx & = & \int 1 \cdot \log x dx & = & x \log x & - & \int x \cdot (1/x) dx \\ \downarrow & & \downarrow & & \downarrow & \downarrow & \downarrow \\ g' & & f & & g & f & g & f' \\ & & & & & & & \\ & & & & & & & = x(\log x) - x. \end{array}$$

The second trick is to use integration by parts to find $\int h$ in terms of $\int h$ again, and then solve for $\int h$. A simple example is the calculation

$$\begin{array}{ccccccc} \int (1/x) \cdot \log x dx & = & \log x \cdot \log x & - & \int (1/x) \cdot \log x dx, \\ \downarrow & & \downarrow & & \downarrow & \downarrow & \downarrow \\ g' & & f & & g & f & f' & g \end{array}$$

which implies that

$$2 \int \frac{1}{x} \log x dx = (\log x)^2$$

or

$$\int \frac{1}{x} \log x \, dx = \frac{(\log x)^2}{2}.$$

A more complicated calculation is often required:

$$\begin{aligned} \int e^x \sin x \, dx &= e^x \cdot (-\cos x) - \int e^x \cdot (-\cos x) \, dx \\ &\quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad f \quad g' \quad f \quad g \quad f' \quad g \\ &= -e^x \cos x + \int e^x \cos x \, dx \\ &\quad \quad \quad \downarrow \quad \downarrow \\ &\quad \quad \quad u \quad v' \\ &= -e^x \cos x + [e^x \cdot (\sin x) - \int e^x (\sin x) \, dx]; \\ &\quad \quad \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad \quad \quad u \quad v \quad u' \quad v \end{aligned}$$

therefore,

$$2 \int e^x \sin x \, dx = e^x (\sin x - \cos x)$$

or

$$\int e^x \sin x \, dx = \frac{e^x (\sin x - \cos x)}{2}.$$

Since integration by parts depends upon recognizing that a function is of the form g' , the more functions you can already integrate, the greater your chances for success. It is frequently reasonable to do a preliminary integration before tackling the main problem. For example, we can use parts to integrate

$$\begin{aligned} \int (\log x)^2 \, dx &= \int (\log x)(\log x) \, dx \\ &\quad \downarrow \quad \downarrow \\ &\quad f \quad g' \end{aligned}$$

if we recall that $\int \log x \, dx = x(\log x) - x$ (this formula was itself derived by integration by parts); we have

$$\begin{aligned} \int (\log x)(\log x) \, dx &= (\log x)[x(\log x) - x] - \int (1/x)[x(\log x) - x] \, dx \\ &\quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad f \quad g' \quad f \quad g \quad f' \quad g \\ &= (\log x)[x(\log x) - x] - \int [\log x - 1] \, dx \\ &= (\log x)[x(\log x) - x] - \int \log x \, dx + \int 1 \, dx \\ &= (\log x)[x(\log x) - x] - [x(\log x) - x] + x \\ &= x(\log x)^2 - 2x(\log x) + 2x. \end{aligned}$$

The most important method of integration is a consequence of the Chain Rule. The use of this method requires considerably more ingenuity than integrating by parts, and even the explanation of the method is more difficult.

We will therefore develop this method in stages, stating the theorem for definite integrals first, and saving the treatment of indefinite integrals for later.

THEOREM 2
(THE SUBSTITUTION FORMULA)

If f and g' are continuous, then

$$\begin{aligned}\int_{g(a)}^{g(b)} f &= \int_a^b (f \circ g) \cdot g' \\ \int_{g(a)}^{g(b)} f(u) \, du &= \int_a^b f(g(x)) \cdot g'(x) \, dx.\end{aligned}$$

PROOF

If F is a primitive of f , then the left side is $F(g(b)) - F(g(a))$. On the other hand,

$$(F \circ g)' = (F' \circ g) \cdot g' = (f \circ g) \cdot g',$$

so $F \circ g$ is a primitive of $(f \circ g) \cdot g'$ and the right side is

$$(F \circ g)(b) - (F \circ g)(a) = F(g(b)) - F(g(a)). \blacksquare$$

The simplest uses of the substitution formula depend upon recognizing that a given function is of the form $(f \circ g) \cdot g'$. For example, the integration of

$$\int_a^b \sin^5 x \cos x \, dx \quad \left(= \int_a^b (\sin x)^5 \cos x \, dx \right)$$

is facilitated by the appearance of the factor $\cos x$, which will be the factor $g'(x)$ for $g(x) = \sin x$; the remaining expression, $(\sin x)^5$, can be written as $(g(x))^5 = f(g(x))$, for $f(u) = u^5$. Thus

$$\begin{aligned}\int_a^b \sin^5 x \cos x \, dx & \quad \begin{bmatrix} g(x) = \sin x \\ f(u) = u^5 \end{bmatrix} \\ &= \int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du \\ &= \int_{\sin a}^{\sin b} u^5 \, du = \frac{\sin^6 b}{6} - \frac{\sin^6 a}{6}.\end{aligned}$$

The integration of $\int_a^b \tan x \, dx$ can be treated similarly if we write

$$\int_a^b \tan x \, dx = - \int_a^b \frac{-\sin x}{\cos x} \, dx.$$

In this case the factor $-\sin x$ is $g'(x)$, where $g(x) = \cos x$; the remaining factor $1/\cos x$ can then be written $f(\cos x)$ for $f(u) = 1/u$. Hence

$$\begin{aligned}\int_a^b \tan x \, dx & \quad \begin{bmatrix} g(x) = \cos x \\ f(u) = \frac{1}{u} \end{bmatrix} \\ &= - \int_a^b f(g(x))g'(x) \, dx = - \int_{g(a)}^{g(b)} f(u) \, du \\ &= - \int_{\cos a}^{\cos b} \frac{1}{u} \, du = \log(\cos a) - \log(\cos b).\end{aligned}$$

Finally, to find

$$\int_a^b \frac{1}{x \log x} dx,$$

notice that $1/x = g'(x)$ where $g(x) = \log x$, and that $1/\log x = f(g(x))$ for $f(u) = 1/u$. Thus

$$\begin{aligned} \int_a^b \frac{1}{x \log x} dx & \quad \left[\begin{array}{l} g(x) = \log x \\ f(u) = \frac{1}{u} \end{array} \right] \\ &= \int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du \\ &= \int_{\log a}^{\log b} \frac{1}{u} du = \log(\log b) - \log(\log a). \end{aligned}$$

Fortunately, these uses of the substitution formula can be shortened considerably. The intermediate steps, which involve writing

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du,$$

can easily be eliminated by noticing the following: To go from the left side to the right side,

$$\begin{aligned} &\text{substitute} \begin{cases} u \text{ for } g(x) \\ du \text{ for } g'(x) dx \end{cases} \\ &(\text{and change the limits of integration}); \end{aligned}$$

the substitutions can be performed directly on the original function (accounting for the name of this theorem). For example,

$$\int_a^b \sin^5 x \cos x dx \left[\text{substitute} \begin{array}{l} u \text{ for } \sin x \\ du \text{ for } \cos x dx \end{array} \right] = \int_{\sin a}^{\sin b} u^5 du,$$

and similarly

$$\int_a^b \frac{-\sin x}{\cos x} dx \left[\text{substitute} \begin{array}{l} u \text{ for } \cos x \\ du \text{ for } -\sin x dx \end{array} \right] = \int_{\cos a}^{\cos b} \frac{1}{u} du.$$

Usually we abbreviate this method even more, and say simply:

$$\begin{aligned} &\text{"Let } u = g(x) \\ &du = g'(x) dx." \end{aligned}$$

Thus

$$\int_a^b \frac{1}{x \log x} dx \left[\begin{array}{l} \text{let } u = \log x \\ du = \frac{1}{x} dx \end{array} \right] = \int_{\log a}^{\log b} \frac{1}{u} du.$$

In this chapter we are usually interested in primitives rather than definite integrals, but if we can find $\int_a^b f(x) dx$ for all a and b , then we can certainly

find $\int f(x) dx$. For example, since

$$\int_a^b \sin^5 x \cos x dx = \frac{\sin^6 b}{6} - \frac{\sin^6 a}{6},$$

it follows that

$$\int \sin^5 x \cos x dx = \frac{\sin^6 x}{6}.$$

Similarly,

$$\begin{aligned} \int \tan x dx &= -\log x \cos x, \\ \int \frac{1}{x \log x} dx &= \log(\log x). \end{aligned}$$

It is quite uneconomical to obtain primitives from the substitution formula by first finding definite integrals. Instead, the two steps can be combined, to yield the following procedure:

(1) Let

$$\begin{aligned} u &= g(x), \\ du &= g'(x) dx; \end{aligned}$$

(after this manipulation only the letter u should appear, *not* the letter x).

(2) Find a primitive (as an expression involving u).

(3) Substitute $g(x)$ back for u .

Thus, to find

$$\int \sin^5 x \cos x dx,$$

(1) let

$$\begin{aligned} u &= \sin x, \\ du &= \cos x dx, \end{aligned}$$

so that we obtain

$$\int u^5 du;$$

(2) evaluate

$$\int u^5 du = \frac{u^6}{6};$$

(3) remember to substitute $\sin x$ back for u , so that

$$\int \sin^5 x \cos x dx = \frac{\sin^6 x}{6}.$$

Similarly, if

$$\begin{aligned} u &= \log x, \\ du &= \frac{1}{x} dx, \end{aligned}$$

then

$$\int \frac{1}{x \log x} dx \quad \text{becomes} \quad \int \frac{1}{u} du = \log u,$$

so that

$$\int \frac{1}{x \log x} dx = \log(\log x).$$

To evaluate

$$\int \frac{x}{1+x^2} dx,$$

let

$$\begin{aligned} u &= 1 + x^2, \\ du &= 2x dx; \end{aligned}$$

the factor 2 which has just popped up causes no problem—the integral becomes

$$\frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \log u,$$

so

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \log(1+x^2).$$

(This result may be combined with integration by parts to yield a formula already mentioned:

$$\begin{aligned} \int 1 \cdot \arctan x \, dx &= x \arctan x - \int \frac{x}{1+x^2} dx \\ &= x \arctan x - \frac{1}{2} \log(1+x^2). \end{aligned}$$

These applications of the substitution formula* illustrate the most straightforward and least interesting types—once the suitable factor $g'(x)$ is recognized, the whole problem may even become simple enough to do mentally. The following three problems require only the information provided by the short table of indefinite integrals at the beginning of the chapter and, of course, the right substitution (the third problem has been disguised a little by some algebraic chicanery).

* The substitution formula is often written in the form

$$\int f(u) \, du = \int f(g(x)) g'(x) \, dx, \quad u = g(x).$$

This formula cannot be taken literally (after all, $\int f(u) \, du$ should mean a primitive of f and $\int f(g(x)) g'(x) \, dx$ should mean a primitive of $(f \circ g) \cdot g'$; these are certainly not equal). However, it may be regarded as a symbolic summary of the procedure which we have developed. If we use Leibnitz's notation, and a little fudging, the formula reads particularly well:

$$\int f(u) \, du = \int f(u) \frac{du}{dx} dx.$$

$$\int \sec^2 x \tan^5 x \, dx,$$

$$\int (\cos x) e^{\sin x} \, dx,$$

$$\int \frac{e^x}{\sqrt{1 - e^{2x}}} \, dx.$$

If you have not succeeded in finding the right substitutions, you should be able to guess them from the answers, which are $(\tan^6 x)/6$, $e^{\sin x}$, and $\arcsin e^x$. At first you may find these problems too hard to do in your head, but at least when g is of the very simple form $g(x) = ax + b$ you should not have to waste time writing out the substitution. The following integrations should all be clear. (The only worrisome detail is the proper positioning of the constant—should the answer to the second be $e^{3x}/3$ or $3e^{3x}$? I always take care of these problems as follows. Clearly $\int e^{3x} \, dx = e^{3x} \cdot (\text{something})$. Now if I differentiate $F(x) = e^{3x}$, I get $F'(x) = 3e^{3x}$, so “something” must be $\frac{1}{3}$, to cancel the 3.)

$$\int \frac{dx}{x+3} = \log(x+3),$$

$$\int e^{3x} \, dx = \frac{e^{3x}}{3},$$

$$\int \cos 4x \, dx = \frac{\sin 4x}{4},$$

$$\int \sin(2x+1) \, dx = \frac{-\cos(2x+1)}{2},$$

$$\int \frac{dx}{1+4x^2} = \frac{\arctan 2x}{2}.$$

More interesting uses of the substitution formula occur when the factor $g'(x)$ does *not* appear. There are two main types of substitutions where this happens. Consider first

$$\int \frac{1+e^x}{1-e^x} \, dx.$$

The prominent appearance of the expression e^x suggests the simplifying substitution

$$\begin{aligned} u &= e^x, \\ du &= e^x \, dx. \end{aligned}$$

Although the expression $e^x \, dx$ does not appear, it can always be put in:

$$\int \frac{1+e^x}{1-e^x} \, dx = \int \frac{1+e^x}{1-e^x} \cdot \frac{1}{e^x} \cdot e^x \, dx.$$

We therefore obtain

$$\int \frac{1+u}{1-u} \cdot \frac{1}{u} du,$$

which can be evaluated by the algebraic trick

$$\int \frac{1+u}{1-u} \cdot \frac{1}{u} du = \int \frac{2}{1-u} + \frac{1}{u} du = -2 \log(1-u) + \log u,$$

so that

$$\int \frac{1+e^x}{1-e^x} dx = -2 \log(1-e^x) + \log e^x = -2 \log(1-e^x) + x.$$

There is an alternative and preferable way of handling this problem, which does not require multiplying and dividing by e^x . If we write

$$u = e^x, \quad x = \log u, \\ dx = \frac{1}{u} du,$$

then

$$\int \frac{1+e^x}{1-e^x} dx \text{ immediately becomes } \int \frac{1+u}{1-u} \cdot \frac{1}{u} du.$$

Most substitution problems are much easier if one resorts to this trick of expressing x in terms of u , and dx in terms of du , instead of vice versa. It is not hard to see why this trick always works (as long as the function expressing u in terms of x is one-one for all x under consideration): If we apply the substitution

$$u = g(x), \quad x = g^{-1}(u) \\ dx = (g^{-1})'(u) du$$

to the integral

$$\int f(g(x)) dx,$$

we obtain

$$(1) \quad \int f(u)(g^{-1})'(u) du.$$

On the other hand, if we apply the straightforward substitution

$$u = g(x) \\ du = g'(x) dx$$

to the same integral,

$$\int f(g(x)) dx = \int f(g(x)) \cdot \frac{1}{g'(x)} \cdot g'(x) dx,$$

we obtain

$$(2) \quad \int f(u) \cdot \frac{1}{g'(g^{-1}(u))} du.$$

The integrals (1) and (2) are identical, since $(g^{-1})'(u) = 1/g'(g^{-1}(u))$.

As another concrete example, consider

$$\int \frac{e^{2x}}{\sqrt{e^x + 1}} dx.$$

In this case we will go whole hog and replace the entire expression $\sqrt{e^x + 1}$ by one letter. Thus we choose the substitution

$$\begin{aligned} u &= \sqrt{e^x + 1}, \\ u^2 &= e^x + 1, \\ u^2 - 1 &= e^x, \quad x = \log(u^2 - 1), \\ dx &= \frac{2u}{u^2 - 1} du. \end{aligned}$$

The integral then becomes

$$\int \frac{(u^2 - 1)^2}{u} \cdot \frac{2u}{u^2 - 1} du = 2 \int u^2 - 1 du = \frac{2u^3}{3} - 2u.$$

Thus

$$\int \frac{e^{2x}}{\sqrt{e^x + 1}} dx = \frac{2}{3} (e^x + 1)^{3/2} - 2(e^x + 1)^{1/2}.$$

Another example, which illustrates the second main type of substitution that can occur, is the integral

$$\int \sqrt{1 - x^2} dx.$$

In this case, instead of replacing a complicated expression by a simpler one, we will replace x by $\sin u$, because $\sqrt{1 - \sin^2 u} = \cos u$. This really means that we are using the substitution $u = \arcsin x$, but it is the expression for x in terms of u which helps us find the expression to be substituted for dx . Thus,

$$\begin{aligned} \text{let } x &= \sin u, \quad [u = \arcsin x] \\ dx &= \cos u du; \end{aligned}$$

then the integral becomes

$$\int \sqrt{1 - \sin^2 u} \cos u du = \int \cos^2 u du.$$

The evaluation of this integral depends on the equation

$$\cos^2 u = \frac{1 + \cos 2u}{2}$$

(see the discussion of trigonometric functions below) so that

$$\int \cos^2 u du = \int \frac{1 + \cos 2u}{2} du = \frac{u}{2} + \frac{\sin 2u}{4},$$

and

$$\begin{aligned}\int \sqrt{1-x^2} dx &= \frac{\arcsin x}{2} + \frac{\sin(2 \arcsin x)}{4} \\ &= \frac{\arcsin x}{2} + \frac{1}{2} \sin(\arcsin x) \cdot \cos(\arcsin x) \\ &= \frac{\arcsin x}{2} + \frac{1}{2} x \sqrt{1-x^2}.\end{aligned}$$

Substitution and integration by parts are the only fundamental methods which you have to learn; with their aid primitives can be found for a large number of functions. Nevertheless, as some of our examples reveal, success often depends upon some additional tricks. The most important are listed below. Using these you should be able to integrate all the functions in Problems 1 to 7 (a few other interesting tricks are explained in some of the remaining problems).

1. TRIGONOMETRIC FUNCTIONS

Since

$$\sin^2 x + \cos^2 x = 1$$

and

$$\cos 2x = \cos^2 x - \sin^2 x,$$

we obtain

$$\begin{aligned}\cos 2x &= \cos^2 x - (1 - \cos^2 x) = 2 \cos^2 x - 1, \\ \cos 2x &= (1 - \sin^2 x) - \sin^2 x = 1 - 2 \sin^2 x,\end{aligned}$$

or

$$\begin{aligned}\sin^2 x &= \frac{1 - \cos 2x}{2}, \\ \cos^2 x &= \frac{1 + \cos 2x}{2}.\end{aligned}$$

These formulas may be used to integrate

$$\begin{aligned}\int \sin^n x dx, \\ \int \cos^n x dx,\end{aligned}$$

if n is even. Substituting

$$\frac{(1 - \cos 2x)}{2} \quad \text{or} \quad \frac{(1 + \cos 2x)}{2}$$

for $\sin^2 x$ or $\cos^2 x$ yields a sum of terms involving lower powers of $\cos$. For example,

$$\int \sin^4 x dx = \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx = \int \frac{1}{4} dx - \int \cos 2x dx + \frac{1}{4} \int \cos^2 2x dx$$

and

$$\int \cos^2 2x \, dx = \int \frac{1 + \cos 4x}{2} \, dx.$$

If n is odd, $n = 2k + 1$, then

$$\int \sin^n x \, dx = \int \sin x (1 - \cos^2 x)^k \, dx;$$

the latter expression, multiplied out, involves terms of the form $\sin x \cos^l x$, all of which can be integrated easily. The integral for $\cos^n x$ is treated similarly. An integral

$$\int \sin^n x \cos^m x \, dx$$

is handled the same way if n or m is odd. If n and m are both even, use the formulas for $\sin^2 x$ and $\cos^2 x$.

A final important trigonometric integral is

$$\int \frac{1}{\cos x} \, dx = \int \sec x \, dx = \log(\sec x + \tan x).$$

Although there are several ways of “deriving” this result, by means of the methods already at our disposal (Problem 10), it is simplest to check this formula by differentiating the right side, and to memorize it.

2. REDUCTION FORMULAS

Integration by parts yields (Problem 17)

$$\begin{aligned} \int \sin^n x \, dx &= -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx, \\ \int \cos^n x \, dx &= \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx, \\ \int \frac{1}{(x^2 + 1)^n} \, dx &= \frac{1}{2n-2} \frac{x}{(x^2 + 1)^{n-1}} + \frac{2n-3}{2n-2} \int \frac{1}{(x^2 + 1)^{n-1}} \, dx, \end{aligned}$$

and many similar formulas. The first two, used repeatedly, give a different method for evaluating primitives of $\sin^n$ or $\cos^n$. The third is very important for integrating a large general class of functions, which will complete our discussion.

3. RATIONAL FUNCTIONS

Consider a rational function p/q where

$$\begin{aligned} p(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0, \\ q(x) &= b_m x^m + b_{m-1} x^{m-1} + \cdots + b_0. \end{aligned}$$

We might as well assume that $a_n = b_m = 1$. Moreover, we can assume that $n < m$, for otherwise we may express p/q as a polynomial function plus a

rational function which is of this form by dividing (for example:

$$\frac{u^2}{u-1} = u + 1 + \frac{1}{u-1}$$

The integration of an arbitrary rational function depends on two facts; the first follows from the "Fundamental Theorem of Algebra" (see Chapter 25, Theorem 2 and Problem 25-3), but the second will not be proved in this book.

THEOREM Every polynomial function

$$q(x) = x^m + b_{m-1}x^{m-1} + \cdots + b_0$$

can be written as a product

$$q(x) = (x - \alpha_1)^{r_1} \cdots (x - \alpha_k)^{r_k} (x^2 + \beta_1x + \gamma_1)^{s_1} \cdots (x^2 + \beta_lx + \gamma_l)^{s_l} \\ \text{(where } r_1 + \cdots + r_k + 2(s_1 + \cdots + s_l) = m \text{)}.$$

(In this expression, identical factors have been collected together, so that all $x - \alpha_i$ and $x^2 + \beta_ix + \gamma_i$ may be assumed distinct. Moreover, we assume that each quadratic factor cannot be factored further. This means that

$$\beta_i^2 - 4\gamma_i < 0,$$

since otherwise we can factor

$$x^2 + \beta_ix + \gamma_i = \left[x - \left(\frac{-\beta_i + \sqrt{\beta_i^2 - 4\gamma_i}}{2} \right) \right] \left[x - \left(\frac{-\beta_i - \sqrt{\beta_i^2 - 4\gamma_i}}{2} \right) \right]$$

into linear factors).

THEOREM If $n < m$ and

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0, \\ q(x) = x^m + b_{m-1}x^{m-1} + \cdots + b_0 \\ = (x - \alpha_1)^{r_1} \cdots (x - \alpha_k)^{r_k} (x^2 + \beta_1x + \gamma_1)^{s_1} \cdots (x^2 + \beta_lx + \gamma_l)^{s_l},$$

then $p(x)/q(x)$ can be written in the form

$$\frac{p(x)}{q(x)} = \left[\frac{a_{1,1}}{(x - \alpha_1)} + \cdots + \frac{a_{1,r_1}}{(x - \alpha_1)^{r_1}} \right] + \cdots \\ + \left[\frac{a_{k,1}}{(x - \alpha_k)} + \cdots + \frac{a_{k,r_k}}{(x - \alpha_k)^{r_k}} \right] \\ + \left[\frac{b_{1,1}x + c_{1,1}}{(x^2 + \beta_1x + \gamma_1)} + \cdots + \frac{b_{1,s_1}x + c_{1,s_1}}{(x^2 + \beta_1x + \gamma_1)^{s_1}} \right] + \cdots \\ + \left[\frac{b_{l,1}x + c_{l,1}}{(x^2 + \beta_lx + \gamma_l)} + \cdots + \frac{b_{l,s_l}x + c_{l,s_l}}{(x^2 + \beta_lx + \gamma_l)^{s_l}} \right]$$

This expression, known as the "partial fraction decomposition" of $p(x)/q(x)$, is so complicated that it is simpler to examine the following example, which

illustrates such an expression and shows how to find it. According to the theorem, it is possible to write

$$\frac{2x^7 + 8x^6 + 13x^5 + 20x^4 + 15x^3 + 16x^2 + 7x + 10}{(x^2 + x + 1)^2(x^2 + 2x + 2)(x - 1)^2} \\ = \frac{a}{x - 1} + \frac{b}{(x - 1)^2} + \frac{cx + d}{x^2 + 2x + 2} + \frac{ex + f}{(x^2 + x + 1)} + \frac{gx + h}{(x^2 + x + 1)^2}.$$

To find the numbers a, b, c, d, e, f, g , and h , write the right side as a polynomial over the common denominator $(x^2 + x + 1)^2(x^2 + 2x + 2)(x - 1)^2$; the numerator becomes

$$a(x - 1)(x^2 + 2x + 2)(x^2 + x + 1)^2 + b(x^2 + 2x + 2)(x^2 + x + 1)^2 \\ + (cx + d)(x - 1)^2(x^2 + x + 1)^2 + (ex + f)(x - 1)^2(x^2 + 2x + 2)(x^2 + x + 1) \\ + (gx + h)(x - 1)^2(x^2 + 2x + 2).$$

Actually multiplying this out (!) we obtain a polynomial of degree 8, whose coefficients are combinations of $a, \dots, h$. Equating these coefficients with the coefficients of $2x^7 + 8x^6 + 13x^5 + 20x^4 + 15x^3 + 16x^2 + 7x + 10$ (the coefficient of x^8 is 0) we obtain 8 equations in the eight unknowns $a, \dots, h$. After heroic calculations these can be solved to give

$$a = 1, \quad b = 2, \quad c = 1, \quad d = 3, \\ e = 0, \quad f = 0, \quad g = 0, \quad h = 1.$$

Thus

$$\int \frac{2x^7 + 8x^6 + 13x^5 + 20x^4 + 21x^3 + 16x^2 + 7x + 4}{(x^2 + x + 1)^2(x^2 + 2x + 2)(x - 1)^2} dx \\ = \int \frac{1}{(x - 1)} dx + \int \frac{2}{(x - 1)^2} dx + \int \frac{1}{(x^2 + x + 1)^2} dx + \int \frac{x + 3}{x^2 + 2x + 2} dx.$$

(In simpler cases the requisite calculations may actually be feasible. I obtained this particular example by *starting* with the partial fraction decomposition and converting it into one fraction.)

We are already in a position to find each of the integrals appearing in the above expression; the calculations will illustrate all the difficulties which arise in integrating rational functions.

The first two integrals are simple:

$$\int \frac{1}{x - 1} dx = \log(x - 1), \\ \int \frac{2}{(x - 1)^2} dx = \frac{-2}{x - 1}.$$

The third integration depends on “completing the square”:

$$x^2 + x + 1 = (x + \tfrac{1}{2})^2 + \tfrac{3}{4} \\ = \tfrac{3}{4} \left[\left(\frac{x + \frac{1}{2}}{\sqrt{\frac{3}{4}}} \right)^2 + 1 \right].$$

(If we had obtained $-\frac{3}{4}$ instead of $\frac{3}{4}$ we could not take the square root, but in this case our original quadratic factor could have been factored into linear factors.) We can now write

$$\int \frac{1}{(x^2 + x + 1)^2} dx = \frac{4}{3} \int \frac{1}{\left[\left(\frac{x + \frac{1}{2}}{\sqrt{\frac{3}{4}}}\right)^2 + 1\right]^2} dx.$$

The substitution

$$u = \frac{x + \frac{1}{2}}{\sqrt{\frac{3}{4}}},$$

$$du = \frac{1}{\sqrt{\frac{3}{4}}} dx,$$

changes this integral to

$$\frac{4}{3} \int \frac{\sqrt{\frac{3}{4}}}{(u^2 + 1)^2} du,$$

which can be computed using the third reduction formula given above.

Finally, to evaluate

$$\int \frac{x + 3}{(x^2 + 2x + 2)} dx$$

we write

$$\int \frac{x + 3}{x^2 + 2x + 2} dx = \frac{1}{2} \int \frac{2x + 2}{x^2 + 2x + 3} dx + \int \frac{2}{(x + 1)^2 + 1} dx.$$

The first integral on the right side has been purposely constructed so that we can evaluate it by using the substitution

$$u = x^2 + 2x + 3,$$

$$du = (2x + 2) dx.$$

The second integral on the right, which is just the difference of the other two, is simply $2 \arctan(x + 1)$. If the original integral were

$$\int \frac{x + 3}{(x^2 + 2x + 2)^n} dx = \frac{1}{2} \int \frac{2x + 2}{(x^2 + 2x + 2)^n} dx + \int \frac{2}{[(x + 1)^2 + 1]^n} dx,$$

the first integral on the right would still be evaluated by the same substitution. The second integral would be evaluated by means of a reduction formula.

This example has probably convinced you that integration of rational functions is a theoretical curiosity only, especially since it is necessary to find the factorization of $q(x)$ before you can even begin. This is only partly true. We have already seen that simple rational functions sometimes arise, as in the integration

$$\int \frac{1 + e^x}{1 - e^x} dx;$$

another important example is the integral

$$\int \frac{1}{x^2 - 1} dx = \int \frac{\frac{1}{2}}{x - 1} - \frac{\frac{1}{2}}{x + 1} dx = \frac{1}{2} \log(x - 1) - \frac{1}{2} \log(x + 1).$$

Moreover, if a problem has been reduced to the integration of a rational function, it is then certain that an elementary primitive exists, even when the difficulty or impossibility of finding the factors of the denominator may preclude writing this primitive explicitly.

PROBLEMS

1. This problem contains some integrals which require little more than algebraic manipulation, and consequently test your ability to discover algebraic tricks, rather than your understanding of the integration processes. Nevertheless, any one of these tricks might be an important preliminary step in an honest integration problem. Moreover, you want to have some feel for which integrals are easy, so that you can see when the end of an integration process is in sight. The answer section, if you resort to it, will only reveal what algebra you should have used.

$$(i) \quad \int \frac{\sqrt[5]{x^3} + \sqrt[6]{x}}{\sqrt{x}} dx.$$

$$(ii) \quad \int \frac{dx}{\sqrt{x-1} + \sqrt{x+1}}.$$

$$(iii) \quad \int \frac{e^x + e^{2x} + e^{3x}}{e^{4x}} dx.$$

$$(iv) \quad \int \frac{a^x}{b^x} dx.$$

$$(v) \quad \int \tan^2 x \, dx. \quad (\text{Trigonometric integrals are always very touchy, because there are so many trigonometric identities that an easy problem can easily look hard.})$$

$$(vi) \quad \int \frac{dx}{a^2 + x^2}.$$

$$(vii) \quad \int \frac{dx}{\sqrt{a^2 - x^2}}.$$

$$(viii) \quad \int \frac{dx}{1 + \sin x}.$$

$$(ix) \quad \int \frac{8x^2 + 6x + 4}{x + 1} dx.$$

$$(x) \quad \int \frac{1}{\sqrt{2x - x^2}} dx.$$

2. The following integrations involve simple substitutions, most of which you should be able to do in your head.

$$(i) \int e^x \sin e^x dx.$$

$$(ii) \int x e^{-x^2} dx.$$

$$(iii) \int \frac{\log x}{x} dx. \quad (\text{In the text this was done by parts.})$$

$$(iv) \int \frac{e^x dx}{e^{2x} + 2e^x + 1}.$$

$$(v) \int e^{e^x} e^x dx.$$

$$(vi) \int \frac{x dx}{\sqrt{1-x^4}}.$$

$$(vii) \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx.$$

$$(viii) \int x \sqrt{1-x^2} dx.$$

$$(ix) \int \log(\cos x) \tan x dx.$$

$$(x) \int \frac{\log(\log x)}{x \log x} dx.$$

3. Integration by parts.

$$(i) \int x^2 e^x dx.$$

$$(ii) \int x^3 e^{x^2} dx.$$

$$(iii) \int e^{ax} \sin bx dx.$$

$$(iv) \int x^2 \sin x dx.$$

$$(v) \int (\log x)^3 dx.$$

$$(vi) \int \frac{\log(\log x)}{x} dx.$$

$$(vii) \int \sec^3 x dx. \quad (\text{This is a tricky and important integral that often comes up. If you do not succeed in evaluating it, be sure to consult the answers.})$$

$$(viii) \int \cos(\log x) dx.$$

$$(ix) \int \sqrt{x} \log x \, dx.$$

$$(x) \int x(\log x)^2 \, dx.$$

4. The following integrations can all be done with substitutions of the form $x = \sin u$, $x = \cos u$, etc. To do some of these you will need to remember that

$$\int \sec x \, dx = \log(\sec x + \tan x)$$

as well as the following formula, which can also be checked by differentiation:

$$\int \csc x \, dx = -\log(\csc x + \cot x).$$

In addition, at this point the derivatives of all the trigonometric functions should be kept handy.

$$(i) \int \frac{dx}{\sqrt{1-x^2}}. \quad (\text{You already know this integral, but use the substitution } x = \sin u \text{ anyway, just to see how it works out.})$$

$$(ii) \int \frac{dx}{\sqrt{1+x^2}}. \quad (\text{Since } \tan^2 u + 1 = \sec^2 u, \text{ you want to use the substitution } x = \tan u.)$$

$$(iii) \int \frac{dx}{\sqrt{x^2-1}}.$$

$$(iv) \int \frac{dx}{x\sqrt{x^2-1}}. \quad (\text{The answer will be a certain inverse function that was given short shrift in the text.})$$

$$(v) \int \frac{dx}{x\sqrt{1-x^2}}.$$

$$(vi) \int \frac{dx}{x\sqrt{1+x^2}}.$$

$$(vii) \int x^3 \sqrt{1-x^2} \, dx. \quad \left. \begin{array}{l} (viii) \int \sqrt{1-x^2} \, dx. \end{array} \right\} \quad (\text{You will need to remember the methods for integrating powers of sin and cos.})$$

$$(ix) \int \sqrt{1+x^2} \, dx.$$

$$(x) \int \sqrt{x^2-1} \, dx.$$

5. The following integrations involve substitutions of various types. There is no substitute for cleverness, but there is a general rule to follow: substitute for an expression which appears frequently or prominently; if two different troublesome expressions appear, try to express them both in terms of some new expression. And don't forget that it usually helps to

express x directly in terms of u , to find out the proper expression to substitute for dx .

$$(i) \int \frac{dx}{1 + \sqrt{x+1}}.$$

$$(ii) \int \frac{dx}{1 + e^x}.$$

$$(iii) \int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}.$$

$$(iv) \int \frac{dx}{\sqrt{1 + e^x}}. \text{ (The substitution } u = e^x \text{ leads to an integral requiring yet another substitution; this is all right, but both substitutions can be done at once.)}$$

$$(v) \int \frac{dx}{2 + \tan x}.$$

$$(vi) \int \frac{dx}{\sqrt{\sqrt{x+1}}}. \text{ (Another place where one substitution can be made to do the work of two.)}$$

$$(vii) \int \frac{4^x + 1}{2^x + 1} dx.$$

$$(viii) \int e^{\sqrt{x}} dx.$$

$$(ix) \int \frac{\sqrt{1-x}}{1-\sqrt{x}} dx. \text{ (In this case two successive substitutions work out best; there are two obvious candidates for the first substitution, and either will work.)}$$

$$*(x) \int \sqrt{\frac{x-1}{x+1}} \cdot \frac{1}{x^2} dx.$$

6. The previous problem provided gratis a haphazard selection of rational functions to be integrated. Here is a more systematic selection.

$$(i) \int \frac{2x^2 + 7x - 1}{x^3 + x^2 - x - 1} dx.$$

$$(ii) \int \frac{2x + 1}{x^3 - 3x^2 + 3x - 1} dx.$$

$$(iii) \int \frac{x^3 + 7x^2 - 5x + 5}{(x-1)^2(x+1)^3} dx.$$

$$(iv) \int \frac{2x^2 + x + 1}{(x+3)(x-1)^2} dx.$$

$$(v) \int \frac{x+4}{x^2+1} dx.$$

$$(vi) \int \frac{x^3 + x + 2}{x^4 + 2x^2 + 1} dx.$$

$$(vii) \int \frac{3x^2 + 3x + 1}{x^3 + 2x^2 + 2x + 1} dx.$$

$$(viii) \int \frac{dx}{x^4 + 1}.$$

$$(ix) \int \frac{2x}{(x^2 + x + 1)^2} dx.$$

$$(x) \int \frac{3x}{(x^2 + x + 1)^3} dx.$$

*7. Potpourri. (No holds barred.) The following integrations involve all the methods of the previous problems.

$$(i) \int \frac{\arctan x}{1 + x^2} dx.$$

$$(ii) \int \frac{x \arctan x}{(1 + x^2)^3} dx.$$

$$(iii) \int \log \sqrt{1 + x^2} dx.$$

$$(iv) \int x \log \sqrt{1 + x^2} dx.$$

$$(v) \int \frac{x^2 - 1}{x^2 + 1} \cdot \frac{1}{\sqrt{1 + x^4}} dx.$$

$$(vi) \int \arcsin \sqrt{x} dx.$$

$$(vii) \int \frac{x}{1 + \sin x} dx.$$

$$(viii) \int e^{\sin x} \cdot \frac{x \cos^3 x - \sin x}{\cos^2 x} dx.$$

$$(ix) \int \sqrt{\tan x} dx.$$

$$(x) \int \frac{dx}{x^6 + 1} \quad (\text{To factor } x^6 + 1, \text{ first factor } y^3 + 1, \text{ using Problem 1-1.})$$

8. If you have done Problem 17-7, the integrals (ii) and (iii) in Problem 4 will look very familiar. In general, the substitution $x = \cosh u$ often works for integrals involving $\sqrt{x^2 - 1}$, while $x = \sinh u$ is the thing to try for integrals involving $\sqrt{x^2 + 1}$. Try these substitutions on the other inte-

grals in Problem 4. (The method is not really recommended; it is easier to stick with trigonometric substitutions.)

***9.** The world's sneakiest substitution is undoubtedly

$$t = \tan \frac{x}{2}, \quad x = 2 \arctan t,$$

$$dx = \frac{2}{1+t^2} dt.$$

As we found in Problem 15-16, this substitution leads to the expressions

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}.$$

This substitution thus transforms any integral which involves only sin and cos, combined by addition, multiplication, and division, into the integral of a rational function. Find

(i) $\int \frac{dx}{1 + \sin x}$. (Compare your answer with Problem 1(viii).)

(ii) $\int \frac{dx}{1 - \sin^2 x}$. (In this case it is better to let $t = \tan x$. Why?)

(iii) $\int \frac{dx}{a \sin x + b \cos x}$. (There is also another way to do this, using Problem 15-7.)

(iv) $\int \sin^2 x \, dx$. (An exercise to convince you that this substitution should be used only as a last resort.)

(v) $\int \frac{dx}{3 + 5 \sin x}$. (A last resort.)

***10.** Derive the formula for $\int \sec x \, dx$ in the following two ways:

(a) By writing

$$\begin{aligned} \frac{1}{\cos x} &= \frac{\cos x}{\cos^2 x} \\ &= \frac{\cos x}{1 - \sin^2 x} \\ &= \frac{1}{2} \left[\frac{\cos x}{1 + \sin x} + \frac{\cos x}{1 - \sin x} \right], \end{aligned}$$

an expression obviously inspired by partial fraction decompositions. Be sure to note that $\int \cos x / (1 - \sin x) \, dx = -\log(1 - \sin x)$; the minus sign is very important. And remember that $\frac{1}{2} \log \alpha = \log \sqrt{\alpha}$. From there on, keep doing algebra, and trust to luck.

- (b) By using the substitution $t = \tan x/2$. Once again, quite a bit of manipulation is required to put the answer in the desired form; the expression $\tan x/2$ can be attacked by using Problem 15-8, or both answers can be expressed in terms of t . There is another expression for $\int \sec x \, dx$, which is less cumbersome than $\log(\sec x + \tan x)$; using Problem 15-8, we obtain

$$\int \sec x \, dx = \log \left(\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right) = \log \left(\tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right).$$

This last expression was actually the one first discovered, and was due, not to any mathematician's cleverness, but to a curious historical accident: In 1599 Wright computed nautical tables that amounted to definite integrals of $\sec$. When the first tables for the logarithms of tangents were produced, the correspondence between the two tables was immediately noticed (but remained unexplained until the invention of calculus).

11. The derivation of $\int e^x \sin x \, dx$ given in the text seems to prove that the only primitive of $f(x) = e^x \sin x$ is $F(x) = e^x(\sin x - \cos x)/2$, whereas $F(x) = e^x(\sin x - \cos x)/2 + C$ is also a primitive for any number C . Where does C come from? (What is the meaning of the equation

$$\int e^x \sin x \, dx = e^x \sin x - e^x \cos x - \int e^x \sin x \, dx?)$$

12. (a) Find $\int \arcsin x \, dx$, using the same trick that worked for $\log$ and $\arctan$.
 *(b) Generalize this trick: Find $\int f^{-1}(x) \, dx$ in terms of $\int f(x) \, dx$. Compare with Problems 12-15 and 14-10.
13. (a) Find $\int \sin^4 x \, dx$ in two different ways: first using the reduction formula, and then using the formula for $\sin^2 x$.
 (b) Combine your answers to obtain an impressive trigonometric identity.
14. Express $\int \log(\log x) \, dx$ in terms of $\int (\log x)^{-1} \, dx$. (Neither is expressible in terms of elementary functions.)
15. Express $\int x^2 e^{-x^2} \, dx$ in terms of $\int e^{-x^2} \, dx$.
16. Prove that the function $f(x) = e^x/(e^{5x} + e^x + 1)$ has an elementary primitive. (Do not try to find it!)
17. Prove the reduction formulas in the text. For the third one write

$$\int \frac{dx}{(1+x^2)^n} = \int \frac{dx}{(1+x^2)^{n-1}} - \int \frac{x^2 dx}{(1+x^2)^n}$$

and work on the last integral. (Another possibility is to use the substitution $x = \tan u$.)

18. Find a reduction formula for

- (a) $\int x^n e^x dx$.
 (b) $\int (\log x)^n dx$.

*19. Prove that

$$\int_1^{\cosh x} \sqrt{t^2 - 1} dt = \frac{\cosh x \sinh x}{2} - \frac{x}{2}.$$

(See Problem 17-4 for the significance of this computation.)

20. Prove that

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx.$$

(A geometric interpretation makes this clear, but it is also a good exercise in the handling of limits of integration during a substitution.)

21. Prove that the area of a circle of radius r is πr^2 . (Naturally you must remember that π is defined as the area of the unit circle.)
 22. Use induction and integration by parts to generalize Problem 14-6:

$$\int_0^x \frac{f(u)(x-u)^n}{n!} du = \int_0^x \left(\int_0^{u_n} \left(\cdots \left(\int_0^{u_1} f(t) dt \right) du_1 \right) \cdots \right) du_n.$$

23. If f' is continuous on $[a, b]$, use integration by parts to prove the Riemann-Lebesgue Lemma for f :

$$\lim_{\lambda \rightarrow \infty} \int_a^b f(t) \sin(\lambda t) dt = 0.$$

This result is just a special case of Problem 15-25, but it can be used to prove the general case (in much the same way that the Riemann-Lebesgue Lemma was derived in Problem 15-25 from the special case in which f is a step function).

24. Prove the following version of integration by parts for improper integrals:

$$\int_a^\infty u'(x)v(x) dx = u(x)v(x) \Big|_a^\infty - \int_a^\infty u(x)v'(x) dx.$$

The first symbol on the right side means, of course,

$$\lim_{x \rightarrow \infty} u(x)v(x) - u(a)v(a).$$

- *25. One of the most important functions in analysis is the gamma function,

$$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt.$$

- (a) Prove that the improper integral $\Gamma(x)$ is defined if $x > 0$.
 (b) Use integration by parts (more precisely, the improper integral version in the previous problem) to prove that

$$\Gamma(x+1) = x\Gamma(x).$$

- (c) Show that $\Gamma(1) = 1$, and conclude that $\Gamma(n) = (n-1)!$ for all natural numbers n .

The gamma function thus provides a simple example of a continuous function which “interpolates” the values of $n!$ for natural numbers x . Of course there are infinitely many continuous functions f with $f(n) = (n-1)!$; there are even infinitely many continuous functions f with $f(x+1) = xf(x)$ for all $x > 0$. However, the gamma function has the important additional property that $\log \circ \Gamma$ is convex, a condition which expresses the extreme smoothness of this function. A beautiful theorem due to Harald Bohr and Johannes Møllerup states that Γ is the only function f with $\log \circ f$ convex, $f(1) = 1$ and $f(x+1) = xf(x)$. See the Suggested Reading for a reference.

- *26. (a) Use the reduction formula for $\int \sin^n x \, dx$ to show that

$$\int_0^{\pi/2} \sin^n x \, dx = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x \, dx.$$

- (b) Now show that

$$\begin{aligned} \int_0^{\pi/2} \sin^{2n+1} x \, dx &= \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n+1}, \\ \int_0^{\pi/2} \sin^{2n} x \, dx &= \frac{\pi}{2} \cdot \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n}, \end{aligned}$$

and conclude that

$$\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \frac{\int_0^{\pi/2} \sin^{2n} x \, dx}{\int_0^{\pi/2} \sin^{2n+1} x \, dx}.$$

- (c) Show that the quotient of the two integrals in this expression is between 1 and $1 + 1/2n$, starting with the inequalities

$$0 < \sin^{2n+1} x \leq \sin^{2n} x \leq \sin^{2n-1} x \quad \text{for } 0 < x < \pi/2.$$

This result, which shows that the products

$$\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n-1} \cdot \frac{2n}{2n+1}$$

can be made as close to $\pi/2$ as desired, is usually written as an infinite product, known as Wallis' product:

$$\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots$$

- (d) Show also that the products

$$\frac{1}{\sqrt{n}} \frac{2 \cdot 4 \cdot 6 \cdots 2n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$$

can be made as close to $\sqrt{\pi}$ as desired. (This fact is used in the next problem and in Problem 26-18.)

****27.** It is an astonishing fact that improper integrals $\int_0^\infty f(x) dx$ can often be computed in cases where ordinary integrals $\int_a^b f(x) dx$ cannot. There is no elementary formula for $\int_a^b e^{-x^2} dx$, but we can find the value of $\int_0^\infty e^{-x^2} dx$ precisely! There are many ways of evaluating this integral, but most require some advanced techniques; the following method involves a fair amount of work, but no facts that you do not already know.

(a) Show that

$$\int_0^1 (1 - x^2)^n dx = \frac{2}{3} \cdot \frac{4}{5} \cdots \frac{2n}{2n+1},$$

$$\int_0^\infty \frac{1}{(1 + x^2)^n} dx = \frac{\pi}{2} \cdot \frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-3}{2n-2}.$$

(This can be done using reduction formulas, or by appropriate substitutions, combined with the previous problem.)

(b) Prove, using the derivative, that

$$1 - x^2 \leq e^{-x^2} \quad \text{for } 0 \leq x \leq 1.$$

$$e^{-x^2} \leq \frac{1}{1 + x^2} \quad \text{for } 0 \leq x.$$

(c) Integrate the n th powers of these inequalities from 0 to 1 and from 0 to ∞ , respectively. Then use the substitution $y = \sqrt{nx}$ to show that

$$\sqrt{n} \frac{2}{3} \cdot \frac{4}{5} \cdots \frac{2n}{2n+1}$$

$$\leq \int_0^{\sqrt{n}} e^{-y^2} dy \leq \int_0^\infty e^{-y^2} dy$$

$$\leq \frac{\pi}{2} \sqrt{n} \frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-3}{2n-2}.$$

(d) Now use Problem 26(c) to show that

$$\int_0^\infty e^{-y^2} dy = \frac{\sqrt{\pi}}{2}.$$

****28.** (a) Use integration by parts to show that

$$\int_a^b \frac{\sin x}{x} dx = \frac{\cos a}{a} - \frac{\cos b}{b} + \int_a^b \frac{\cos x}{x^2} dx,$$

and conclude that $\int_0^\infty (\sin x)/x dx$ exists. (Use the left side to investigate the limit as $a \rightarrow 0^+$ and the right side for the limit as $b \rightarrow \infty$.)

- (b) Use Problem 15-31 to show that

$$\int_0^\pi \frac{\sin(n + \frac{1}{2})t}{\sin \frac{t}{2}} dt = \pi$$

for any natural number n .

- (c) Prove that

$$\lim_{\lambda \rightarrow \infty} \int_0^\pi \sin(\lambda + \frac{1}{2})t \left[\frac{2}{t} - \frac{1}{\sin \frac{t}{2}} \right] dt = 0.$$

Hint: The term in brackets is bounded by Problem 15-2(vi); the Riemann-Lebesgue Lemma then applies.

- (d) Use the substitution
- $u = (\lambda + \frac{1}{2})t$
- and part (b) to show that

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

- *29. (a) Use the substitution
- $u = t^x$
- to show that

$$\Gamma(x) = \frac{1}{x} \int_0^\infty e^{-u^{1/x}} dx.$$

- (b) Find
- $\Gamma(\frac{1}{2})$
- .

PART **4**
INFINITE
SEQUENCES
AND
INFINITE
SERIES

One of the most remarkable series of algebraic analysis is the following:

$$\begin{aligned}
 &1 + \frac{m}{1} x + \frac{m(m-1)}{1 \cdot 2} x^2 \\
 &+ \frac{m(m-1)(m-2)}{1 \cdot 2 \cdot 3} x^3 + \dots \\
 &+ \frac{m(m-1) \dots [m-(n-1)]}{1 \cdot 2 \dots n} x^n \\
 &+ \dots
 \end{aligned}$$

when m is a positive whole number the sum of the series, which is then finite, can be expressed, as is known, by $(1+x)^m$. When m is not an integer, the series goes on to infinity, and it will converge or diverge according as the quantities m and x have this or that value. In this case, one writes the same equality

$$\begin{aligned}
 (1+x)^m &= 1 + \frac{m}{1} x \\
 &+ \frac{m(m-1)}{1 \cdot 2} x^2 + \dots \text{etc.}
 \end{aligned}$$

... It is assumed that the numerical equality will always occur whenever the series is convergent, but this has never yet been proved.

NIELS HENRIK ABEL

There is one sense in which the “elementary functions” are not elementary at all. If p is a polynomial function,

$$p(x) = a_0 + a_1x + \cdots + a_nx^n,$$

then $p(x)$ can be computed easily for any number x . This is not at all true for functions like $\sin$, $\log$, or $\exp$. At present, to find $\log x = \int_1^x 1/t \, dt$ approximately, we must compute some upper or lower sums, and make certain that the error involved in accepting such a sum for $\log x$ is not too great. Computing $e^x = \log^{-1}(x)$ would be even more difficult: we would have to compute $\log a$ for many values of a until we found a number a such that $\log a$ is approximately x —then a would be approximately e^x .

In this chapter we will obtain important theoretical results which reduce the computation of $f(x)$, for many functions f , to the evaluation of polynomial functions. The method depends on finding polynomial functions which are close approximations to f . In order to guess a polynomial which is appropriate, it is useful to first examine polynomial functions themselves more thoroughly.

Suppose that

$$p(x) = a_0 + a_1x + \cdots + a_nx^n.$$

It is interesting, and for our purposes very important, to note that the coefficients a_i can be expressed in terms of the value of p and its various derivatives at 0. To begin with, note that

$$p(0) = a_0.$$

Differentiating the original expression for $p(x)$ yields

$$p'(x) = a_1 + 2a_2x + \cdots + na_nx^{n-1}.$$

Therefore,

$$p'(0) = p^{(1)}(0) = a_1.$$

Differentiating again we obtain

$$p''(x) = 2a_2 + 3 \cdot 2 \cdot a_3x + \cdots + n(n-1) \cdot a_nx^{n-2}.$$

Therefore,

$$p''(0) = p^{(2)}(0) = 2a_2.$$

In general, we will have

$$p^{(k)}(0) = k!a_k \quad \text{or} \quad a_k = \frac{p^{(k)}(0)}{k!}.$$

If we agree to define $0! = 1$, and recall the notation $p^{(0)} = p$, then this formula holds for $k = 0$ also.

If we had begun with a function p that was written as a “polynomial in $(x - a)$,”

$$p(x) = a_0 + a_1(x - a) + \cdots + a_n(x - a)^n,$$

then a similar argument would show that

$$a_k = \frac{p^{(k)}(a)}{k!}.$$

Suppose now that f is a function (not necessarily a polynomial) such that

$$f^{(1)}(a), \dots, f^{(n)}(a)$$

all exist. Let

$$a_k = \frac{f^{(k)}(a)}{k!}, \quad 0 \leq k \leq n,$$

and define

$$P_{n,a}(x) = a_0 + a_1(x - a) + \cdots + a_n(x - a)^n.$$

The polynomial $P_{n,a}$ is called the **Taylor polynomial of degree n for f at a** . (Strictly speaking, we should use an even more complicated expression, like $P_{n,a,f}$, to indicate the dependence on f ; at times this more precise notation will be useful.) The Taylor polynomial has been defined so that

$$P_{n,a}^{(k)}(a) = f^{(k)}(a) \quad \text{for } 0 \leq k \leq n;$$

in fact, it is clearly the only polynomial of degree $\leq n$ with this property.

Although the coefficients of $P_{n,a,f}$ seem to depend upon f in a fairly complicated way, the most important elementary functions have extremely simple Taylor polynomials. Consider first the function $\sin$. We have

$$\begin{aligned} \sin(0) &= 0, \\ \sin'(0) &= \cos 0 = 1, \\ \sin''(0) &= -\sin 0 = 0, \\ \sin'''(0) &= -\cos 0 = -1, \\ \sin^{(4)}(0) &= \sin 0 = 0. \end{aligned}$$

From this point on, the derivatives repeat in a cycle of 4. The numbers

$$a_k = \frac{\sin^{(k)}(0)}{k!}$$

are

$$0, 1, 0, -\frac{1}{3!}, 0, \frac{1}{5!}, 0, -\frac{1}{7!}, 0, \frac{1}{9!}, \dots$$

Therefore the Taylor polynomial $P_{2n+1,0}$ of degree $2n + 1$ for $\sin$ at 0 is

$$P_{2n+1,0}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

(Of course, $P_{2n+1,0} = P_{2n+2,0}$).

The Taylor polynomial $P_{2n,0}$ of degree $2n$ for $\cos$ at 0 is (the computations are left to you)

$$P_{2n,0}(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!}.$$

The Taylor polynomial for $\exp$ is especially easy to compute. Since $\exp^{(k)}(0) = \exp(0) = 1$ for all k , the Taylor polynomial of degree n at 0 is

$$P_{n,0}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!}.$$

The Taylor polynomial for $\log$ must be computed at some point $a \neq 0$, since $\log$ is not even defined at 0. The standard choice is $a = 1$. Then

$$\log'(x) = \frac{1}{x}, \quad \log'(1) = 1;$$

$$\log''(x) = -\frac{1}{x^2}, \quad \log''(1) = -1;$$

$$\log'''(x) = \frac{2}{x^3}, \quad \log'''(1) = 2;$$

in general

$$\log^{(k)}(x) = \frac{(-1)^{k-1}(k-1)!}{x^k}, \quad \log^{(k)}(1) = (-1)^{k-1}(k-1)!.$$

Therefore the Taylor polynomial of degree n for $\log$ at 1 is

$$P_{n,1}(x) = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} + \cdots + \frac{(-1)^{n-1}(x-1)^n}{n}.$$

It is often more convenient to consider the function $f(x) = \log(1+x)$. In this case we can choose $a = 0$. We have

$$f^{(k)}(x) = \log^{(k)}(1+x),$$

so

$$f^{(k)}(0) = \log^{(k)}(1) = (-1)^{k-1}(k-1)!.$$

Therefore the Taylor polynomial of degree n for f at 0 is

$$P_{n,0}(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{(-1)^{n-1}x^n}{n}.$$

There is one other elementary function whose Taylor polynomial is important—arctan. The computations of the derivatives begin

$$\arctan'(x) = \frac{1}{1+x^2}, \quad \arctan'(0) = 1;$$

$$\arctan''(x) = \frac{-2x}{(1+x^2)^2}, \quad \arctan''(0) = 0;$$

$$\arctan'''(x) = \frac{(1+x^2)^2 \cdot (-2) + 2x \cdot 2(1+x^2) \cdot 2x}{(1+x^2)^4}, \quad \arctan'''(0) = -2.$$

It is clear that this brute force computation will never do. However, the Taylor polynomials of $\arctan$ will be easy to find after we have examined the properties of Taylor polynomials more closely—although the Taylor polynomial $P_{n,a,f}$ was simply defined so as to have the same first n derivatives at a as f , the connection between f and $P_{n,a,f}$ will actually turn out to be much deeper.

One line of evidence for a closer connection between f and the Taylor polynomials for f may be uncovered by examining the Taylor polynomial of degree 1, which is

$$P_{1,a}(x) = f(a) + f'(a)(x - a).$$

Notice that

$$\frac{f(x) - P_{1,a}(x)}{x - a} = \frac{f(x) - f(a)}{x - a} - f'(a).$$

Now, by the definition of $f'(a)$ we have

$$\lim_{x \rightarrow a} \frac{f(x) - P_{1,a}(x)}{x - a} = 0.$$

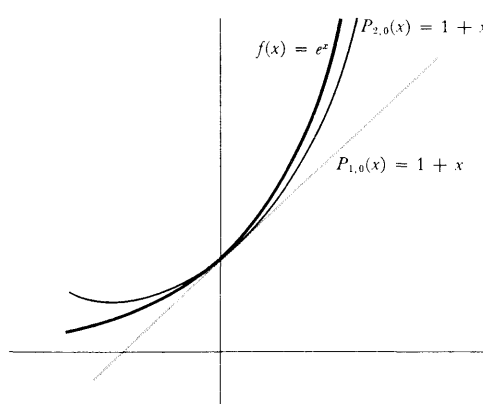


FIGURE 1

In other words, as x approaches a the difference $f(x) - P_{1,a}(x)$ not only becomes small, but actually becomes small even compared to $x - a$. Figure 1 illustrates the graph of $f(x) = e^x$ and of

$$P_{1,0}(x) = f(0) + f'(0)x = 1 + x,$$

which is the Taylor polynomial of degree 1 for f at 0. The diagram also shows the graph of

$$P_{2,0}(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 = 1 + x + \frac{x^2}{2},$$

which is the Taylor polynomial of degree 2 for f at 0. As x approaches 0, the difference $f(x) - P_{2,0}(x)$ seems to be getting small even faster than the

difference $f(x) - P_{1,0}(x)$. As it stands, this assertion is not very precise, but we are now prepared to give it a definite meaning. We have just noted that in general

$$\lim_{x \rightarrow a} \frac{f(x) - P_{1,a}(x)}{x - a} = 0.$$

For $f(x) = e^x$ and $a = 0$ this means that

$$\lim_{x \rightarrow 0} \frac{f(x) - P_{1,0}(x)}{x} = \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x} = 0.$$

On the other hand, an easy double application of l'Hôpital's Rule shows that

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2} \neq 0.$$

Thus, although $f(x) - P_{1,0}(x)$ becomes small compared to x , as x approaches 0, it does *not* become small compared to x^2 . For $P_{2,0}(x)$ the situation is quite different; the extra term $x^2/2$ provides just the right compensation:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^2} &= \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{2x} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{2} = 0. \end{aligned}$$

This result holds in general—if $f'(a)$ and $f''(a)$ exist, then

$$\lim_{x \rightarrow a} \frac{f(x) - P_{2,a}(x)}{(x - a)^2} = 0;$$

in fact, the analogous assertion for $P_{n,a}$ is also true.

THEOREM 1 Suppose that f is a function for which

$$f'(a), \dots, f^{(n)}(a)$$

all exist. Let

$$a_k = \frac{f^{(k)}(a)}{k!}, \quad 0 \leq k \leq n,$$

and define

$$P_{n,a}(x) = a_0 + a_1(x - a) + \dots + a_n(x - a)^n.$$

Then

$$\lim_{x \rightarrow a} \frac{f(x) - P_{n,a}(x)}{(x - a)^n} = 0.$$

PROOF Writing out $P_{n,a}(x)$ explicitly, we obtain

$$\frac{f(x) - P_{n,a}(x)}{(x-a)^n} = \frac{f(x) - \sum_{i=0}^{n-1} \frac{f^{(i)}(a)}{i!} (x-a)^i}{(x-a)^n} - \frac{f^{(n)}(a)}{n!}.$$

It will help to introduce the new functions

$$Q(x) = \sum_{i=0}^{n-1} \frac{f^{(i)}(a)}{i!} (x-a)^i \quad \text{and} \quad g(x) = (x-a)^n;$$

now we must prove that

$$\lim_{x \rightarrow a} \frac{f(x) - Q(x)}{g(x)} = \frac{f^{(n)}(a)}{n!}.$$

Notice that

$$\begin{aligned} Q^{(k)}(a) &= f^{(k)}(a), \quad k \leq n-1, \\ g^{(k)}(x) &= n!(x-a)^{n-k}/(n-k)!. \end{aligned}$$

Thus

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) - Q(x)] &= f(a) - Q(a) = 0, \\ \lim_{x \rightarrow a} [f'(x) - Q'(x)] &= f'(a) - Q'(a) = 0, \\ &\vdots \end{aligned}$$

$$\lim_{x \rightarrow a} [f^{(n-2)}(x) - Q^{(n-2)}(x)] = f^{(n-2)}(a) - Q^{(n-2)}(a) = 0,$$

and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} g'(x) = \cdots = \lim_{x \rightarrow a} g^{(n-2)}(x) = 0.$$

We may therefore apply l'Hôpital's Rule $n-1$ times to obtain

$$\lim_{x \rightarrow a} \frac{f(x) - Q(x)}{(x-a)^n} = \lim_{x \rightarrow a} \frac{f^{(n-1)}(x) - Q^{(n-1)}(x)}{n!(x-a)}.$$

Since Q is a polynomial of degree $n-1$, its $(n-1)$ st derivative is a constant; in fact, $Q^{(n-1)}(x) = f^{(n-1)}(a)$. Thus

$$\lim_{x \rightarrow a} \frac{f(x) - Q(x)}{(x-a)^n} = \lim_{x \rightarrow a} \frac{f^{(n-1)}(x) - f^{(n-1)}(a)}{n!(x-a)},$$

and this last limit is $f^{(n)}(a)/n!$ by definition of $f^{(n)}(a)$. ■

One simple consequence of Theorem 1 allows us to perfect the test for local maxima and minima which was developed in Chapter 11. If a is a critical point of f , then, according to Theorem 11-5, the function f has a local minimum at a if $f''(a) > 0$, and a local maximum at a if $f''(a) < 0$. If $f''(a) = 0$ no conclusion was possible, but it is conceivable that the sign of $f'''(a)$ might give further information; and if $f'''(a) = 0$, then the sign of $f^{(4)}(a)$ might be signifi-

cant. Even more generally, we can ask what happens when

$$(*) \quad f'(a) = f''(a) = \cdots = f^{(n-1)}(a) = 0, \\ f^{(n)}(a) \neq 0.$$

The situation in this case can be guessed by examining the functions

$$f(x) = (x - a)^n, \\ g(x) = -(x - a)^n,$$

which satisfy (*). Notice (Figure 2) that if n is odd, then a is neither a local maximum nor a local minimum point for f or g . On the other hand, if n is even, then f , with a positive n th derivative, has a local minimum at a , while g , with a negative n th derivative, has a local maximum at a . Of all functions satisfying (*), these are about the simplest available; nevertheless they indicate the general situation exactly. In fact, the whole point of the next proof is that any function satisfying (*) looks very much like one of these functions, in a sense that is made precise by Theorem 1.

THEOREM 2 Suppose that

$$f'(a) = \cdots = f^{(n-1)}(a) = 0, \\ f^{(n)}(a) \neq 0.$$

- (1) If n is even and $f^{(n)}(a) > 0$, then f has a local minimum at a .
- (2) If n is even and $f^{(n)}(a) < 0$, then f has a local maximum at a .
- (3) If n is odd, then f has neither a local maximum nor minimum at a .

PROOF

There is clearly no loss of generality in assuming that $f(a) = 0$, since neither the hypotheses nor the conclusion are affected if f is replaced by $f - f(a)$. Then, since the first $n - 1$ derivatives of f at a are 0, the Taylor polynomial $P_{n,a}$ of f is

$$P_{n,a}(x) = f(a) + \frac{f'(a)}{1!}(x - a) + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n \\ = \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

Thus, Theorem 1 states that

$$0 = \lim_{x \rightarrow a} \frac{f(x) - P_{n,a}(x)}{(x - a)^n} = \lim_{x \rightarrow a} \left[\frac{f(x)}{(x - a)^n} - \frac{f^{(n)}(a)}{n!} \right].$$

Consequently, if x is sufficiently close to a , then

$$\frac{f(x)}{(x - a)^n} \text{ has the same sign as } \frac{f^{(n)}(a)}{n!}.$$

Suppose now that n is even. In this case $(x - a)^n > 0$ for all $x \neq a$. Since $f(x)/(x - a)^n$ has the same sign as $f^{(n)}(a)/n!$ for x sufficiently close to a , it

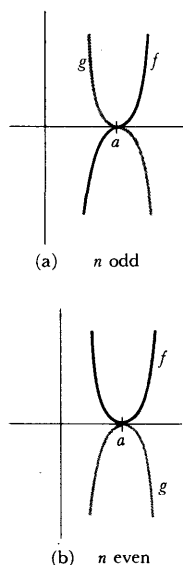


FIGURE 2

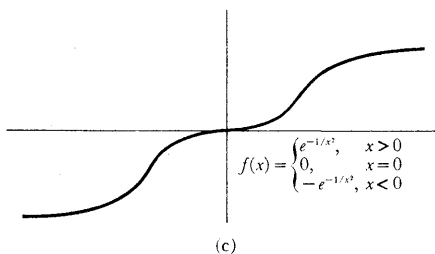
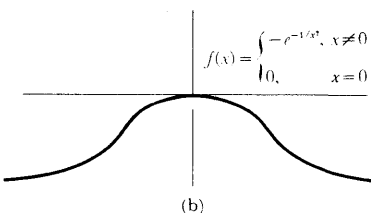
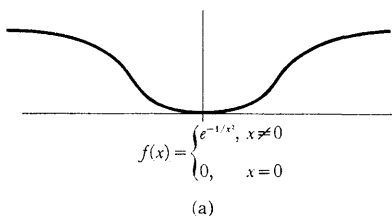


FIGURE 3

follows that $f(x)$ itself has the same sign as $f^n(a)/n!$ for x sufficiently close to a . If $f^{(n)}(a) > 0$, this means that

$$f(x) > 0 = f(a)$$

for x close to a . Consequently, f has a local minimum at a . A similar proof works for the case $f^{(n)}(a) < 0$.

Now suppose that n is odd. The same argument as before shows that if x is sufficiently close to a , then

$$\frac{f(x)}{(x-a)^n} \text{ always has the same sign.}$$

But $(x-a)^n > 0$ for $x > a$ and $(x-a)^n < 0$ for $x < a$. Therefore $f(x)$ has different signs for $x > a$ and $x < a$. This proves that f has neither a local maximum nor a local minimum at a . ■

Although Theorem 2 will settle the question of local maxima and minima for just about any function which arises in practice, it does have some theoretical limitations, because $f^{(k)}(a)$ may be 0 for *all* k . This happens (Figure 3(a)) for the function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

which has a minimum at 0, and also for the negative of this function (Figure 3(b)), which has a maximum at 0. Moreover (Figure 3(c)), if

$$f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x = 0 \\ -e^{-1/x^2}, & x < 0, \end{cases}$$

then $f^{(k)}(0) = 0$ for all k , but f has neither a local minimum nor a local maximum at 0.

The conclusion of Theorem 1 is often expressed in terms of an important concept of “order of equality.” Two functions f and g are **equal up to order n at a** if

$$\lim_{x \rightarrow a} \frac{f(x) - g(x)}{(x-a)^n} = 0.$$

In the language of this definition, Theorem 1 says that the Taylor polynomial $P_{n,a,f}$ equals f up to order n at a . The Taylor polynomial might very well have been designed to make this fact true, because there is at most one polynomial of degree $\leq n$ with this property. This assertion is a consequence of the following elementary theorem.

THEOREM 3

Let P and Q be two polynomials in $(x-a)$, of degree $\leq n$, and suppose that P and Q are equal up to order n at a . Then $P = Q$.

PROOF

Let $R = P - Q$. Since R is a polynomial of degree $\leq n$, it is only necessary to

prove that if

$$R(x) = b_0 + \cdots + b_n(x-a)^n$$

satisfies

$$\lim_{x \rightarrow a} \frac{R(x)}{(x-a)^n} = 0,$$

then $R = 0$. Now the hypotheses on R surely imply that

$$\lim_{x \rightarrow a} \frac{R(x)}{(x-a)^i} = 0 \quad \text{for } 0 \leq i \leq n.$$

For $i = 0$ this condition reads simply $\lim_{x \rightarrow a} R(x) = 0$; on the other hand,

$$\begin{aligned} \lim_{x \rightarrow a} R(x) &= \lim_{x \rightarrow a} [b_0 + b_1(x-a) + \cdots + b_n(x-a)^n] \\ &= b_0. \end{aligned}$$

Thus $b_0 = 0$ and

$$R(x) = b_1(x-a) + \cdots + b_n(x-a)^n.$$

Therefore,

$$\frac{R(x)}{x-a} = b_1 + b_2(x-a) + \cdots + b_n(x-a)^{n-1}$$

and

$$\lim_{x \rightarrow a} \frac{R(x)}{x-a} = b_1.$$

Thus $b_1 = 0$ and

$$R(x) = b_2(x-a)^2 + \cdots + b_n(x-a)^n.$$

Continuing in this way, we find that

$$b_0 = \cdots = b_n = 0. \blacksquare$$

COROLLARY Let f be n -times differentiable at a , and suppose that P is a polynomial in $(x-a)$ of degree $\leq n$, which equals f up to order n at a . Then $P = P_{n,a}$.

PROOF Since P and $P_{n,a}$ both equal f up to order n at a , it is easy to see that P equals $P_{n,a}$ up to order n at a . Consequently, $P = P_{n,a}$ by the Theorem. $\blacksquare$

At first sight this corollary appears to have unnecessarily complicated hypotheses; it might seem that the existence of the polynomial P would automatically imply that f is sufficiently differentiable for $P_{n,a}$ to exist. But in fact this is not so. For example (Figure 4), suppose that

$$f(x) = \begin{cases} x^{n+1}, & x \text{ irrational} \\ 0, & x \text{ rational.} \end{cases}$$

If $P(x) = 0$, then P is certainly a polynomial of degree $\leq n$ which equals f up to order n at 0. On the other hand, $f'(a)$ does not exist for any $a \neq 0$, so $f''(0)$ is undefined.

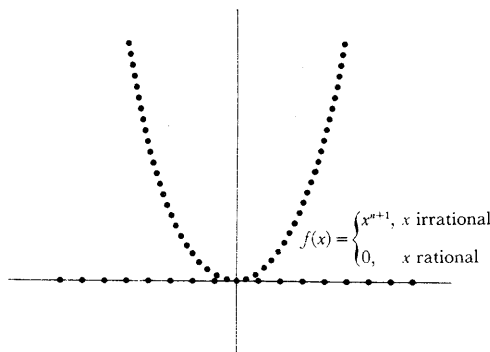


FIGURE 4

When f does have n derivatives at a , however, the corollary may provide a useful method for finding the Taylor polynomial of f . In particular, remember that our first attempt to find the Taylor polynomial for $\arctan$ ended in failure. The equation

$$\arctan x = \int_0^x \frac{1}{1+t^2} dt$$

suggests a promising method of finding a polynomial close to $\arctan$ —divide 1 by $1+t^2$, to obtain a polynomial plus a remainder:

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \cdots + (-1)^n t^{2n} + \frac{(-1)^{n+1} t^{2n+2}}{1+t^2}.$$

This formula, which can be checked easily by multiplying both sides by $1+t^2$, shows that

$$\begin{aligned} \arctan x &= \int_0^x 1 - t^2 + t^4 - \cdots + (-1)^n t^{2n} dt + (-1)^{n+1} \int_0^x \frac{t^{2n+2}}{1+t^2} dt \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + (-1)^{n+1} \int_0^x \frac{t^{2n+2}}{1+t^2} dt. \end{aligned}$$

According to our corollary, the polynomial which appears here will be the Taylor polynomial of degree $2n+1$ for $\arctan$ at 0, provided that

$$\lim_{x \rightarrow 0} \frac{\int_0^x \frac{t^{2n+2}}{1+t^2} dt}{x^{2n+1}} = 0.$$

Since

$$\left| \int_0^x \frac{t^{2n+2}}{1+t^2} dt \right| \leq \left| \int_0^x t^{2n+2} dt \right| = \frac{|x|^{2n+3}}{2n+3},$$

this is clearly true. Thus we have found that the Taylor polynomial of degree $2n+1$ for $\arctan$ at 0 is

$$P_{2n+1,0}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1}.$$

By the way, now that we have discovered the Taylor polynomials of $\arctan$, it is possible to work backwards and find $\arctan^{(k)}(0)$ for all k : Since

$$P_{2n+1,0}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1},$$

and since this polynomial is, by definition,

$$\arctan^{(0)}(0) + \arctan^{(1)}(0)x + \frac{\arctan^{(2)}(0)}{2!}x^2 + \cdots + \frac{\arctan^{(2n+1)}(0)}{(2n+1)!}x^{2n+1},$$

we can find $\arctan^{(k)}(0)$ by simply equating the coefficients of x^k in these two polynomials:

$$\frac{\arctan^{(k)}(0)}{k!} = 0 \quad \text{if } k \text{ is even,}$$

$$\frac{\arctan^{(2l+1)}(0)}{(2l+1)!} = \frac{(-1)^l}{2l+1} \quad \text{or} \quad \arctan^{(2l+1)}(0) = (-1)^l \cdot (2l)!.$$

A much more interesting fact emerges if we go back to the original equation

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + (-1)^{n+1} \int_0^x \frac{t^{2n+2}}{1+t^2} dt,$$

and remember the estimate

$$\left| \int_0^x \frac{t^{2n+2}}{1+t^2} dt \right| \leq \frac{|x|^{2n+3}}{2n+3}.$$

When $|x| \leq 1$, this expression is at most $1/(2n+3)$, and we can make this as small as we like simply by choosing n large enough. In other words, for $|x| \leq 1$ we can use the Taylor polynomials for $\arctan$ to compute $\arctan x$ as accurately as we like. The most important theorems about Taylor polynomials extend this isolated result to other functions, and the Taylor polynomials will soon play quite a new role. The theorems proved so far have always examined the behavior of the Taylor polynomial $P_{n,a}$ for fixed n , as x approaches a . Henceforth we will compare Taylor polynomials $P_{n,a}$ for fixed x , and different n . In anticipation of the coming theorem we introduce some new notation.

If f is a function for which $P_{n,a}(x)$ exists, we define the **remainder term** $R_{n,a}(x)$ by

$$\begin{aligned} f(x) &= P_{n,a}(x) + R_{n,a}(x) \\ &= f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_{n,a}(x). \end{aligned}$$

We would like to have an expression for $R_{n,a}(x)$ whose size is easy to estimate. There is such an expression, involving an integral, just as in the case for $\arctan$. One way to guess this expression is to begin with the case $n = 0$:

$$f(x) = f(a) + R_{0,a}(x).$$

The Fundamental Theorem of Calculus enables us to write

$$f(x) = f(a) + \int_a^x f'(t) dt,$$

so that

$$R_{0,a}(x) = \int_a^x f'(t) dt.$$

A similar expression for $R_{1,a}(x)$ can be derived from this formula using integration by parts in a rather tricky way: Let

$$u(t) = f'(t) \quad \text{and} \quad v(t) = t - x$$

(notice that x represents some fixed number in the expression for $v(t)$, so $v'(t) = 1$); then

$$\begin{aligned} \int_a^x f'(t) dt &= \int_a^x \underbrace{f'(t)}_{u(t)} \cdot \underbrace{1}_{v'(t)} dt \\ &= u(t)v(t) \Big|_a^x - \int_a^x \underbrace{f''(t)}_{u'(t)} \underbrace{(t-x)}_{v(t)} dt. \end{aligned}$$

Since $v(x) = 0$, we obtain

$$\begin{aligned} f(x) &= f(a) + \int_a^x f'(t) dt \\ &= f(a) - u(a)v(a) + \int_a^x f''(t)(x-t) dt \\ &= f(a) + f'(a)(x-a) + \int_a^x f''(t)(x-t) dt. \end{aligned}$$

Thus

$$R_{1,a}(x) = \int_a^x f''(t)(x-t) dt.$$

It is hard to give any motivation for choosing $v(t) = t - x$, rather than $v(t) = t$. It just happens to be the choice which works out, the sort of thing one might discover after sufficiently many similar but futile manipulations. However, it is now easy to guess the formula for $R_{2,a}(x)$. If

$$u(t) = f''(t) \quad \text{and} \quad v(t) = \frac{-(x-t)^2}{2},$$

then $v'(t) = (x-t)$, so

$$\begin{aligned} \int_a^x f''(t)(x-t) dt &= u(t)v(t) \Big|_a^x - \int_a^x f'''(t) \cdot \frac{-(x-t)^2}{2} dt \\ &= \frac{f''(a)(x-a)^2}{2} + \int_a^x \frac{f'''(t)}{2} (x-t)^2 dt. \end{aligned}$$

This shows that

$$R_{2,a}(x) = \int_a^x \frac{f^{(3)}(t)}{2} (x-t)^2 dt.$$

You should now have little difficulty giving a rigorous proof, by induction,

that if $f^{(n+1)}$ is continuous on $[a, x]$, then

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt.$$

From this formula, which is called the integral form of the remainder, it is possible (Problem 13) to derive two other important expressions for $R_{n,a}(x)$: the Cauchy form of the remainder,

$$R_{n,a}(x) = \frac{f^{(n+1)}(t)}{n!} (x-t)^n (x-a) \quad \text{for some } t \text{ in } (a, x),$$

and the Lagrange form of the remainder,

$$R_{n,a}(x) = \frac{f^{(n+1)}(t)}{(n+1)!} (x-a)^{n+1} \quad \text{for some } t \text{ in } (a, x).$$

In the proof of the next theorem (Taylor's Theorem) we will derive all three forms of the remainder in an entirely different way. One virtue of this proof (aside from its cleverness) is the fact that the Cauchy and Lagrange forms of the remainder will be proved without assuming the extra hypothesis that $f^{(n+1)}$ is continuous. In this way Taylor's Theorem appears as a direct generalization of the Mean Value Theorem, to which it reduces for $n = 0$, and which is the crucial tool used in the proof.

These remarks may suggest a strategy for proving Taylor's Theorem. Since $R_{n,a}(a) = 0$, we might try to apply the Mean Value Theorem to the expression

$$\frac{R_{n,a}(x)}{x-a} = \frac{R_{n,a}(x) - R_{n,a}(a)}{x-a}.$$

On second thought, however, this idea does not look very promising, since it is not at all clear how $f^{(n+1)}(t)$ is ever going to be involved in the answer. Indeed, if we take the most straightforward route, and differentiate both sides of the equation which defines $R_{n,a}$, we obtain

$$f'(x) = f'(a) + f''(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{(n-1)!} (x-a)^{n-1} + R_{n,a}'(x),$$

which is useless. The proper application of the Mean Value Theorem has a lot in common with the integration by parts proof outlined above. This proof involved the derivative of a function in which x denoted a number which was fixed. This is just how x will be treated in the following proof.

THEOREM 4 (TAYLOR'S THEOREM)

Suppose that $f', \dots, f^{(n+1)}$ are defined on $[a, x]$, and that $R_{n,a}(x)$ is defined by

$$f(x) = f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!} (x-a)^n + R_{n,a}(x).$$

Then

$$(1) R_{n,a}(x) = \frac{f^{(n+1)}(t)}{n!} (x-t)^n (x-a) \quad \text{for some } t \text{ in } (a, x).$$

$$(2) R_{n,a}(x) = \frac{f^{(n+1)}(t)}{(n+1)!} (x-a)^{n+1} \quad \text{for some } t \text{ in } (a, x).$$

Moreover, if $f^{(n+1)}$ is integrable on $[a, x]$, then

$$(3) R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt.$$

(If $x < a$, then the hypothesis should state that f is $(n+1)$ -times differentiable on $[x, a]$; the number t in (1) and (2) will then be in (x, a) , while (3) will remain true as stated, provided that $f^{(n+1)}$ is integrable on $[x, a]$.)

PROOF For each number t in $[a, x]$ we have

$$f(x) = f(t) + f'(t)(x-t) + \cdots + \frac{f^{(n)}(t)}{n!} (x-t)^n + R_{n,t}(x).$$

Let us denote the number $R_{n,t}(x)$ by $S(t)$; the function S is defined on $[a, x]$, and we have

$$(*) \quad f(x) = f(t) + f'(t)(x-t) + \cdots + \frac{f^{(n)}(t)}{n!} (x-t)^n + S(t)$$

for all t in $[a, x]$.

We will now differentiate both sides of this equation, which asserts that the function whose value at t is $f(x)$, equals the function whose value at t is

$$f(t) + \cdots + \frac{f^{(n)}(t)}{n!} (x-t)^n + S(t).$$

(In common parlance we are considering both sides of $(*)$ "as a function of t ." Just to make sure that the letter x causes no confusion, notice that if

$$g(t) = f(x) \quad \text{for all } t,$$

then

$$g'(t) = 0 \quad \text{for all } t;$$

and if

$$g(t) = \frac{f^{(k)}(t)}{k!} (x-t)^k,$$

then

$$\begin{aligned} g'(t) &= \frac{f^{(k)}(t)}{k!} k(x-t)^{k-1}(-1) + \frac{f^{(k+1)}(t)}{k!} (x-t)^k \\ &= -\frac{f^{(k)}(t)}{(k-1)!} (x-t)^{k-1} + \frac{f^{(k+1)}(t)}{k!} (x-t)^k. \end{aligned}$$

Applying these formulas to each term of (*), we obtain

$$0 = f'(t) + \left[-f'(t) + \frac{f''(t)}{1!} (x-t) \right] + \left[\frac{-f''(t)}{1!} (x-t) + \frac{f^{(3)}(t)}{2!} (x-t)^2 \right] \\ + \cdots + \left[\frac{-f^{(n)}(t)}{(n-1)!} (x-t)^{n-1} + \frac{f^{(n+1)}(t)}{n!} (x-t)^n \right] + S'(t).$$

In this beautiful formula practically everything in sight cancels out, and we obtain

$$S'(t) = -\frac{f^{(n+1)}(t)}{n!} (x-t)^n.$$

Now we can apply the Mean Value Theorem to the function S on $[a, x]$: there is some t in (a, x) such that

$$\frac{S(x) - S(a)}{x - a} = S'(t) = -\frac{f^{(n+1)}(t)}{n!} (x-t)^n.$$

Remember that

$$S(t) = R_{n,t}(x);$$

this means in particular that

$$\begin{aligned} S(x) &= R_{n,x}(x) = 0, \\ S(a) &= R_{n,a}(x). \end{aligned}$$

Thus

$$\frac{0 - R_{n,a}(x)}{x - a} = -\frac{f^{(n+1)}(t)}{n!} (x-t)^n$$

or

$$R_{n,a}(x) = \frac{f^{(n+1)}(t)}{n!} (x-t)^n (x-a);$$

this is the Cauchy form of the remainder.

To derive the Lagrange form we apply the Cauchy Mean Value Theorem to the functions S and $g(t) = (x-t)^{n+1}$: there is some t in (a, x) such that

$$\frac{S(x) - S(a)}{g(x) - g(a)} = \frac{S'(t)}{g'(t)} = \frac{-\frac{f^{(n+1)}(t)}{n!} (x-t)^n}{-(n+1)(x-t)^n}.$$

Thus

$$\frac{R_{n,a}(x)}{(x-a)^{n+1}} = \frac{f^{(n+1)}(t)}{(n+1)!}$$

or

$$R_{n,a}(x) = \frac{f^{(n+1)}(t)}{(n+1)!} (x-a)^{n+1},$$

which is the Lagrange form.

Finally, if $f^{(n+1)}$ is integrable on $[a, x]$, then

$$S(x) - S(a) = \int_a^x S'(t) dt = - \int_a^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt$$

or

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt. \blacksquare$$

Although the Lagrange and Cauchy forms of the remainder are more than theoretical curiosities (see, e.g., Problem 22-16), the integral form of the remainder will usually be quite adequate. If this form is applied to the functions $\sin$, $\cos$, and $\exp$, with $a = 0$, Taylor's Theorem yields the following formulas:

$$\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ &\quad + \int_0^x \frac{\sin^{(2n+2)}(t)}{(2n+1)!} (x-t)^{2n+1} dt, \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \int_0^x \frac{\cos^{(2n+1)}(t)}{(2n)!} (x-t)^{2n} dt, \\ e^x &= 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \int_0^x \frac{e^t}{n!} (x-t)^n dt. \end{aligned}$$

To evaluate any of these integrals explicitly would be supreme foolishness—the answer of course will be exactly the difference of the left side and all the other terms on the right side! To *estimate* these integrals, however, is both easy and worthwhile.

The first two integrals are especially easy. Since

$$|\sin^{(2n+2)}(t)| \leq 1 \quad \text{for all } t,$$

we have

$$\left| \int_0^x \frac{\sin^{(2n+2)}(t)}{(2n+1)!} (x-t)^{2n+1} dt \right| \leq \frac{1}{(2n+1)!} \left| \int_0^x (x-t)^{2n+1} dt \right|.$$

Since

$$\begin{aligned} \int_0^x (x-t)^{2n+1} dt &= \left. \frac{-(x-t)^{2n+2}}{2n+2} \right|_{t=0}^{t=x} \\ &= \frac{x^{2n+2}}{2n+2}, \end{aligned}$$

we conclude that

$$\left| \int_0^x \frac{\sin^{(2n+2)}(t)}{(2n+1)!} (x-t)^{2n+1} dt \right| \leq \frac{|x|^{2n+2}}{(2n+2)!}.$$

Similarly, we can show that

$$\left| \int_0^x \frac{\cos^{(2n+1)}(t)}{(2n)!} (x-t)^{2n} dt \right| \leq \frac{|x|^{2n+1}}{(2n+1)!}.$$

These estimates are particularly interesting, because (as proved in Chapter 16) for any $\varepsilon > 0$ we can make

$$\frac{x^n}{n!} < \varepsilon$$

by choosing n large enough (how large n must be will depend on x). This enables us to compute $\sin x$ to any degree of accuracy desired simply by evaluating the proper Taylor polynomial $P_{n,0}(x)$. For example, suppose we wish to compute $\sin 2$ with an error of less than 10^{-4} . Since

$$\sin 2 = P_{2n+1,0}(2) + R, \quad \text{where } |R| \leq \frac{2^{2n+2}}{(2n+2)!}$$

we can use $P_{2n+1,0}(2)$ as our answer, provided that

$$\frac{2^{2n+2}}{(2n+2)!} < 10^{-4}.$$

A number n with this property can be found by a straightforward search—it obviously helps to have a table of values for $n!$ and 2^n (see page 356). In this case it happens that $n = 5$ works, so that

$$\begin{aligned} \sin 2 &= P_{11,0}(2) + R \\ &= 2 - \frac{2^3}{3!} + \frac{2^5}{5!} - \frac{2^7}{7!} + \frac{2^9}{9!} - \frac{2^{11}}{11!} + R, \\ &\quad \text{where } |R| < 10^{-4}. \end{aligned}$$

It is even easier to calculate $\sin 1$ approximately, since

$$\sin 1 = P_{2n+1,0}(1) + R, \quad \text{where } |R| < \frac{1}{(2n+2)!}.$$

To obtain an error less than ε we need only find an n such that

$$\frac{1}{(2n+2)!} < \varepsilon,$$

and this requires only a brief glance at a table of factorials. (Moreover, the individual terms of $P_{2n+1,0}(1)$ will be easier to handle.)

For very small x the estimates will be even easier. For example,

$$\sin \frac{1}{10} = P_{2n+1,0}\left(\frac{1}{10}\right) + R, \quad \text{where } |R| < \frac{1}{10^{2n+2}(2n+2)!}.$$

To obtain $|R| < 10^{-10}$ we can clearly take $n = 4$ (and we could even get away with $n = 3$). These methods are actually used to compute tables of $\sin$ and $\cos$. A high-speed computer can compute $P_{2n+1,0}(x)$ for many different x in almost no time at all.

Estimating the remainder for e^x is only slightly harder. For simplicity assume that $x \geq 0$ (the estimates for $x \leq 0$ are obtained in Problem 9). On

the interval $[0, x]$ the maximum value of e^t is e^x , since $\exp$ is increasing, so

$$\int_0^x \frac{e^t}{n!} (x-t)^n dt \leq \frac{e^x}{n!} \int_0^x (x-t)^n dt = \frac{e^x x^{n+1}}{(n+1)!}.$$

Since we already know that $e < 4$, we have

$$\frac{e^x x^{n+1}}{(n+1)!} < \frac{4^x x^{n+1}}{(n+1)!},$$

which can be made as small as desired by choosing n sufficiently large. How large n must be will depend on x (and the factor 4^x will make things more difficult). Once again, the estimates are easier for small x . If $0 \leq x \leq 1$, then

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + R, \quad \text{where } 0 < R < \frac{4}{(n+1)!}.$$

(The inequality $0 < R$ follows immediately from the integral form for R .) In particular, if $n = 4$, then

$$0 < R < \frac{4}{5!} < \frac{1}{10},$$

so

$$\begin{aligned} e = e^1 &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + R, \quad \text{where } 0 < R < \frac{1}{10} \\ &= \frac{65}{24} + R \\ &= 2 + \frac{17}{24} + R, \end{aligned}$$

which shows that

$$2 < e < 3.$$

(This allows us to improve our estimate of R slightly:

$$0 < R < \frac{3^x x^{n+1}}{(n+1)!}.)$$

By taking $n = 7$ you can compute that the first 3 decimals for e are

$$e = 2.718 \dots$$

(you should check that $n = 7$ does give this degree of accuracy, but it would be cruel to insist that you actually do the computations).

The function $\arctan$ is also important but, as you may recall, an expression for $\arctan^{(k)}(x)$ is hopelessly complicated, so that the integral form of the remainder is useless. On the other hand, our derivation of the Taylor polynomial for $\arctan$ automatically provided a formula for the remainder:

$$\arctan x = x - \frac{x^3}{3} + \cdots + \frac{(-1)^n x^{2n+1}}{2n+1} + \int_0^x \frac{(-1)^{n+1} t^{2n+2}}{1+t^2} dt.$$

As we have already estimated,

$$\left| \int_0^x \frac{(-1)^{n+1} t^{2n+2}}{1+t^2} dt \right| \leq \left| \int_0^x t^{2n+2} dt \right| = \frac{|x|^{2n+3}}{2n+3}.$$

For the moment we will consider only numbers x with $|x| \leq 1$. In this case, the remainder term can clearly be made as small as desired by choosing n sufficiently large. In particular,

$$\arctan 1 = 1 - \frac{1}{3} + \frac{1}{5} - \cdots + \frac{(-1)^n}{2n+1} + R, \quad \text{where } |R| < \frac{1}{2n+3}.$$

With this estimate it is easy to find an n which will make the remainder less than any preassigned number; on the other hand, n will usually have to be so large as to make computations hopelessly long. To obtain a remainder $< 10^{-4}$, for example, we must take $n > (10^4 - 3)/2$. This is really a shame, because $\arctan 1 = \pi/4$, so the Taylor polynomial for $\arctan$ should allow us to compute π . Fortunately, there are some clever tricks which enable us to surmount these difficulties. Since

$$|R_{2n+1,0}(x)| < \frac{|x|^{2n+3}}{2n+3},$$

much smaller n 's will work for only somewhat smaller x 's. The trick for computing π is to express $\arctan 1$ in terms of $\arctan x$ for smaller x ; Problem 5 shows how this can be done in a convenient way.

The Taylor polynomial for the function $f(x) = \log(x+1)$ at $a=1$ is best handled in the same manner as the Taylor polynomial for $\arctan$. Although the integral form of the remainder for f is not hard to write down, it is difficult to estimate. On the other hand, we obtain a simple formula if we begin with the equation

$$\frac{1}{1+t} = 1 - t + t^2 - \cdots + (-1)^{n-1} t^{n-1} + \frac{(-1)^n t^n}{1+t};$$

this implies that

$$\begin{aligned} \log(1+x) &= \int_0^x \frac{1}{1+t} dt = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} \\ &\quad + (-1)^n \int_0^x \frac{t^n}{1+t} dt, \end{aligned}$$

for all $x > -1$. If $x \geq 0$, then

$$\int_0^x \frac{t^n}{1+t} dt \leq \int_0^x t^n dt = \frac{x^{n+1}}{n+1},$$

and there is a slightly more complicated estimate when $-1 < x < 0$ (Problem 10). For this function the remainder term can be made as small as desired by choosing n sufficiently large, provided that $-1 < x \leq 1$.

The behavior of the remainder terms for $\arctan$ and $f(x) = \log(x+1)$ is quite another matter when $|x| > 1$. In this case, the estimates

$$|R_{2n+1,0}(x)| < \frac{|x|^{2n+3}}{2n+3} \quad \text{for } \arctan,$$

$$|R_{n,0}(x)| < \frac{x^{n+1}}{n+1} \quad (x > 0) \text{ for } f,$$

are of no use, because when $|x| > 1$ the bounds x^m/m become large as m becomes large. This predicament is unavoidable, and is not just a deficiency of our estimates. It is easy to get estimates in the other direction which show that the remainders actually do remain large. To obtain such an estimate for $\arctan$, note that if t is in $[0, x]$ (or in $[x, 0]$ if $x < 0$), then

$$1 + t^2 \leq 1 + x^2 \leq 2x^2, \quad \text{if } |x| \geq 1,$$

so

$$\left| \int_0^x \frac{t^{2n+2}}{1+t^2} dt \right| \geq \frac{1}{2x^2} \left| \int_0^x t^{2n+2} dt \right| = \frac{|x|^{2n+1}}{4n+6}.$$

Similarly, if $x > 0$, then for t in $[0, x]$ we have

$$1 + t \leq 1 + x \leq 2x, \quad \text{if } x \geq 1,$$

so

$$\int_0^x \frac{t^n}{t+1} dt \geq \frac{1}{2x} \int_0^x t^n dt = \frac{x^n}{2n+2}.$$

These estimates show that if $|x| > 1$, then the remainder terms become large as n becomes large. In other words, for $|x| > 1$, the Taylor polynomials for $\arctan$ and f are of no use whatsoever in computing $\arctan x$ and $\log(x+1)$. This is no tragedy, because the values of these functions can be found for any x once they are known for all x with $|x| < 1$.

This same situation occurs in a spectacular way for the function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

We have already seen that $f^{(k)}(0) = 0$ for every natural number k . This means that the Taylor polynomial $P_{n,0}$ for f is

$$\begin{aligned} P_{n,0}(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n \\ &= 0. \end{aligned}$$

In other words, the remainder term $R_{n,0}(x)$ always equals $f(x)$, and the Taylor polynomial is useless for computing $f(x)$, except for $x = 0$. Eventually we will be able to offer some explanation for the behavior of this function, which is such a disconcerting illustration of the limitations of Taylor's Theorem.

The word "compute" has been used so often in connection with our estimates for the remainder term, that the significance of Taylor's Theorem might

be misconstrued. It is true that Taylor's Theorem is an almost ideal computational aid (despite its ignominious failure in the previous example), but it has equally important theoretical consequences. Most of these will be developed in succeeding chapters, but two proofs will illustrate some ways in which Taylor's Theorem may be used. The first illustration will be particularly impressive to those who have waded through the proof, in Chapter 16, that π is irrational.

THEOREM 5 e is irrational.

PROOF We know that, for any n ,

$$e = e^1 = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + R_n, \quad \text{where } 0 < R_n < \frac{3}{(n+1)!}.$$

Suppose that e were rational, say $e = a/b$, where a and b are positive integers. Choose $n > b$ and also $n > 3$. Then

$$\frac{a}{b} = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!} + R_n,$$

so

$$\frac{n!a}{b} = n! + n! + \frac{n!}{2!} + \cdots + \frac{n!}{n!} + n!R_n.$$

Every term in this equation other than $n!R_n$ is an integer (the left side is an integer because $n > b$). Consequently, $n!R_n$ must be an integer also. But

$$0 < R_n < \frac{3}{(n+1)!},$$

so

$$0 < n!R_n < \frac{3}{n+1} < \frac{3}{4} < 1,$$

which is impossible for an integer. ■

The second illustration is merely a straightforward demonstration of a fact proved in Chapter 15: If

$$\begin{aligned} f'' + f &= 0, \\ f(0) &= 0, \\ f'(0) &= 0, \end{aligned}$$

then $f = 0$. To prove this, observe first that $f^{(k)}$ exists for every k ; in fact

$$\begin{aligned} f^{(3)} &= (f'')' = -f', \\ f^{(4)} &= (f^{(3)})' = (-f')' = -f'' = f, \\ f^{(5)} &= (f^{(4)})' = f', \\ &\text{etc.} \end{aligned}$$

This shows, not only that all $f^{(k)}$ exist, but also that there are at most 4 different ones: $f, f', -f, -f'$. Since $f(0) = f'(0) = 0$, all $f^{(k)}(0)$ are 0. Now Taylor's Theorem states, for any n , that

$$f(x) = \int_0^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt.$$

Each function $f^{(n+1)}$ is continuous (since $f^{(n+2)}$ exists), so for any particular x there is a number M such that

$$|f^{(n+1)}(t)| \leq M \quad \text{for } 0 \leq t \leq x, \text{ and all } n$$

(we can add the phrase “and all n ” because there are only four different $f^{(k)}$). Thus

$$|f(x)| \leq M \left| \int_0^x \frac{(x-t)^n}{n!} dt \right| = \frac{M|x|^{n+1}}{(n+1)!}.$$

Since this is true for every n , and since $x^n/n!$ can be made as small as desired by choosing n sufficiently large, this shows that $|f(x)| < \epsilon$ for any $\epsilon > 0$; consequently, $f(x) = 0$.

The other uses to which Taylor's Theorem will be put in succeeding chapters are closely related to the computational considerations which have concerned us for much of this chapter. If the remainder term $R_{n,a}(x)$ can be made as small as desired by choosing n sufficiently large, then $f(x)$ can be computed to any degree of accuracy desired by using the polynomials $P_{n,a}(x)$. As we require greater and greater accuracy we must add on more and more terms. If we are willing to add up infinitely many terms (in theory at least!), then we ought to be able to ignore the remainder completely. There should be “infinite sums” like

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots,$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots,$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \quad \text{if } |x| \leq 1,$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{if } -1 < x \leq 1.$$

We are almost completely prepared for this step. Only one obstacle remains—we have never even defined an infinite sum. Chapters 21 and 22 contain the necessary definitions.

PROBLEMS

1. Find the Taylor polynomials (of the indicated degree, and at the indicated point) for the following functions.
 - (i) $f(x) = e^{e^x}$; degree 3, at 0.
 - (ii) $f(x) = e^{\sin x}$; degree 3, at 0.
 - (iii) $\sin$; degree $2n$, at $\frac{\pi}{2}$.
 - (iv) $\cos$; degree $2n$, at π .
 - (v) $\exp$; degree n , at 1.
 - (vi) $\log$; degree n , at 2.
 - (vii) $f(x) = x^5 + x^3 + x$; degree 4, at 0.
 - (viii) $f(x) = x^5 + x^3 + x$; degree 4, at 1.
 - (ix) $f(x) = \frac{1}{1+x^2}$; degree $2n+1$, at 0.
 - (x) $f(x) = \frac{1}{1+x}$; degree n , at 0.
2. Write each of the following polynomials in x as a polynomial in $(x-3)$. (It is only necessary to compute the Taylor polynomial at 3, of the same degree as the original polynomial. Why?)
 - (i) $x^2 - 4x - 9$.
 - (ii) $x^4 - 12x^3 + 44x^2 + 2x + 1$.
 - (iii) x^5 .
 - (iv) $ax^2 + bx + c$.
3. Write down a sum (using Σ notation) which equals each of the following numbers to within the specified accuracy. To minimize needless computation, consult the tables for 2^n and $n!$ on the next page.
 - (i) $\sin 1$; error $< 10^{-17}$.
 - (ii) $\sin 2$; error $< 10^{-12}$.
 - (iii) $\sin \frac{1}{2}$; error $< 10^{-20}$.
 - (iv) e ; error $< 10^{-4}$.
 - (v) e^2 ; error $< 10^{-5}$.

n	2^n	$n!$
1	2	1
2	4	2
3	8	6
4	16	24
5	32	120
6	64	720
7	128	5,040
8	256	40,320
9	512	362,880
10	1,024	3,628,800
11	2,048	39,916,800
12	4,096	479,001,600
13	8,192	6,227,020,800
14	16,384	87,178,291,200
15	32,768	1,307,674,368,000
16	65,536	20,922,789,888,000
17	131,072	355,687,428,096,000
18	262,144	6,402,373,705,728,000
19	524,288	121,645,100,408,832,000
20	1,048,576	2,432,902,008,176,640,000

- *4. This problem is similar to the previous one, except that the errors demanded are so small that the tables cannot be used. You will have to do a little thinking, and in some cases it may be necessary to consult the proof, in Chapter 16, that $x^n/n!$ can be made small by choosing n large—the proof actually provides a method for finding the appropriate n . In the previous problem it was possible to find rather short sums; in fact, it was possible to find the smallest n which makes the estimate of the remainder given by Taylor's Theorem less than the desired error. But in this problem finding *any* specific sum is a moral victory (provided you can demonstrate that the sum works).

- (i) $\sin 1$; error $< 10^{-(10^{10})}$.
- (ii) e ; error $< 10^{-1,000}$.
- (iii) $\sin 10$; error $< 10^{-20}$.
- (iv) e^{10} ; error $< 10^{-30}$.
- (v) $\arctan \frac{1}{10}$; error $< 10^{-(10^{10})}$.

5. (a) Prove, using Problem 15-8, that

$$\frac{\pi}{4} = \arctan \frac{1}{2} + \arctan \frac{1}{3},$$

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}.$$

- (b) Show that $\pi = 3.14159 \dots$ (Every young man should verify a few decimals of π for himself, but the purpose of this exercise is not to set you off on an immense calculation. If the second expression in part (a) is used, the first 5 decimals for π can be computed with remarkably little work.)
6. For every number α , and every nonnegative integer n , we define the "binomial coefficient"

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!}.$$

If α is not an integer, then $\binom{\alpha}{n}$ is never 0, and alternates in sign for $n > \alpha$. Show that the Taylor polynomial of degree n for $f(x) = (1+x)^\alpha$ at 0 is $P_{n,0}(x) = \sum_{k=0}^n \binom{\alpha}{k} x^k$, and that the Cauchy and Lagrange forms

of the remainder are the following:

Cauchy form:

$$\begin{aligned} R_{n,0}(x) &= \frac{\alpha(\alpha-1) \cdots (\alpha-n)}{n!} x(x-t)^n (1+t)^{\alpha-n-1} \\ &= \frac{\alpha(\alpha-1) \cdots (\alpha-n)}{n!} x(1+t)^{\alpha-1} \left(\frac{x-t}{1+t}\right)^n \\ &= (n+1) \binom{\alpha}{n+1} x(1+t)^{\alpha-1} \left(\frac{x-t}{1+t}\right)^n, \quad t \text{ in } [0, x] \text{ or } [x, 0]. \end{aligned}$$

Lagrange form:

$$\begin{aligned} R_{n,0}(x) &= \frac{\alpha(\alpha-1) \cdots (\alpha-n)}{(n+1)!} x^{n+1} (1+t)^{\alpha-n-1} \\ &= \binom{\alpha}{n+1} x^{n+1} (1+t)^{\alpha-n-1}, \quad t \text{ in } [0, x] \text{ or } [x, 0]. \end{aligned}$$

Estimates for these remainder terms are rather difficult to handle, and are postponed to Problem 22-16.

7. Suppose that a_i and b_i are the coefficients in the Taylor polynomials at a of f and g , respectively. In other words, $a_i = f^{(i)}(a)/i!$ and $b_i = g^{(i)}(a)/i!$. Find the coefficients c_i of the Taylor polynomials at a of the following functions, in terms of the a_i 's and b_i 's.
- $f + g$.
 - fg .
 - f' .
 - $h(x) = \int_a^x f(t) dt$.
 - $k(x) = \int_0^x f(t) dt$.

- *8. (a) Prove that the Taylor polynomial of $f(x) = \sin(x^2)$ of degree $4n + 1$ at 0 is

$$x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \cdots + (-1)^n \frac{x^{4n}}{(2n)!}$$

Hint: If P is the Taylor polynomial of degree $2n + 1$ for $\sin$ at 0, then $\sin x = P(x) + R(x)$, where $\lim_{x \rightarrow 0} R(x)/x^{2n+1} = 0$. What does this imply about $\lim_{x \rightarrow 0} R(x^2)/x^{4n+1}$?

- (b) Find $f^{(k)}(0)$ for all k .
 (c) In general, if $f(x) = g(x^n)$, find $f^{(k)}(0)$ in terms of the derivatives of g at 0.
9. Prove that if $x \leq 0$, then

$$\left| \int_0^x \frac{e^t}{n!} (x-t)^n dt \right| \leq \frac{|x|^{n+1}}{(n+1)!}.$$

10. Prove that if $-1 < x \leq 0$, then

$$\left| \int_0^x \frac{t^n}{1+t} dt \right| \leq \frac{|x|^{n+1}}{(1+x)(n+1)}.$$

- *11. (a) Show that if $|g'(x)| \leq M|x-a|^n$ for $|x-a| < \delta$, then $|g(x)| \leq M|x-a|^{n+1}/(n+1)$ for $|x-a| < \delta$.
 (b) Use part (a) to show that if $\lim_{x \rightarrow a} g'(x)/(x-a)^n = 0$, then

$$\lim_{x \rightarrow a} g(x)/(x-a)^{n+1} = 0.$$

- (c) Show that if $g(x) = f(x) - P_{n,a,f}(x)$, then $g'(x) = f'(x) - P_{n-1,a,f'}(x)$.
 (d) Give an inductive proof of Theorem 1, without using l'Hôpital's Rule.
12. Deduce Theorem 1 as a corollary of Taylor's Theorem, with any form of the remainder. (The catch is that it will be necessary to assume one more derivative than in the hypotheses for Theorem 1.)
13. Deduce the Cauchy and Lagrange forms of the remainder from the integral form, using Problems 13-27 and 13-28. There will be the same catch as in Problem 12.
14. (a) Prove that if $f''(a)$ exists, then

$$f''(a) = \lim_{h \rightarrow 0} \frac{f(a+h) + f(a-h) - 2f(a)}{h^2}.$$

Hint: Use the Taylor polynomial of degree 2 with $x = a + h$ and with $x = a - h$.

(b) Let $f(x) = x^2$ for $x \geq 0$, and $-x^2$ for $x \leq 0$. Show that

$$\lim_{h \rightarrow 0} \frac{f(0+h) + f(0-h) - 2f(0)}{h^2}$$

exists, even though $f''(0)$ does not.

15. Use the Taylor polynomial $P_{1,a,f}$, together with the remainder, to prove a weak form of Theorem 2 of the Appendix to Chapter 11: If $f'' > 0$, then the graph of f always lies above the tangent line of f , except at the point of contact.
- *16. Problem 17-32 presented a rather complicated proof that $f = 0$ if $f'' - f = 0$ and $f(0) = f'(0) = 0$. Give another proof, using Taylor's Theorem. (This problem is really a preliminary skirmish before doing battle with the general case in Problem 17, and is meant to convince you that Taylor's Theorem is a good tool for tackling such problems, even though tricks work out more neatly for special cases.)
- **17. Consider a function f which satisfies the differential equation

$$f^{(n)} = \sum_{j=0}^{n-1} a_j f^{(j)},$$

for certain numbers $a_0, \dots, a_{n-1}$. Several special cases have already received detailed treatment, either in the text or in other problems; in particular, we have found all functions satisfying $f' = f$, or $f'' + f = 0$, or $f'' - f = 0$. The trick in Problem 17-31 enables us to find many solutions for such equations, but doesn't say whether these are the only solutions. This requires a *uniqueness* result, which will be supplied by this problem. At the end you will find some (necessarily sketchy) remarks about the general solution.

(a) Derive the following formula for $f^{(n+1)}$ (let us agree that " a_{-1} " will be 0):

$$f^{(n+1)} = \sum_{j=0}^{n-1} (a_{j-1} + a_{n-1}a_j)f^{(j)}.$$

(b) Deduce a formula for $f^{(n+2)}$.

The formula in part (b) is not going to be used; it was inserted only to convince you that a general formula for $f^{(n+k)}$ is out of the question. On the other hand, as part (c) shows, it is not very hard to obtain estimates on the size of $f^{(n+k)}(x)$.

(c) Let $N = \max(1, |a_0|, \dots, |a_{n-1}|)$. Then $|a_{j-1} + a_{n-1}a_j| \leq 2N^2$; this means that

$$f^{(n+1)} = \sum_{j=0}^{n-1} b_j f^{(j)}, \quad \text{where } |b_j| \leq 2N^2.$$

Show that

$$f^{(n+2)} = \sum_{j=0}^{n-1} b_j^2 f^{(j)}, \quad \text{where } |b_j^2| \leq 4N^3,$$

and, more generally,

$$f^{(n+k)} = \sum_{j=0}^{n-1} b_j^k f^{(j)}, \quad \text{where } |b_j^k| \leq 2^k N^{k+1}$$

- (d) Conclude from part (c) that, for any particular number x , there is a number M such that

$$|f^{(n+k)}(x)| \leq M \cdot 2^k N^{k+1} \quad \text{for all } k.$$

- (e) Now suppose that $f(0) = f'(0) = \cdots = f^{(n-1)}(0) = 0$. Show that

$$\begin{aligned} |f(x)| &\leq \frac{M \cdot 2^k N^{k+1} |x|^{n+k+1}}{(n+k+1)!} \\ &\leq \frac{M \cdot |2Nx|^{n+k+1}}{(n+k+1)!}, \end{aligned}$$

and conclude that $f = 0$.

- (f) Show that if f_1 and f_2 are both solutions of the differential equation

$$f^{(n)} = \sum_{j=0}^{n-1} a_j f^{(j)},$$

and $f_1^{(j)}(0) = f_2^{(j)}(0)$ for $0 \leq j \leq n-1$, then $f_1 = f_2$.

In other words, the solutions of this differential equation are determined by the “initial conditions” (the values $f^{(j)}(0)$ for $0 \leq j \leq n-1$). This means that we can find *all* solutions once we can find enough solutions to obtain any given set of initial conditions. If the equation

$$x^n - a_{n-1}x^{n-1} - \cdots - a_0 = 0$$

has n distinct roots $\alpha_1, \dots, \alpha_n$, then any function of the form

$$f(x) = c_1 e^{\alpha_1 x} + \cdots + c_n e^{\alpha_n x}$$

is a solution, and

$$\begin{aligned} f(0) &= c_1 + \cdots + c_n, \\ f'(0) &= \alpha_1 c_1 + \cdots + \alpha_n c_n, \\ &\vdots \\ f^{(n-1)}(0) &= \alpha_1^{n-1} c_1 + \cdots + \alpha_n^{n-1} c_n. \end{aligned}$$

As a matter of fact, every solution is of this form, because we can obtain any set of numbers on the left side by choosing the c 's

properly, but we will not try to prove this last assertion. (It is a purely algebraic fact, which you can easily check for $n = 2$ or 3 .) These remarks are also true if some of the roots are multiple roots, and even in the more general situation considered in Chapter 26.

- *18.** (a) Let $f(x) = x^4 \sin 1/x^2$ for $x \neq 0$, and $f(0) = 0$. Show that $f = 0$ up to order 2 at 0, even though $f''(0)$ does not exist.

This example is slightly more complex, but also slightly more impressive, than the example in the text, because both $f'(a)$ and $f''(a)$ exist for $a \neq 0$. Thus, for each number a there is another number $m(a)$ such that

$$(*) \quad f(x) = f(a) + f'(a)(x - a) + \frac{m(a)}{2} (x - a)^2 + R_a(x),$$

$$\text{where } \lim_{x \rightarrow a} \frac{R_a(x)}{(x - a)^2} = 0;$$

namely, $m(a) = f''(a)$ for $a \neq 0$, and $m(0) = 0$. Notice that the function m defined in this way is not continuous.

- (b) Let f be a differentiable function. Suppose that there is a function m such that $(*)$ holds for all numbers a , and that m is continuous. Prove that $f''(a)$ exists for all a and equals $m(a)$. Hint: Write $(*)$ for $x = a + h$, $a = a$, and also for $x = a$, $a = a + h$.

- *** (c) Suppose that there is a continuous function m such that

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2} (x - a)^2 + \frac{m(a)}{3!} (x - a)^3 + R_a(x),$$

$$\text{where } \lim_{x \rightarrow a} \frac{R_a(x)}{(x - a)^3} = 0.$$

Does it follow that $f'''(a) = m(a)$?

The irrationality of e was so easy to prove that in this optional chapter we will attempt a more difficult feat, and prove that the number e is not merely irrational, but actually much worse. Just how a number might be even worse than irrational is suggested by a slight rewording of definitions. A number x is irrational if it is not possible to write $x = a/b$ for any integers a and b , with $b \neq 0$. This is the same as saying that x does not satisfy any equation

$$bx - a = 0$$

for integers a and b , except for $a = 0, b = 0$. Viewed in this light, the irrationality of $\sqrt{2}$ does not seem to be such a terrible deficiency; rather, it appears that $\sqrt{2}$ just barely manages to be irrational—although $\sqrt{2}$ is not the solution of an equation

$$a_1x + a_0 = 0,$$

it *is* the solution of the equation

$$x^2 - 2 = 0,$$

of one higher degree. Problem 2-17 shows how to produce many irrational numbers x which satisfy higher-degree equations

$$a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0,$$

where the a_i are integers and $a_0 \neq 0$ (this condition rules out the possibility that all $a_i = 0$). A number which satisfies an “algebraic” equation of this sort is called an **algebraic number**, and practically every number we have ever encountered is defined in terms of solutions of algebraic equations (π and e are the great exceptions in our limited mathematical experience). All roots, such as

$$\sqrt{2}, \sqrt[10]{3}, \sqrt[4]{7},$$

are clearly algebraic numbers, and even complicated combinations, like

$$\sqrt[3]{3 + \sqrt{5} + \sqrt[4]{1 + \sqrt{2} + \sqrt[5]{6}}}$$

are algebraic (although we will not try to prove this). Numbers which cannot be obtained by the process of solving algebraic equations are called **transcendental**; the main result of this chapter states that e is a number of this anomalous sort.

The proof that e is transcendental is well within our grasp, and was theoretically possible even before Chapter 19. Nevertheless, with the inclusion of this proof, we can justifiably classify ourselves as something more than novices in the study of higher mathematics; while many irrationality proofs depend

only on elementary properties of numbers, the proof that a number is transcendental usually involves some really high-powered mathematics. Even the dates connected with the transcendence of e are impressively recent—the first proof that e is transcendental, due to Hermite, dates from 1873. The proof that we will give is a simplification, due to Hilbert.

Before tackling the proof itself, it is a good idea to map out the strategy, which depends on an idea used even in the proof that e is irrational. Two features of the expression

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + R_n$$

were important for the proof that e is irrational: On the one hand, the number

$$1 + \frac{1}{1!} + \cdots + \frac{1}{n!}$$

can be written as a fraction p/q with $q \leq n!$ (so that $n!(p/q)$ is an integer); on the other hand, $0 < R_n < 3/(n+1)!$ (so $n!R_n$ is not an integer). These two facts show that e can be approximated particularly well by rational numbers. Of course, every number x can be approximated arbitrarily closely by rational numbers—if $\varepsilon > 0$ there is a rational number r with $|x - r| < \varepsilon$; the catch, however, is that it may be necessary to allow a very large denominator for r , as large as $1/\varepsilon$ perhaps. For e we are assured that this is not the case: there is a fraction p/q within $3/(n+1)!$ of e , whose denominator q is at most $n!$. If you look carefully at the proof that e is irrational, you will see that only this fact about e is ever used. The number e is by no means unique in this respect: generally speaking, the *better* a number can be approximated by rational numbers, the *worse* it is (some evidence for this assertion is presented in Problem 3). The proof that e is transcendental depends on a natural extension of this idea: not only e , but any finite number of powers $e, e^2, \dots, e^n$, can be simultaneously approximated especially well by rational numbers. In our proof we will begin by assuming that e is algebraic, so that

$$(*) \quad a_n e^n + \cdots + a_1 e + a_0 = 0, \quad a_0 \neq 0$$

for some integers $a_0, \dots, a_n$. In order to reach a contradiction we will then find certain integers $M, M_1, \dots, M_n$ and certain “small” numbers $\epsilon_1, \dots, \epsilon_n$ such that

$$\begin{aligned} e^1 &= \frac{M_1 + \epsilon_1}{M}, \\ e^2 &= \frac{M_2 + \epsilon_2}{M}, \\ &\vdots \\ e^n &= \frac{M_n + \epsilon_n}{M}. \end{aligned}$$

Just how small the ϵ 's must be will appear when these expressions are substituted into the assumed equation (*). After multiplying through by M we obtain

$$[a_0M + a_1M_1 + \cdots + a_nM_n] + [\epsilon_1a_1 + \cdots + \epsilon_na_n] = 0.$$

The first term in brackets is an integer, and we will choose the M 's so that it will necessarily be a *nonzero* integer. We will also manage to find ϵ 's so small that

$$|\epsilon_1a_1 + \cdots + \epsilon_na_n| < \frac{1}{2};$$

this will lead to the desired contradiction—the sum of a nonzero integer and a number of absolute value less than $\frac{1}{2}$ cannot be zero!

As a basic strategy this is all very reasonable and quite straightforward. The remarkable part of the proof will be the way that the M 's and ϵ 's are defined. In order to read the proof you will need to know about the gamma function! (This function was introduced in Problem 18-25.)

THEOREM 1 e is transcendental.

PROOF Suppose there were integers $a_0, \dots, a_n$, with $a_0 \neq 0$, such that

$$(*) \quad a_ne^n + a_{n-1}e^{n-1} + \cdots + a_0 = 0.$$

Define numbers $M, M_1, \dots, M_n$ and $\epsilon_1, \dots, \epsilon_n$ as follows.

$$\begin{aligned} M &= \int_0^\infty \frac{x^{p-1}[(x-1) \cdots (x-n)]^pe^{-x}}{(p-1)!} dx, \\ M_k &= e^k \int_k^\infty \frac{x^{p-1}[(x-1) \cdots (x-n)]^pe^{-x}}{(p-1)!} dx, \\ \epsilon_k &= e^k \int_0^k \frac{x^{p-1}[(x-1) \cdots (x-n)]^pe^{-x}}{(p-1)!} dx. \end{aligned}$$

The unspecified number p represents a prime number* which we will choose later. Despite the forbidding aspect of these three expressions, with a little work they will appear much more reasonable. We concentrate on M first. If the expression in brackets,

$$[(x-1) \cdots (x-n)],$$

is actually multiplied out, we obtain a polynomial

$$x^n + \cdots \pm n!$$

* The term "prime number" was defined in Problem 2-16. An important fact about prime numbers will be used in the proof, although it is not proved in this book: If p is a prime number which does not divide the integer a , and which does not divide the integer b , then p also does not divide ab . The Suggested Reading mentions references for this theorem (which is crucial in proving that the factorization of an integer into primes is unique). We will also use the result of Problem 2-16(d), that there are infinitely many primes—the reader is asked to determine at precisely which points this information is required.

with integer coefficients. When raised to the p th power this becomes an even more complicated polynomial

$$x^{np} + \cdots \pm (n!)^p.$$

Thus M can be written in the form

$$M = \sum_{\alpha=0}^{np} \frac{1}{(p-1)!} C_{\alpha} \int_0^{\infty} x^{p-1+\alpha} e^{-x} dx,$$

where the C_{α} are certain integers, and $C_0 = \pm (n!)^p$. But

$$\int_0^{\infty} x^k e^{-x} dx = k!.$$

Thus

$$M = \sum_{\alpha=0}^{np} C_{\alpha} \frac{(p-1+\alpha)!}{(p-1)!}.$$

Now, for $\alpha = 0$ we obtain the term

$$\pm (n!)^p \frac{(p-1)!}{(p-1)!} = \pm (n!)^p.$$

We will now consider only primes $p > n$; then this term is an integer which is *not* divisible by p . On the other hand, if $\alpha > 0$, then

$$C_{\alpha} \frac{(p-1+\alpha)!}{(p-1)!} = C_{\alpha} (p+\alpha-1)(p+\alpha-2) \cdots p,$$

which *is* divisible by p . Therefore M itself is an integer which is *not* divisible by p .

Now consider M_k . We have

$$\begin{aligned} M_k &= e^k \int_k^{\infty} \frac{x^{p-1} [(x-1) \cdots (x-n)]^p e^{-x}}{(p-1)!} dx \\ &= \int_k^{\infty} \frac{x^{p-1} [(x-1) \cdots (x-n)]^p e^{-(x-k)}}{(p-1)!} dx. \end{aligned}$$

This can be transformed into an expression looking very much like M by the substitution

$$\begin{aligned} u &= x - k, \\ du &= dx. \end{aligned}$$

The limits of integration are changed to 0 and ∞ , and

$$M_k = \int_0^{\infty} \frac{(u+k)^{p-1} [(u+k-1) \cdots (u+k-n)]^p e^{-u}}{(p-1)!} du.$$

There is one very significant difference between this expression and that for M . The term in brackets contains the factor u in the k th place. Thus the p th power contains the factor u^p . This means that the entire expression

$$(u+k)^{p-1} [(u+k-1) \cdots (u+k-n)]^p$$

is a polynomial with integer coefficients, every term of which has degree at least p . Thus

$$M_k = \sum_{\alpha=1}^{np} \frac{1}{(p-1)!} D_{\alpha} \int_0^{\infty} u^{p-1+\alpha} e^{-u} du = \sum_{\alpha=1}^{np} D_{\alpha} \frac{(p-1+\alpha)!}{(p-1)!},$$

where the D_{α} are certain integers. Notice that the summation begins with $\alpha = 1$; in this case every term in the sum is divisible by p . Thus each M_k is an integer which is divisible by p .

Now it is clear that

$$e^k = \frac{M_k + \epsilon_k}{M}, \quad k = 1, \dots, n.$$

Substituting into (*) and multiplying by M we obtain

$$[a_0 M + a_1 M_1 + \dots + a_n M_n] + [a_1 \epsilon_1 + \dots + a_n \epsilon_n] = 0.$$

In addition to requiring that $p > n$ let us also stipulate that $p > |a_0|$. This means that both M and a_0 are not divisible by p , so $a_0 M$ is also not divisible by p . Since each M_k is divisible by p , it follows that

$$a_0 M + a_1 M_1 + \dots + a_n M_n$$

is not divisible by p . In particular it is a nonzero integer.

In order to obtain a contradiction to the assumed equation (*), and thereby prove that e is transcendental, it is only necessary to show that

$$|a_1 \epsilon_1 + \dots + a_n \epsilon_n|$$

can be made as small as desired, by choosing p large enough; it is clearly sufficient to show that each $|\epsilon_k|$ can be made as small as desired. This requires nothing more than some simple estimates; for the remainder of the argument remember that n is a certain fixed number (the degree of the assumed polynomial equation (*)). To begin with, if $1 \leq k \leq n$, then

$$\begin{aligned} |\epsilon_k| &\leq e^k \int_0^k \frac{|x^{p-1}[(x-1) \cdots (x-n)]^p| e^{-x}}{(p-1)!} dx \\ &\leq e^n \int_0^n \frac{n^{p-1}|[(x-1) \cdots (x-n)]^p| e^{-x}}{(p-1)!} dx. \end{aligned}$$

Now let A be the maximum of $|(x-1) \cdots (x-n)|$ for x in $[0, n]$. Then

$$\begin{aligned} |\epsilon_k| &\leq \frac{e^n n^{p-1} A^p}{(p-1)!} \int_0^n e^{-x} dx \\ &\leq \frac{e^n n^{p-1} A^p}{(p-1)!} \int_0^{\infty} e^{-x} dx \\ &= \frac{e^n n^{p-1} A^p}{(p-1)!} \\ &\leq \frac{e^n n^p A^p}{(p-1)!} = \frac{e^n (nA)^p}{(p-1)!}. \end{aligned}$$

But n and A are fixed; thus $(nA)^p/(p-1)!$ can be made as small as desired by making p sufficiently large. ■

This proof, like the proof that π is irrational, deserves some philosophic afterthoughts. At first sight, the argument seems quite “advanced”—after all, we use integrals, and integrals from 0 to ∞ at that. Actually, as many mathematicians have observed, integrals can be eliminated from the argument completely; the only integrals essential to the proof are of the form

$$\int_0^\infty x^k e^{-x} dx,$$

for integral k , and these integrals can be replaced by $k!$ whenever they occur. Thus M , for example, could have been defined initially as

$$M = \sum_{\alpha=0}^{np} C_\alpha \frac{(p-1+\alpha)!}{(p-1)!},$$

where C_α are the coefficients of the polynomial

$$[(x-1) \cdots (x-n)]^p.$$

If this idea is developed consistently, one obtains a “completely elementary” proof that e is transcendental, depending only on the fact that

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

Unfortunately, this “elementary” proof is harder to understand than the original one—the whole structure of the proof must be hidden just to eliminate a few integral signs! This situation is by no means peculiar to this specific theorem—“elementary” arguments are frequently more difficult than “advanced” ones. Our proof that π is irrational is a case in point. You probably remember nothing about this proof except that it involves quite a few complicated functions. There is actually a more advanced, but much more conceptual proof, which shows that π is *transcendental*, a fact which is of great historical, as well as intrinsic, interest. One of the classical problems of Greek mathematics was to construct, with compass and straightedge alone, a square whose area is that of a circle of radius 1. This requires the construction of a line segment whose length is $\sqrt{\pi}$, which can be accomplished if a line segment of length π is constructible. The Greeks were totally unable to decide whether such a line segment could be constructed, and even the full resources of modern mathematics were unable to settle this question until 1882. In that year Lindemann proved that π is transcendental; since the length of any segment that can be constructed with straightedge and compass can be written in terms of $+$, $\cdot$, $-$, $\div$, and $\sqrt{}$, and is therefore algebraic, this proves that a line segment of length π cannot be constructed.

The proof that π is transcendental requires a sizable amount of mathematics

which is too advanced to be reached in this book. Nevertheless, the proof is not much more difficult than the proof that e is transcendental. In fact, the proof for π is practically the same as the proof for e . This last statement should certainly surprise you. The proof that e is transcendental seems to depend so thoroughly on particular properties of e that it is almost inconceivable how any modifications could ever be used for π ; after all, what does e have to do with π ? Just wait and see!

PROBLEMS

1. (a) Prove that if $\alpha > 0$ is algebraic, then $\sqrt{\alpha}$ is algebraic.
 (b) Prove that if α is algebraic and r is rational, then $\alpha + r$ and αr are algebraic.

Part (b) can actually be strengthened considerably: the sum, product, and quotient of algebraic numbers is algebraic. This fact is too difficult for us to prove here, but some special cases can be examined:

2. Prove that $\sqrt{2} + \sqrt{3}$ and $\sqrt{2}(1 + \sqrt{3})$ are algebraic, by actually finding algebraic equations which they satisfy. (You will need equations of degree 4.)
- *3. (a) Let α be an algebraic number which is not rational. Suppose that α satisfies the polynomial equation

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = 0,$$

and that no polynomial function of lower degree has this property. Show that $f(p/q) \neq 0$ for any rational number p/q . Hint: Use Problem 3-7(b).

- (b) Now show that $|f(p/q)| \geq 1/q^n$ for all rational numbers p/q with $q > 0$. Hint: Write $f(p/q)$ as a fraction over the common denominator q^n .
- (c) Let $M = \sup\{|f'(x)|: |x - \alpha| < 1\}$. Use the Mean Value Theorem to prove that if p/q is a rational number with $|\alpha - p/q| < 1$, then $|\alpha - p/q| > 1/Mq^n$. (It follows that for $c = \max(1, 1/M)$ we have $|\alpha - p/q| > c/q^n$ for all rational p/q .)

- *4. Let

$$\alpha = 0.1100100000000000000000001000 \dots,$$

where the 1's occur in the $n!$ place, for each n . Use Problem 3 to prove that α is transcendental. (For each n , show that α is not the root of an equation of degree n .)

Although Problem 4 mentions only one specific transcendental number, it should be clear that one can easily construct infinitely many other numbers α which do not satisfy $|\alpha - p/q| > c/q^n$ for any c and n . Such numbers were first considered by Liouville (1809–1882), and the inequality in Problem 3 is

often called Liouville's inequality. None of the transcendental numbers constructed in this way happens to be particularly interesting, but for a long time Liouville's transcendental numbers were the only ones known. This situation was changed quite radically by the work of Cantor (1845–1918), who showed, without exhibiting a single transcendental number, that *most* numbers are transcendental. The next two problems provide an introduction to the ideas that allow us to make sense of such statements. The basic definition with which we must work is the following: A set A is called **countable** if its elements can be arranged in a sequence

$$a_1, a_2, a_3, a_4, \dots$$

The obvious example (in fact, more or less the Platonic ideal of) a countable set is $\mathbf{N}$, the set of natural numbers; clearly the set of even natural numbers is also countable:

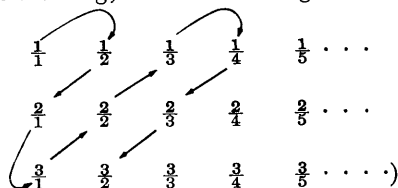
$$2, 4, 6, 8, \dots$$

It is a little more surprising to learn that $\mathbf{Z}$, the set of all integers (positive, negative, and 0) is also countable, but seeing is believing:

$$0, 1, -1, 2, -2, 3, -3, \dots$$

The next two problems, which outline the basic features of countable sets, are really a series of examples to show that (1) a lot more sets are countable than one might think and (2) nevertheless, some sets are not countable.

- *5. (a) Show that if A and B are countable, then so is $A \cup B = \{x: x \text{ is in } A \text{ or } x \text{ is in } B\}$. Hint: Use the same trick that worked for $\mathbf{Z}$.
 (b) Show that the set of positive rational numbers is countable. (This is really quite startling; use the following look-and-see proof:



- (c) Show that the set of all pairs (m, n) of integers is countable. (This is practically the same as part (b).)
 (d) If $A_1, A_2, A_3, \dots$ are each countable, prove that

$$A_1 \cup A_2 \cup A_3 \cup \dots$$

is also countable. (Again use the same trick as in part (b).)

- (e) Prove that the set of all triples (l, m, n) of integers is countable. (A triple (l, m, n) can be described by a pair (l, m) and a number n .)
 (f) Prove that the set of all n -tuples $(a_1, a_2, \dots, a_n)$ are countable. (If you have done part (e), you can do this, using induction.)

- (g) Prove that the set of all roots of polynomial functions of degree n is countable. (Part (f) shows that the set of all polynomial functions of degree n can be arranged in a sequence, and each of these functions has at most n roots.)
- (h) Now use parts (d) and (g) to prove that the set of all algebraic numbers is countable.
- *6. Since so many sets turn out to be countable, it is important to note that the set of all real numbers between 0 and 1 is *not* countable. In other words, there is no way of listing all these real numbers in a sequence

$$\begin{aligned}\alpha_1 &= 0.a_1^1 a_2^1 a_3^1 a_4^1 \cdot \cdot \cdot \\ \alpha_2 &= 0.a_1^2 a_2^2 a_3^2 a_4^2 \cdot \cdot \cdot \\ \alpha_3 &= 0.a_1^3 a_2^3 a_3^3 a_4^3 \cdot \cdot \cdot \\ &\vdots\end{aligned}$$

(decimal notation is being used on the right). To prove that this is so, suppose such a list were possible and consider the decimal

$$0.\bar{a}_1^1 \bar{a}_2^2 \bar{a}_3^3 \bar{a}_4^4 \cdot \cdot \cdot ,$$

where $\bar{a}_n^n = 5$ if $a_n^n \neq 5$ and $\bar{a}_n^n = 6$ if $a_n^n = 5$. Show that this number cannot possibly be in the list, thus obtaining a contradiction.

Problems 5 and 6 can be summed up as follows. The set of algebraic numbers is countable. If the set of transcendental numbers were also countable, then the set of all real numbers would be countable, by Problem 5(a), and consequently the set of real numbers between 0 and 1 would be countable. But this is false. Thus, the set of algebraic numbers is countable and the set of transcendental numbers is not ("there are more transcendental numbers than algebraic numbers"). The remaining two problems illustrate further how important it can be to distinguish between sets which are countable and sets which are not.

- *7. Let f be a nondecreasing function on $[0, 1]$. Recall (Problem 8-8) that $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ both exist.
- (a) For any $\varepsilon > 0$ prove that there are only finitely many numbers a in $[0, 1]$ with $\lim_{x \rightarrow a^+} f(x) - \lim_{x \rightarrow a^-} f(x) > \varepsilon$. Hint: There are, in fact, at most $[f(1) - f(0)]/\varepsilon$ of them.
- (b) Prove that the set of points at which f is discontinuous is countable. Hint: If $\lim_{x \rightarrow a^+} f(x) - \lim_{x \rightarrow a^-} f(x) > 0$, then it is $> 1/n$ for some natural number n .

This problem shows that a nondecreasing function is automatically continuous at most points. For differentiability the situation is more difficult to analyze and also more interesting. A nondecreasing func-

tion can fail to be differentiable at a set of points which is not countable, but it is still true that nondecreasing functions are differentiable at most points (in a different sense of the word "most"). Reference [33] of the Suggested Reading gives a beautiful proof, using the Rising Sun Lemma of Problem 8-20. For those who have done Problem 9 of the Appendix to Chapter 11, it is possible to provide at least one application to differentiability of the ideas already developed in this problem set: If f is convex, then f is differentiable except at those points where its right-hand derivative f_+' is discontinuous; but the function f_+' is increasing, so a convex function is automatically differentiable except at a countable set of points.

- *8. (a) Problem 11-50 showed that if every point is a local maximum point for a *continuous* function f , then f is a constant function. Suppose now that the hypothesis of continuity is dropped. Prove that f takes on only a countable set of values. Hint: For each x choose *rational* numbers a_x and b_x such that $a_x < x < b_x$ and x is a maximum point for f on (a_x, b_x) . Then every value $f(x)$ is the maximum value of f on some interval (a_x, b_x) . How many such intervals are there?
- (b) Now deduce Problem 11-50 as a corollary of part (a).
- (c) Suppose that every point of f is either a local maximum or a local minimum point for f . Prove that f takes on only a countable set of values. (This is a minor variant of part (a).)
- (d) Now suppose that f is continuous, and that every point is either a local maximum or a local minimum point for f . Show that f is a constant function. (Although this *statement* is only a minor variant of Problem 11-50, the only *proof* I know uses part (c), and thus relies on very different ideas than the ones in Chapter 11.)

CHAPTER 21 INFINITE SEQUENCES

The idea of an infinite sequence is so natural a concept that it is tempting to dispense with a definition altogether. One frequently writes simply “an infinite sequence

$$a_1, a_2, a_3, a_4, a_5, \dots,$$

the three dots indicating that the numbers a_i continue to the right “forever.” A rigorous definition of an infinite sequence is not hard to formulate, however; the important point about an infinite sequence is that for each natural number, n , there is a real number a_n . This sort of correspondence is precisely what functions are meant to formalize.

DEFINITION

An **infinite sequence** of real numbers is a function whose domain is $\mathbf{N}$.

From the point of view of this definition, a sequence should be designated by a single letter like a , and particular values by

$$a(1), a(2), a(3), \dots,$$

but the subscript notation

$$a_1, a_2, a_3, \dots$$

is almost always used instead, and the sequence itself is usually denoted by a symbol like $\{a_n\}$. Thus $\{n\}$, $\{(-1)^n\}$, and $\{1/n\}$ denote the sequences α , β , and γ defined by

$$\begin{aligned}\alpha_n &= n, \\ \beta_n &= (-1)^n, \\ \gamma_n &= \frac{1}{n}.\end{aligned}$$

A sequence, like any function, can be graphed (Figure 1) but the graph is usually rather unrevealing, since most of the function cannot be fit on the page.

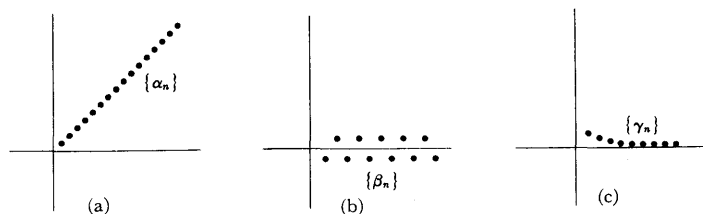


FIGURE 1

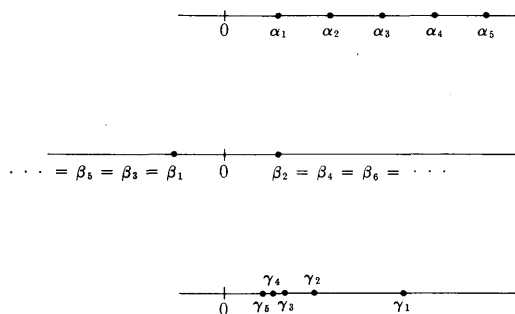


FIGURE 2

A more convenient representation of a sequence is obtained by simply labeling the points $a_1, a_2, a_3, \dots$ on a line (Figure 2). This sort of picture shows where the sequence “is going.” The sequence $\{\alpha_n\}$ “goes out to infinity,” the sequence $\{\beta_n\}$ “jumps back and forth between -1 and 1 ,” and the sequence $\{\gamma_n\}$ “converges to 0 .” Of the three phrases in quotation marks, the last is the crucial concept associated with sequences, and will be defined precisely (the definition is illustrated in Figure 3).

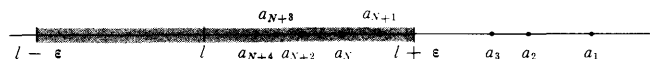


FIGURE 3

DEFINITION

A sequence $\{a_n\}$ **converges to l** (in symbols $\lim_{n \rightarrow \infty} a_n = l$) if for every $\epsilon > 0$ there is a natural number N such that, for all natural numbers n ,

if $n > N$, then $|a_n - l| < \epsilon$.

In addition to the terminology introduced in this definition, we sometimes say that the sequence $\{a_n\}$ **approaches l** or has the **limit l** . A sequence $\{a_n\}$ is said to **converge** if it converges to l for some l , and to **diverge** if it does not converge.

To show that the sequence $\{\gamma_n\}$ converges to 0 , it suffices to observe the following. If $\epsilon > 0$, there is a natural number N such that $1/N < \epsilon$. Then, if $n > N$ we have

$$\gamma_n = \frac{1}{n} < \frac{1}{N} < \epsilon, \quad \text{so } |\gamma_n - 0| < \epsilon.$$

The limit

$$\lim_{n \rightarrow \infty} \sqrt{n+1} - \sqrt{n} = 0$$

will probably seem reasonable after a little reflection (it just says that $\sqrt{n+1}$ is practically the same as $\sqrt{n}$ for large n), but a mathematical proof might not

be so obvious. To estimate $\sqrt{n+1} - \sqrt{n}$ we can use an algebraic trick:

$$\begin{aligned}\sqrt{n+1} - \sqrt{n} &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{n+1-n}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{1}{\sqrt{n+1} + \sqrt{n}}.\end{aligned}$$

It is also possible to estimate $\sqrt{n+1} - \sqrt{n}$ by applying the Mean Value Theorem to the function $f(x) = \sqrt{x}$ on the interval $[n, n+1]$. We obtain

$$\begin{aligned}\frac{\sqrt{n+1} - \sqrt{n}}{1} &= f'(x) \\ &= \frac{1}{2\sqrt{x}}, \quad \text{for some } x \text{ in } (n, n+1) \\ &< \frac{1}{2\sqrt{n}}.\end{aligned}$$

Either of these estimates may be used to prove the above limit; the detailed proof is left to you, as a simple but valuable exercise.

The limit

$$\lim_{n \rightarrow \infty} \frac{3n^3 + 7n^2 + 1}{4n^3 - 8n + 63} = \frac{3}{4}$$

should also seem reasonable, because the terms involving n^3 are the most important when n is large. If you remember the proof of Theorem 7-9 you will be able to guess the trick that translates this idea into a proof—dividing top and bottom by n^3 yields

$$\frac{3n^3 + 7n^2 + 1}{4n^3 - 8n + 63} = \frac{3 + \frac{7}{n} + \frac{1}{n^3}}{4 - \frac{8}{n^2} + \frac{63}{n^3}}.$$

Using this expression, the proof of the above limit is not difficult, especially if one uses the following facts:

If $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ both exist, then

$$\begin{aligned}\lim_{n \rightarrow \infty} (a_n + b_n) &= \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n, \\ \lim_{n \rightarrow \infty} (a_n \cdot b_n) &= \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n;\end{aligned}$$

moreover, if $\lim_{n \rightarrow \infty} b_n \neq 0$, then $b_n \neq 0$ for all n greater than some N , and

$$\lim_{n \rightarrow \infty} a_n/b_n = \lim_{n \rightarrow \infty} a_n / \lim_{n \rightarrow \infty} b_n.$$

(If we wanted to be utterly precise, the third statement would have to be even more complicated. As it stands, we are considering the limit of the sequence $\{c_n\} = \{a_n/b_n\}$, where the numbers c_n might not even be defined for certain $n < N$. This doesn't really matter—we could define c_n any way we liked for such n —because the limit of a sequence is not changed if we change the sequence at a finite number of points.)

Although these facts are very useful, we will not bother stating them as a theorem—you should have no difficulty proving these results for yourself, because the definition of $\lim_{n \rightarrow \infty} a_n = l$ is so similar to previous definitions of limits, especially $\lim_{x \rightarrow \infty} f(x) = l$.

The similarity between the definitions of $\lim_{n \rightarrow \infty} a_n = l$ and $\lim_{x \rightarrow \infty} f(x) = l$ is actually closer than mere analogy; it is possible to define the first in terms of the second. If f is the function whose graph (Figure 4) consists of line segments

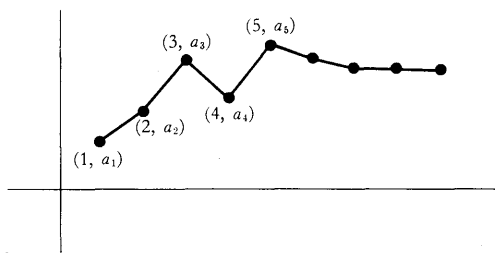


FIGURE 4

joining the points in the graph of the sequence $\{a_n\}$, so that

$$f(x) = (a_{n+1} - a_n)(x - n) + a_n, \quad n \leq x \leq n + 1,$$

then

$$\lim_{n \rightarrow \infty} a_n = l \quad \text{if and only if} \quad \lim_{x \rightarrow \infty} f(x) = l.$$

This observation is frequently very useful. For example, suppose that $0 < a < 1$. Then

$$\lim_{n \rightarrow \infty} a^n = 0.$$

To prove this we note that

$$\begin{aligned} \lim_{x \rightarrow \infty} a^x &= \lim_{x \rightarrow \infty} e^{x \log a} \\ &= 0, \end{aligned}$$

since $\log a < 0$, so that $x \log a$ is negative and large in absolute value for large x . Notice that we actually have

$$\lim_{n \rightarrow \infty} a^n = 0 \quad \text{if } |a| < 1;$$

for if $a < 0$ we can write

$$\lim_{n \rightarrow \infty} a^n = \lim_{n \rightarrow \infty} (-1)^n |a|^n = 0.$$

The behavior of the logarithm function also shows that if $a > 1$, then a^n becomes arbitrarily large as n becomes large. This assertion is often written

$$\lim_{n \rightarrow \infty} a^n = \infty, \quad a > 1,$$

and it is sometimes even said that $\{a^n\}$ approaches ∞ . We also write equations like

$$\lim_{n \rightarrow \infty} -a^n = -\infty,$$

and say that $\{-a^n\}$ approaches $-\infty$. Notice, however, that if $a < -1$, then $\lim_{n \rightarrow \infty} a^n$ does not exist, even in this extended sense.

Despite this connection with a familiar concept, it is more important to picture convergence in terms of the picture of a sequence as points on a line (Figure 3). There is another connection between limits of functions and limits of sequences which is related to *this* picture. This connection is somewhat less obvious, but considerably more interesting, than the one previously mentioned—instead of defining limits of sequences in terms of limits of functions, it is possible to reverse the procedure.

THEOREM 1 Let f be a function defined in an open interval containing c , except perhaps at c itself, with

$$\lim_{x \rightarrow c} f(x) = l.$$

Suppose that $\{a_n\}$ is a sequence such that

- (1) each a_n is in the domain of f ,
- (2) each $a_n \neq c$,
- (3) $\lim_{n \rightarrow \infty} a_n = c$.

Then the sequence $\{f(a_n)\}$ satisfies

$$\lim_{n \rightarrow \infty} f(a_n) = l.$$

Conversely, if this is true for every sequence $\{a_n\}$ satisfying the above conditions, then $\lim_{x \rightarrow c} f(x) = l$.

PROOF Suppose first that $\lim_{x \rightarrow c} f(x) = l$. Then for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - c| < \delta, \text{ then } |f(x) - l| < \varepsilon.$$

If the sequence $\{a_n\}$ satisfies $\lim_{n \rightarrow \infty} a_n = c$, then (Figure 3) there is a natural number N such that,

$$\text{if } n > N, \text{ then } |a_n - c| < \delta.$$

By our choice of δ , this means that

$$|f(a_n) - l| < \varepsilon,$$

showing that

$$\lim_{n \rightarrow \infty} f(a_n) = l.$$

Suppose, conversely, that $\lim_{n \rightarrow \infty} f(a_n) = l$ for every sequence $\{a_n\}$ with $\lim_{n \rightarrow \infty} a_n = c$. If $\lim_{x \rightarrow c} f(x) = l$ were *not* true, there would be some $\varepsilon > 0$ such that for every $\delta > 0$ there is an x with

$$0 < |x - c| < \delta \quad \text{but} \quad |f(x) - l| > \varepsilon.$$

In particular, for each n there would be a number x_n such that

$$0 < |x_n - c| < \frac{1}{n} \quad \text{but} \quad |f(x_n) - l| > \varepsilon.$$

Now the sequence $\{x_n\}$ clearly converges to c but, since $|f(x_n) - l| > \varepsilon$ for all n , the sequence $\{f(x_n)\}$ does not converge to l . This contradicts the hypothesis, so $\lim_{x \rightarrow c} f(x) = l$ must be true. ■

Theorem 1 provides many examples of convergent sequences. For example, the sequences $\{a_n\}$ and $\{b_n\}$ defined by

$$a_n = \sin\left(13 + \frac{1}{n^2}\right)$$

$$b_n = \cos\left(\sin\left(1 + (-1)^n \cdot \frac{1}{n}\right)\right),$$

clearly converge to $\sin(13)$ and $\cos(\sin(1))$, respectively. It is important, however, to have some criteria guaranteeing convergence of sequences which are not obviously of this sort. There is one important criterion which is very easy to prove, but which is the basis for all other results. This criterion is stated in terms of concepts defined for functions, which therefore apply also to sequences: a sequence $\{a_n\}$ is **increasing** if $a_{n+1} > a_n$ for all n , **nondecreasing** if $a_{n+1} \geq a_n$ for all n , and **bounded above** if there is a number M such that $a_n \leq M$ for all n ; there are similar definitions for sequences which are decreasing, nonincreasing, and bounded below.

THEOREM 2 If $\{a_n\}$ is nondecreasing and bounded above, then $\{a_n\}$ converges (a similar statement is true if $\{a_n\}$ is nonincreasing and bounded below).

PROOF The set A consisting of all numbers a_n is, by assumption, bounded above, so A has a least upper bound α . We claim that $\lim_{n \rightarrow \infty} a_n = \alpha$ (Figure 5). In fact, if $\varepsilon > 0$, there is some a_N satisfying $\alpha - a_N < \varepsilon$, since α is the least upper bound of A . Then if $n > N$ we have

$$a_n \geq a_N, \quad \text{so} \quad \alpha - a_n \leq \alpha - a_N < \varepsilon.$$

This proves that $\lim_{n \rightarrow \infty} a_n = \alpha$. ■

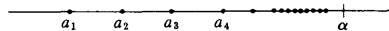


FIGURE 5